

Lecture Notes for University Physics

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August 23, 2022

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Chapter 1

Force and motion

1.1 Introduction and review

Scientific method

To do science we need a series of *scientific methods*. Theory and experiment are the basic building blocks of science that being a cycle of them. Theory should both predict and explain, experiment should be reproducible. Experimenters must agree the same observational results of an experiment no matter who perform it in anywhere and at anytime.

A theory being scientific does not necessarily imply the theory being true. For example, Newtonian mechanics is scientific that it is merely valid in some specific circumstances of domain. Newton's theory does not handle and explain mechanics correctly such as in the cases of high speed moving objects, physics in some cosmological phenomena, and describing the behaviours of small particles. Einstein's special relativity is able to correct most of the deficiencies of Newton's theory in inertial system in the circumstance that gravity is neglected.

Can we prove a theory correct?

Supposed a proposed theory is saying that all living things on earth cannot survive in boiling water. If we want to prove the theory true we probably have to perform a series of experiments, such as to put all living creatures into boiling water to see if they are able to survive. Obviously, it does not sound a good idea and probably is impossible to demonstrate in reality.

On the contrary we may be able to show a theory false. In the example, if we could find some living things which might be able to survive in boiling water to falsify the theory. In fact, scientists have already found some special crabs, shrimps and plants living around the deep sea volcanoes capable of surviving in boiling sea water.

Note. Physical theory could only be **falsified**. A not yet falsified logical and mathematical consistent theory is a tentative or a viable theory.

What is physics?

Physics is a branch of science by using scientific methods to study the fundamental principles governing the universe, and to discover the implications of those laws. Usually the not so mature and developed engineering or technological application problems also belong to physics. After 1900 to understand physics by using the traditional conceptual framework was gradually changed.

Deterministic and probabilistic

Laplace suggested that we could theoretically predict the happenings of future events if we have the adequate and enough quantities of information, parameters, initial or boundary conditions, etc., to satisfy the corresponding mathematical models. Lets say considering a body moving in a horizontal frictionless plane with a velocity v , the displacement s of the body at a given time t is easily to be calculated by the formula

$$v = \frac{ds}{dt}$$

with the help of some given boundary conditions such as $v = v_0$ when $t = t_0$. The theory is *deterministic*. Nevertheless, not all physical theory is like this. A new fundamental framework constituting the basic building block of physics was born in the beginning of 20th century—quantum theory is *probabilistic*. At the quantum level,¹ the best we are able to obtain is only to predict the probabilities of the occurrences of some physical states of some test quantum particles. In short, there are two big categories of physical theory:

- Classical physics
 - e.g. Newtonian mechanics, electrodynamics, Einstein’s special and general relativity
- Quantum physics
 - e.g. quantum mechanics, atomic and nuclear physics

Note. Most of mechanical properties and behaviours (except gravity and nuclear physics) of bodies can be interpreted in terms of electromagnetic theory. For example, frictional force is the electromagnetic interactions of electrons and nuclei between the contacting surfaces of substances.

How to learn physics

This question surely follows to one of the standard answers is you being told to work hard. Indeed, having an effective way to work hard is even better.

To undergo a formal training of physics you will unavoidably face a large number of equations. It would be difficult and sometimes would be boring to memorize those formulas unless you are very fond of doing this or perhaps memorizing things is your hobbit. The best way that clearly is to understand and digest the physical content of the

¹Here it implies doing science in a scale smaller than molecular size. Some topics of biology and medicine also require quantum physics if the object of investigation is small enough.

principles and concepts, use your understanding to derive and write down the proposed physical formulas. Reading few more different styles of reference books is definitely helpful. Sometimes it is frustrated that many physical theories are not supposed to be understood at college level, actually some of them have not yet been solved or explained. Conversely, it is good if you really find that the explanations of those texts of college physics do not satisfy your demand and curiosity.

Do not just memorize the mathematical symbols of formulas although some of them are pretty standard. One should try to understand the meaning of equations. Recalling the example velocity is defined by

$$v = \frac{ds}{dt}.$$

A better way to memorize the formula is to understand “velocity being the rate of change of displacement with respect to time.” You are free to use any symbols for the equations as long as the symbols are consistent throughout the context.

Example 1.1. Interpret

$$F \propto \frac{m_1 m_2}{r^2}$$

where F is the Newton gravitational force between masses m_1 and m_2 separated by a distance r .

Solution. Gravitational force is directly proportional to the masses m_1 and m_2 , and inversely proportional to the square of the distance between the centres of the masses of the objects.

Either m_1 or m_2 increases, F increases *linearly*. Increasing r tends to decrease F *quadratically*, so that steadily increasing the distance r resulting the decrease of the force F is effective than steadily increasing one of the masses for the increase of F at the same pace.

Note. The expression $F \propto \frac{1}{r^2}$ is called the *inverse square law*.

Metric system

Metric which has the sense of a scale is pre-supposed to be linear.² The official name of the metric system is the *Système International*, or SI.

Table 1.2 shows the common metric prefixes. As a special example, the prefix centi- is reserved for length only, i.e.,

$$1 \text{ centimetre} = 1 \text{ cm} = 10^{-2} \text{ m}$$

Dimensional analysis

The fundamental units of physical quantities most often used are **length**, **mass** and **time** (and electric charge). Others may be written as the combinations of them. Since the dimensions of the combined units of an equation are always kept invariant (no change), we use the feature to find the unknown units of some physical quantities in a mathematical expression. This technique is called *dimensional analysis*.

²A quantity to measure the scales of length and time is also called a metric which is defined as a distance function of the space and time coordinates by using the Pythagorean theorem.

ISQ base quantity	SI base unit
Length	metre (m)
Mass	kilogram (kg)
Time	second (s)
Electrical current	ampere (A)
Thermodynamic temperature	kelvin (K)
Amount of substance	mole (mol)
Luminous intensity	candela (cd)

Table 1.1: ISQ is the International System of Quantities

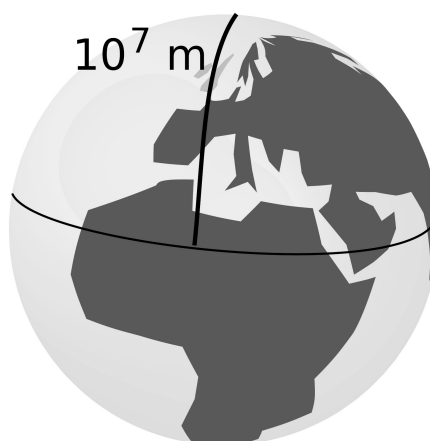


Figure 1.1: One metre is defined as equal to one ten-millionth of the geodesic distance between the north pole and the equator intersecting Paris. Diagram copied from B Crowell 2017 [The Light and Matter](#)

Example 1.2 (Dimensional analysis). Newton's gravitational equation is

$$F = \frac{Gm_1m_2}{r^2}.$$

What is the unit of the *gravitational constant* G ?

Solution. We have to find the unit of F first. Since, assuming you know the Newton's second law,

$$\text{force} = \text{mass} \times \text{acceleration}$$

where the dimension of mass is M , acceleration is the rate of change of the rate of change of the displacement, together i.e. $M \times L/T^2$. Thus, the relation amongst the dimensions of the gravitational equation is

$$\frac{ML}{T^2} = [G] \frac{M^2}{L^2} \quad \Rightarrow \quad [G] = \frac{L^3}{MT^2}.$$

resulting the unit of G being $\text{m}^3\text{kg}^{-1}\text{s}^{-2}$

10^{-9}	nano	n
10^{-6}	micro	μ
10^{-3}	milli	m
10^3	kilo	k
10^6	mega	M
10^9	giga	G

Table 1.2: Common metric prefixes

Figures and estimations

The *order of magnitude* of a number is the **power** of 10 that most closely approximates it. The examples show each group having the same order of magnitude;

$$10^{4.7} \sim 10^5 \sim 10^{4.9}, \quad 10^{-3.2} \sim 10^{-3} \sim 10^{-3.4}$$

To write any number x in a power form simply take the logarithm then exponentiate of it with the same base

$$x = 10^{\log x}$$

where $\log = \log_{10}$.

Example 1.3.

$$250 = 10^{\log 250} \approx 10^{2.397}$$

which may round to 10^2 for expressing the order of magnitude of the number.

Definition 1.1. A number in *scientific notation* is expressed by

$$a \times 10^b \quad \forall a, b \in \mathbb{R}$$

For examples, 2.4×10^3 , 5.667×10^{-12} etc.

Since $\sqrt{10} = 10^{0.5} \approx 3$ is halfway between $1 = 10^0$ and $10 = 10^1$ on a log base-10 scale, any number of the leading term $a \geq 3$ is regarded as 10^1 , $a < 3$ is regarded as 10^0 for the order of magnitude approximation scheme. Thus, the orders of magnitude of the examples are $2.4 \times 10^3 \sim 10^3$ and $5.667 \times 10^{-12} \sim 10^{-11}$, respectively.

The *uncertainty* of a measurement is related to the discrepancy of the accepted reference value A . It is denoted as ΔA .

Definition 1.2. The *percent uncertainty* is defined by

$$\frac{\Delta A}{A} \times 100\% \quad (1.1)$$

1.2 Scaling and order-of-magnitude estimates

1.2.1 Scaling of area and volume

Area and volume

Consider a cube with side length L . The surface area and the volume are

$$A = 6L^2 \quad \text{and} \quad V = L^3.$$

The area-volume ratio is

$$\frac{A}{V} = \frac{6L^2}{L^3} \propto \frac{1}{L}$$

which varies depending on L . In general, area-volume ratio does affect the physical properties of an object. As an example, the mass of a cube depends on the volume as with the assumption that the density is uniformly distributed, however, the force exerted on the cube by pressure depends on the surface area. If we change the A/V ratio the two stated physical properties will also be changed.

Example 1.4. A toy fire engine is $1/30$ the size of the real one, but it is constructed from the same metal with the same proportions. How many times smaller is its weight? How many times less red paint would be needed to paint it?

Solution. Suppose that the fire engine is a cube with length L . The volume ratio of the toy to the real one is

$$\left(\frac{L}{30}\right)^3 \times \left(\frac{1}{L}\right)^3 = \frac{1}{27000}.$$

The weight of an object is directly proportional to the volume, hence the volume ratio just equals to the weight ratio. On the other hand, the area ratio of the toy to the real one is

$$\left(\frac{L}{30}\right)^2 \times \left(\frac{1}{L}\right)^2 = \frac{1}{900}.$$

Thus, 900 times less red paint would be needed.

1.2.2 Scaling applied to biology

Suppose that all animals have the same average density, the body mass is

$$m = \text{density} \times \text{volume} \Rightarrow m \propto L^3$$

where L is the dimension of the body. Thus

$$m^{1/3} \propto L \Rightarrow m^{2/3} \propto L^2 \Rightarrow m^{2/3} \propto \text{area}$$

Increasing the mass of a body with a fixed shape surely causes the increasing of the surface area, however the latter is not fast enough to catch up the former.

Note. Heat absorption and emission of an animal mainly rely on the surface area of the body skin. Larger area-mass ratio animals (small animals) tend to absorb and transmit heat energy faster than the animals with smaller area-mass ratio (larger animals).

Changes in shape to accommodate changes in size

Animals must be able to support their own weights. The strength of a rigid body depends on the *cross sectional area* A . The ratio of strength to weight is

$$\frac{\text{strength}}{\text{weight}} \propto \frac{A}{V} \propto \frac{1}{L} \quad (1.2)$$

Thus, the ability of an object to support its own weight decreases by inversely proportional to its linear dimensions.

Example 1.5. Single-cell animals must passively absorb nutrients and oxygen from their surroundings, unlike humans who have lungs to pump air in and out and a heart to distribute the oxygenated blood throughout their bodies. Even the cells composing the bodies of multicellular animals must absorb oxygen from a nearby capillary through their surfaces. Based on these facts, explain why cells are always microscopic in size.

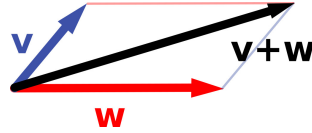
Solution. The high area-volume ratio of microscopic size of cell provides a rigid structure against the weight of the cell, as well as greatly increase the total surface area of a group of cells.

1.3 Vectors

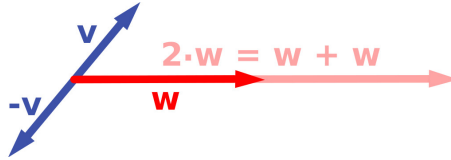
Vector is a mathematical quantity consisting of **magnitude** and **direction**, where the magnitude is a *scalar*.

There are many ways to write a vector, e.g., $\mathbf{V}$, $\vec{V}$, $|V\rangle$ or simply just V . A vector connecting two points A, B in Euclidean space is a displacement vector V_{AB} .

Addition of two vectors is given by the parallelogram rule:



As examples doubling the length of $\mathbf{w}$ and the addition of $\mathbf{v}$ with it's *inverse* $-\mathbf{v}$ may be shown as



Definition 1.3. A *unit vector* is a vector whose magnitude is always unity (i.e., 1). A vector divided by its magnitude gives the unit vector

$$\hat{\mathbf{V}} = \frac{\mathbf{V}}{|\mathbf{V}|} \quad (1.3)$$

Example of the 3D unit vectors are $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ for Cartesian coordinate systems, or the general *orthonormal* basis $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3, \dots$

A *vector space* $\mathbb{V}$ is a set on which *vector addition* and *scalar multiplication* are defined, satisfying the axioms for $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{V}$ and $\alpha, \beta \in \mathbb{R}$:

1. $\mathbf{u} + \mathbf{v} \in \mathbb{V}$
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
4. $\mathbf{u} + \mathbf{0} = \mathbf{u}$ where $\mathbf{0} \in \mathbb{V}$ is a zero vector
5. $\forall \mathbf{u} \in \mathbb{V}, \exists (-\mathbf{u}) \in \mathbb{V}$ s.t. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

6. $\alpha \mathbf{u} \in \mathbb{V}$
7. $\alpha(\mathbf{u} + \mathbf{v}) = \alpha \mathbf{u} + \alpha \mathbf{v}$
8. $(\alpha + \beta)\mathbf{u} = \alpha \mathbf{u} + \beta \mathbf{u}$
9. $\alpha(\beta \mathbf{u}) = (\alpha\beta)\mathbf{u}$
10. $1(\mathbf{u}) = \mathbf{u}$

1.3.1 Dot product, components and basis

To find the magnitude of the product of two vectors **along the same direction**, the *dot product* of them

$$\mathbf{v} \cdot \mathbf{u} = |\mathbf{v}||\mathbf{u}| \cos \theta \quad (1.4)$$

is a **scalar** quantity, where $\mathbf{v}, \mathbf{u} \in \mathbb{V}$, θ is the angle between the vectors.

The implicit meaning of (1.4) is that a dot product is a measure of the “relation” between two vectors. If they are perpendicular to each other, i.e., $\theta = \pi/2$, the dot product is zero $\mathbf{v} \cdot \mathbf{u} = 0$. On the other hand, if $\theta = 0$, then $\mathbf{v} \cdot \mathbf{u} = |\mathbf{v}||\mathbf{u}|$, giving a maximum positive number of the vectors relate to. Therefore, $\mathbf{v} \cdot \mathbf{u} = 0$ implies that $\mathbf{v}$ and $\mathbf{u}$ are unrelated, or independent to each other.

Two vectors $\mathbf{v}, \mathbf{u}$ are independent to each others if $\mathbf{v} \cdot \mathbf{u} = 0$, i.e., 90° in Euclidean space.

Linear combination

A vector $\mathbf{v} \in \mathbb{V}$ is a linear combination of the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ in $\mathbb{V}$ such that

$$\mathbf{v} = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k$$

where $c_1, c_2, \dots, c_k \in \mathbb{R}$.

Definition 1.4. A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in $\mathbb{V}$ is *linearly independent* when

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k = \mathbf{0}$$

has only the *trivial solution* $c_1 = 0, c_2 = 0, \dots, c_k = 0$. Otherwise, S is *linear dependent*.

Components and basis

A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{V}$ is a basis for $\mathbb{V}$ when

- S spans $\mathbb{V}$
- S is linearly independent

A set S of vectors in an inner (dot) product space $\mathbb{V}$ is *orthogonal* when every pair of vectors in S is orthogonal. If, in addition, each vector in the set is a unit vector, then S is *orthonormal*.

Thus, in an n -dimensional Euclidean space a vector $\mathbb{V}$ is written as

$$\mathbf{V} = V^1 \mathbf{e}_1 + V^2 \mathbf{e}_2 + \dots + V^n \mathbf{e}_n = \sum_{i=1}^n V^i \mathbf{e}_i \quad (1.5)$$

where V^i and $\mathbf{e}_i$ are the components and the corresponding *orthonormal basis*.

In mathematical physics, we often omit the summation sign for the repeated indices of the superscripts and subscripts as

$$\mathbf{V} = \sum_{i=1}^n V^i \mathbf{e}_i := V^i \mathbf{e}_i.$$

The abbreviation is known as *Einstein's summation*.

Example 1.6. A vector $\mathbf{F}$ in 2-dimensional Euclidean space can be resolved written in terms of two orthogonal directions (i.e. $\perp$),

$$\mathbf{F} = F_x \mathbf{e}_x + F_y \mathbf{e}_y$$

where $F_x = |\mathbf{F}| \cos \theta$ and $F_y = |\mathbf{F}| \sin \theta$ are the *components*, $\mathbf{e}_x$ and $\mathbf{e}_y$ are the unit vectors or orthonormal basis along the x and y axes of a *Cartesian coordinate system*. The popular alternative notation for the system is $\mathbf{F} = F_x \hat{\mathbf{i}} + F_y \hat{\mathbf{j}}$.

The basis vectors obey the dot product rules, for any orthonormal basis $\mathbf{e}_x, \mathbf{e}_y$,

$$\mathbf{e}_x \cdot \mathbf{e}_y = \mathbf{e}_y \cdot \mathbf{e}_x = 0, \quad \mathbf{e}_x \cdot \mathbf{e}_x = \mathbf{e}_y \cdot \mathbf{e}_y = 1.$$

We may use the *Kronecker delta* to generalize the orthonormal basis properties for n -dimensional space

$$\mathbf{e}_a \cdot \mathbf{e}_b = \delta_{ab} = \begin{cases} 1 & a = b, \\ 0 & \text{otherwise.} \end{cases}$$

Example 1.7. Two 3D vectors $\mathbf{A} = A_x \mathbf{e}_x + A_y \mathbf{e}_y + A_z \mathbf{e}_z$ and $\mathbf{B} = B_x \mathbf{e}_x + B_y \mathbf{e}_y + B_z \mathbf{e}_z$ in the same Cartesian coordinate system gives the dot product

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= (A_x \mathbf{e}_x + A_y \mathbf{e}_y + A_z \mathbf{e}_z) \cdot (B_x \mathbf{e}_x + B_y \mathbf{e}_y + B_z \mathbf{e}_z) \\ &= A_x B_x + A_y B_y + A_z B_z \end{aligned}$$

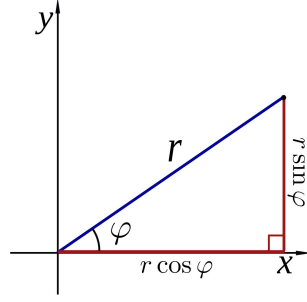
Definition 1.5. The magnitude of a vector is known as the *norm* (e.g. the length) defined by $\|\mathbf{V}\| = \sqrt{\mathbf{V} \cdot \mathbf{V}}$.

Example 1.8. The unit vector of $\mathbf{A} = 2\mathbf{e}_x - 3\mathbf{e}_y + \mathbf{e}_z$ is

$$\hat{\mathbf{A}} = \frac{2\mathbf{e}_x - 3\mathbf{e}_y + \mathbf{e}_z}{\sqrt{2^2 + (-3)^2 + 1^2}} = \frac{2}{\sqrt{14}}\mathbf{e}_x - \frac{3}{\sqrt{14}}\mathbf{e}_y + \frac{1}{\sqrt{14}}\mathbf{e}_z$$

Example 1.9 (Polar coordinate system). It is a 2D coordinate system in which each point on a plane is determined by a distance from a reference point and an angle from a reference direction. The coordinates can be converted by

$$x = r \cos \varphi, \quad y = r \sin \varphi$$



Given a *position vector* $\mathbf{r}$, its length is simply $r = \sqrt{x^2 + y^2}$. The two orthonormal basis for polar coordinate system are

$$\mathbf{e}_r, \quad \mathbf{e}_\varphi$$

where $\mathbf{e}_r$ is in the *radial* direction lying along $\mathbf{r}$, $\mathbf{e}_\varphi$ is chosen to be positive perpendicular to $\mathbf{e}_r$ and away from $\mathbf{e}_r$ in an anti-clockwise way on the xy -plane.

The alternative symbols,

$$\mathbf{e}_r = \hat{\mathbf{r}}, \quad \mathbf{e}_\varphi = \hat{\boldsymbol{\varphi}}$$

In circular motion, $\hat{\boldsymbol{\varphi}} = \hat{\mathbf{t}}$ is in the tangential direction.

1.3.2 Cross product

Cross product is a special vector product only working in 3-dimensional Euclidean space. The result is a vector. As a vector comprises the direction, the order of a cross product between two vectors is important;

$$\mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}$$

The cross product of $\mathbf{v}$, $\mathbf{u}$ is defined by

$$\mathbf{v} \times \mathbf{u} := |\mathbf{v}||\mathbf{u}| \sin \theta \mathbf{e}_w \quad (1.6)$$

where $\mathbf{e}_w$ is the unit vector of the resultant, which must be perpendicular to $\mathbf{v}$ and $\mathbf{u}$.

The cross products among orthonormal basis are

$$\mathbf{e}_x \times \mathbf{e}_y = \mathbf{e}_z, \quad \mathbf{e}_y \times \mathbf{e}_z = \mathbf{e}_x, \quad \mathbf{e}_z \times \mathbf{e}_x = \mathbf{e}_y,$$

$$\mathbf{e}_x \times \mathbf{e}_x = \mathbf{e}_y \times \mathbf{e}_y = \mathbf{e}_z \times \mathbf{e}_z = 0$$

As by the definition cycling the order $x \rightarrow y \rightarrow z$ is positive, a negative sign appears if swapping the positions of any two basis.

Example 1.10. Express the cross product of two 3D vectors $\mathbf{A}$ and $\mathbf{B}$ in coordinates.

As the previous,

$$\mathbf{A} = A_x \mathbf{e}_x + A_y \mathbf{e}_y + A_z \mathbf{e}_z, \quad \mathbf{B} = B_x \mathbf{e}_x + B_y \mathbf{e}_y + B_z \mathbf{e}_z$$

The cross product is

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A_x \mathbf{e}_x + A_y \mathbf{e}_y + A_z \mathbf{e}_z) \times (B_x \mathbf{e}_x + B_y \mathbf{e}_y + B_z \mathbf{e}_z) \\ &= (A_y B_z - A_z B_y) \mathbf{e}_x + (A_z B_x - A_x B_z) \mathbf{e}_y + (A_x B_y - A_y B_x) \mathbf{e}_z \end{aligned}$$

1.4 Motion along a straight line

Mechanics is roughly described by two branches; *dynamics* and *kinematics*. The former is concerned with the causes of motion. In Newton's laws of motion it involves force. On the other hand, kinematics involves describing motion through properties such as position, time, velocity, and acceleration without explicitly referring to what causes it.

Centre of mass

Any physical motion of a rigid body can be analyzed into linear translation and its rotation.

As the classical mechanics is only good for point particles, the linear translation and the rotation of a rigid body are regarded as the motion of and rotating about a special point—*centre of mass* of the body. That is, a point somewhere inside or outside of a rigid body possesses the physical properties as if the object were a point particle.

Example 1.11. How to find the centre of mass of a rigid body?

Solution. Suspending a rigid body by a string under gravity to get the first vertical line in which the centre of mass of the body must lie along the line. Suspend the object by choosing two more different arbitrary suspending positions to get the second and the third straight lines for a 3-dimensional object. The centre of mass of the body is at the intersection of the lines.

1.4.1 Distance and time

Distance which is a *scalar* describes the magnitude of the distance separation between two points. The vector quantity—*displacement* includes both the pointing direction and the magnitude of the length. Distance is one-dimensional.

Time Opposite to Newton's idea, time is supposed being a predefined parameter of a metric scale for helping us to define the process of evolution of an event. It is also one-dimensional by definition. Nevertheless, we adopt Newton's absolute time concept throughout the book except the chapter relating to Einstein's relativistic theory.

Frames of reference

How do we know a body in motion? As an example we can see the movement of a train if we are not staying inside. Inside there it appears motionless, supposed that all the windows are closed and the train moves smoothly and quietly. Motion is relative. To define a relative motion we conveniently prefer to put a *frame of reference* with coordinates to measure the change of the relative distance (with respect to the time interval) between the event and the frame. A typical example of reference frames is a Cartesian coordinate system.

1.4.2 Velocity and speed

Speed is a scalar. It is the rate of change of distance per unit time.

Velocity is a vector defined by the rate of change of displacement. Taking the absolute value of a velocity gives the corresponding speed.

Definition 1.6. The instantaneous velocity $\mathbf{v}$ is the rate of change of displacement. Mathematically,

$$\mathbf{v} = \frac{d\mathbf{s}}{dt}$$

where $\mathbf{s}$ and t are the variables of displacement and time.

Sometimes we do not know the detail of the changes of displacements over a time interval of a body. We may calculate the *average velocity* of the body over the time interval Δt :

$$\mathbf{v} = \frac{\Delta \mathbf{s}}{\Delta t}$$

where $\Delta \mathbf{s}$ is the change of displacement. Average velocity becomes the instantaneous velocity when $\Delta t \rightarrow 0$:

$$\frac{d\mathbf{s}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{s}}{\Delta t}$$

Relative velocity

Velocity is *relative*. Different *observer* measures different velocity depending on the reference system. Suppose that there are two bodies moving with $\mathbf{v}_1$ and $\mathbf{v}_2$ relative to some reference, the relative velocity of the separation between the bodies relative to $\mathbf{v}_2$ is $\mathbf{v}_1 - \mathbf{v}_2$, i.e., the relative velocity between the bodies. Note that the result which is observed by the second moving object $\mathbf{v}_2$ does not depend on other reference system.

Example 1.12. A simple way to show the relative velocity of two vectors is suppose $\mathbf{V}_{OA}, \mathbf{V}_{OB}$ are the velocities of two points A and B relative to the origin O . The relative velocity of A to B denoted by $\mathbf{V}_{BA}$ is simply

$$\mathbf{V}_{BA} = \mathbf{V}_{BO} + \mathbf{V}_{OA} = (-\mathbf{V}_{OB}) + \mathbf{V}_{OA} = \mathbf{V}_{OA} - \mathbf{V}_{OB}$$

You may draw a simple triangle to see the relationships.

Example 1.13. Catherine is riding on her horse at a velocity of 11 m/s. She twists around in her saddle and fires an arrow backward. Her bow fires arrows at 25 m/s. Find the speed of the arrow relative to the ground.

Solution. If we set the riding direction positive, the backward direction of the fired arrow is negative. The speed of the fired arrow relative to the ground is

$$11 + (-25) = -14 \text{ ms}^{-1}.$$

The velocity of the arrow is a lot slower viewed by the people on ground.

Example 1.14. If an object had a straight-line motion graph with $\Delta x = 0$ and $\Delta t \neq 0$, what would be true about its velocity? What would this look like on a graph? What about $\Delta t = 0$ and $\Delta x \neq 0$?

Solution. The object is motionless in the first case, and can be represented as a horizontal line, i.e. $v = 0$. In the second situation, it is a vertical line which is unphysical since it implies to the body moving from one to other places with no time taken. Mathematically, $v \rightarrow \infty$.

1.4.3 The principle of inertia

Theorem 1 (Galileo’s principle of inertia). *No force is required to maintain motion with constant velocity in a straight line, and absolute motion does not cause any observable physical effects.*

Example 1.15. A passenger on a cruise ship finds, while the ship is docked, that he can leap off of the upper deck and just barely make it into the pool on the lower deck. If the ship leaves dock and is cruising rapidly, will this adrenaline junkie still be able to make it?

Solution. As long as the ship is moving in constant velocity (wind is ignored), he can jump to the same position of the pool as if the ship is docked motionless.

1.4.4 Acceleration

Definition 1.7. Acceleration is defined by the rate of change of velocity.

Velocity $\mathbf{v}$ is the rate of change of displacement $\mathbf{s}$, thus acceleration $\mathbf{a}$ is the rate of change of the rate of change of displacement—the second derivative of displacement with respect to time t . That is,

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{s}}{dt^2}$$

in the unit of ‘metre per square second’.

Calculus were developed by Newton in order to formulate his theory of mechanics. The other main contributor of this mathematical aspect is Leibniz.

Example 1.16. Can an object have a velocity that is positive and increasing at the same time that its acceleration is decreasing?

Solution. Consider a velocity function $v(t)$ of time. Positive $v(t)$ means that it is above the t -axis. Increasing means the curve $v(t)$ is moving upward according to the three different cases that the curve $v(t)$ may be bending as concave upward $\cup$, a straight line $/$, or concave downward $\cap$. Acceleration is the derivative of $v(t)$ with respect to time. The bending concave downward increasing curve $v(t)$ shows the decreasing of acceleration $d^2v/dt^2 < 0$. Thus, the answer is yes.

Gravitational acceleration

The story tells Galileo Galilei dropping different objects in various sizes and compositions at the top of Leaning Tower of Pisa to demonstrate gravity.

Theorem 2 (Galileo’s principle of equivalence). *All object falls at the same rate under gravity.*

The dropping ‘rate’ is by definition the *gravitational acceleration* (gravitational field) later defined by Newton. The average value of gravitational acceleration on earth’s surface is about 9.8 ms^{-2} denoted by g .

1.4.5 Motion with constant acceleration

In many daily applications with in good approximations, we make the assumption that acceleration is constant implying that the average and instantaneous accelerations are equal.

Setting the initial time $t = 0$, the initial velocity u (i.e., initial conditions) with **constant acceleration** a , we may derive a set of useful *kinematic equations* for solving simple daily live problems. We perform an anti-derivative of the definition of velocity gives

$$v = \frac{ds}{dt} \Rightarrow \int_0^t v dt = \int_0^s ds \Rightarrow vt = s. \quad (1.7)$$

The conditions that we have imposed on the definite integrals are called *boundary conditions* or simply *physical conditions*. Similarly,

$$\begin{aligned} a &= \frac{dv}{dt} \\ \int_0^t a dt &= \int_u^v dv \\ at &= v - u \\ \int_0^t at dt &= \int_0^t (v - u) dt \\ \frac{1}{2}at^2 &= \int_0^t \frac{ds}{dt} dt - ut \end{aligned} \quad (1.8)$$

where u being a fixed value for the initial condition does not depend on t . Therefore,

$$s = ut + \frac{1}{2}at^2. \quad (1.9)$$

Use the chain rule to find the expression without t ,

$$a = \frac{dv}{dt} = \frac{dv}{ds} \frac{ds}{dt} = \frac{dv}{ds} v.$$

Integrating the equation we get

$$\int_0^s a ds = \int_u^v v dv \Rightarrow as = \frac{1}{2}(v^2 - u^2) \Rightarrow v^2 = u^2 + 2as. \quad (1.10)$$

As this set of equations involves the common notations s, u, v, a and t , they are also simply called as “suvat-equations”.

Example 1.17. Two people stand on the edge of a cliff. As they lean over the edge, one person throws a rock down, while the other throws one straight up with an exactly opposite initial velocity. Compare the speeds of the rocks on impact at the bottom of the cliff.

Solution. Suppose that the initial velocity of the upward rock is u , the downward rock is $-u$. The displacement s of the two cases are the same (from the cliff to the bottom). We see that they give the same final velocity v by (1.10) when the rocks hit the bottom of the cliff.

Example 1.18. You climb half-way up a tree, and drop a rock. Then you climb to the top, and another rock. How many times greater is the velocity of the second rock on impact?

Solution. Let $v_h, v_{h/2}$ be the velocities of the rocks on impact, let h be the height of the tree. Using (1.10) $v^2 = u^2 + 2gs$ with the initial velocity $u = 0$ we have the ratio

$$\left(\frac{v_h}{v_{h/2}}\right)^2 = \frac{2gh}{2gh/2} = 2.$$

Therefore, $v_h = v_{h/2}\sqrt{2}$. The velocity of the second rock is $\sqrt{2}$ times faster than the first rock.

Example 1.19. A person fires a pistol straight up. The bullet leaves the gun at a speed of 100 m/s. How long does it take before the bullet hits the ground?

Solution. For simplicity, we ignore air resistance. Setting the upward direction positive we have

$$u = 100, \quad g = -9.8, \quad s = 0.$$

Directly use (1.9) for solving t ,

$$\begin{aligned} s &= ut + \frac{1}{2}gt^2 \\ 0 &= 100t + \frac{1}{2}(-9.8)t^2 \\ t &= 20.4. \end{aligned}$$

The bullet takes about 20 seconds before hitting the ground (We have rejected $t = 0$ case).

1.5 Motion in 2 & 3 dimensions

1.5.1 Projectile motion

A projectile motion is the motion of an object thrown or projected into the air, subject only to acceleration as a result of gravity. Figure 1.2 showing a *projectile* is launched at an angle θ_i and with initial velocity v_i . A solution is required for the range R , the distance at which the projectile hits the ground.

This type of projectile motion moves in a 2-dimensional plane of a 3-dimensional space (and time) as we may only need to handle the vertical (y -axis) and the horizontal (x -axis) components without any spatial ‘thickness’ here. Once we resolve the motion into these two independent directions we have to apply Newton’s absolute time idea to fix a single universal time t . As gravitational acceleration is assumed being constant we write down the corresponding suvat-equations that the horizontal is in uniform motion and the vertical is in free fall due to gravity.

Example 1.20. Find the equation for the range R and the maximum.

Solution. Resolve the velocity v_i into the vertical and horizontal directions by

Vertical: $s = v_i t \sin \theta_i + \frac{1}{2}(-g)t^2$

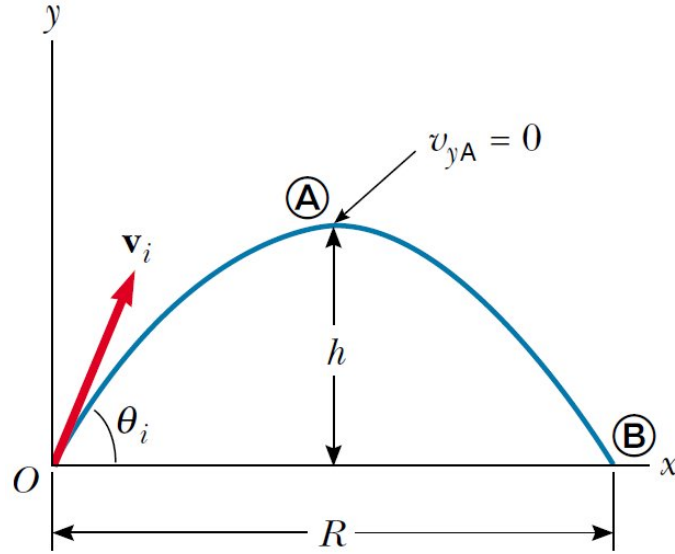


Figure 1.2: A particle is launched at an angle θ_i with the launching velocity $\mathbf{v}_i$.

Horizontal: $R = v_i t \cos \theta_i$

Putting $s = 0$ and eliminate t we obtain

$$R = \frac{2v_i^2 \sin \theta_i \cos \theta_i}{g} = \frac{v_i^2 \sin 2\theta_i}{g}.$$

For the maximum range, putting $\sin 2\theta_i = 1$ gives $\theta_i = \pi/4$. Hence

$$R_{\max} = \frac{v_i^2}{g}.$$

Note. Gravity only affects the vertical direction. The horizontal component is solved by the regular average velocity equation.

1.5.2 Circular motion

Uniform circular motion

Circular motion always involves a change in the direction of the velocity, but it is also possible for the magnitude of the velocity (i.e. the speed) to change at the same time. The motion is referred to as uniform if $|\mathbf{v}|$ is constant, and nonuniform if it is changing.

In Figure 1.3, the change of the velocity is $\Delta \mathbf{v} = \mathbf{v}_f - \mathbf{v}_i$. If $|\mathbf{v}_f| = |\mathbf{v}_i|$, i.e., uniform circular motion, the acceleration

$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt}$$

is pointing to the centre O . This inward acceleration is called *centripetal acceleration*. When $\Delta \mathbf{v}$ is small, the distance traveled by the object approximately equals to the arc length Δs , i.e.

$$\Delta \theta = \frac{\Delta s}{r} = \frac{v \Delta t}{r}$$

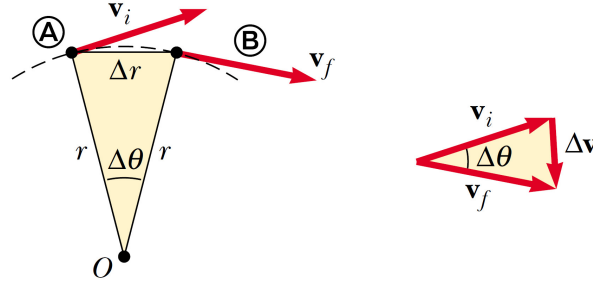


Figure 1.3: The centripetal acceleration is the rate of change of velocity.

where $\Delta\theta$ is the change of the angle during Δt , thus the speed

$$v = r \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = r \frac{d\theta}{dt} = r\dot{\theta} \quad (1.11)$$

where $\dot{\theta}$, or written as ω , is the *angular speed* (angular velocity), where the $\dot{}$ stands for the derivative of something w.r.t. time.

Non-uniform circular motion

If centripetal acceleration increases, the radius of an circular moving object will decrease accordingly, to move inward spirally. It will collide with the centre of the circular motion system at a finite time.

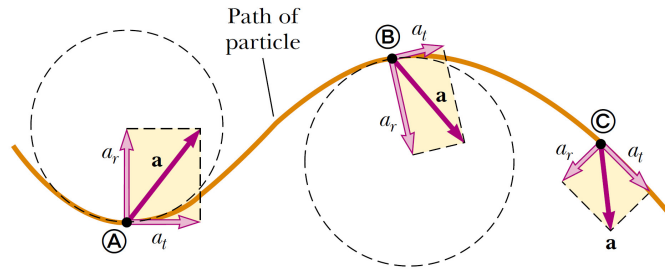


Figure 1.4: The motion of a particle along an arbitrary curved path lying in the xy plane.

At any instant the position of a moving point particle may be regarded as the origin in that instantaneous frame. Therefore at any moment a particle undergoes a non-uniform circular motion. The velocity vector $\mathbf{v}$ is always tangent to the path, but the direction of $\mathbf{a}$ changes from point to point. Figure 1.4 shows the acceleration being resolved into the radial $\mathbf{a}_r$ and the tangential $\mathbf{a}_t$. It is the vector sum

$$\mathbf{a} = \mathbf{a}_r + \mathbf{a}_t.$$

The tangential acceleration causes the change in the speed of the particle. It is parallel to the velocity, the magnitude is

$$a_t = \frac{d|\mathbf{v}|}{dt}.$$

The radial acceleration a_r which arises from the change in direction of the velocity vector has an absolute magnitude. It follows that

$$a = \sqrt{a_r^2 + a_t^2}.$$

1.5.3 Velocity and acceleration vectors

Velocity vector

Figure 1.5(a) shows a *position vector* $\mathbf{r}$ written in the direction of the radial unit vector $\mathbf{e}_r$ which being perpendicular to the $\mathbf{e}_\theta$. The whole frame “ $\mathbf{e}_r, \mathbf{e}_\theta$ ” moves along the trajectory (i.e., the path) as the change of the position vector $\mathbf{r}$. A *velocity vector* is the time

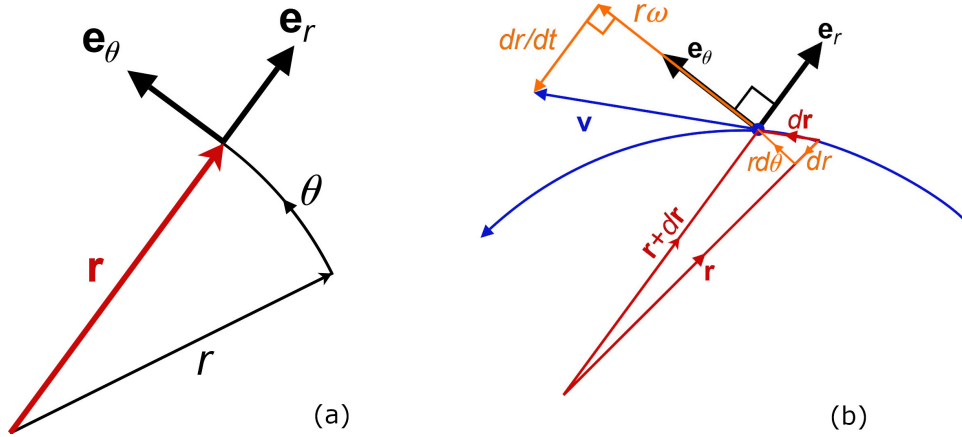


Figure 1.5: (a) Position vector, (b) Velocity vector. (Wikipedia)

derivative of a *position vector*. Since written in terms of the component and basis $\mathbf{r} = r\mathbf{e}_r$, the velocity vector becomes

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}(r\mathbf{e}_r) = \frac{dr}{dt}\mathbf{e}_r + r\frac{d\mathbf{e}_r}{dt} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r$$

where $\mathbf{e}_r$ is the orthonormal basis (unit vector) along $\mathbf{r}$, the ‘dot’ stands for the derivative w.r.t. time for convenience. But what is $\dot{\mathbf{e}}_r$? Since the change of $\mathbf{e}_r$ must be pointing to the direction (angular direction) that perpendicular to itself, i.e., $\mathbf{e}_\theta$, and the magnitude of the rate of change of it is the *angular velocity* (*angular frequency*):

$$\left|\frac{d}{dt}\mathbf{e}_r\right| = \frac{d}{dt}(1\theta) = \frac{d\theta}{dt} = \dot{\theta} \quad (1.12)$$

Direction and magnitude together giving $\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta$, the velocity vector is thus

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta. \quad (1.13)$$

Figure 1.5(b) shows the geometrical relation of the velocity vector and the basis. For uniform circular motion, the radius r is a constant, i.e., $\dot{r} = 0$, the angular velocity must be constant as well, with the magnitude (1.11).

Note. θ has no unit, the unit of $\dot{\theta}$ is interpreted as the number of cycles per second (s^{-1}), also known as *angular frequency*.

Acceleration vector

To writing the acceleration vector $\mathbf{a}$ in terms of the polar coordinates, differentiate the velocity vector (1.13) w.r.t. time

$$\begin{aligned}\mathbf{a} &= \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(\dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta) = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta \\ &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta \\ &= \ddot{r}\mathbf{e}_r + 2\dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta\end{aligned}$$

in which the “double dots” represents the second derivatives w.r.t time. Again, what is $\dot{\mathbf{e}}_\theta$?

The direction of $\dot{\mathbf{e}}_\theta$ must be in 180° anti-clockwise from $\mathbf{e}_r$. With the derivative of the angle we have $\dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r$. We obtain

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\mathbf{e}_\theta \quad (1.14)$$

In uniform circular motion; $\dot{r} = 0$, $\ddot{r} = 0$, $\ddot{\theta} = 0$, only the centripetal acceleration exists

$$\mathbf{a} = -r\dot{\theta}^2\mathbf{e}_r = -\frac{v^2}{r}\mathbf{e}_r, \quad \text{in magnitude } a = -\frac{v^2}{r} \quad (1.15)$$

where the negative sign shows the direction of the acceleration.

To verify this result for uniform circular motion by using our previous simple analysis, we have

$$a = \lim_{\Delta t \rightarrow 0} \frac{v_f - v_i}{\Delta t} = - \lim_{\Delta t \rightarrow 0} \frac{r_f - r_i}{\Delta t} \dot{\theta} = -v\dot{\theta} = -r\dot{\theta}^2.$$

Note. The directions of $\mathbf{a}_r$ and $\mathbf{a}_t$ in the previous are not the same as $\mathbf{e}_r$ and $\mathbf{e}_\theta$ which are always referring to the reference point $(0,0)$ of the x - y plane.

1.6 Force and motion

1.6.1 Newton's laws of motion

Newton's first law

If the total force on an object is zero, its centre of mass continues in the same state of motion.

That is, if there is no external force acting upon an object, the object will keep its state of motion with no change. Motionless and uniform motion is relative, only depending on the inertial reference system that one chooses.

Newton's second law

The state of motion of an object can be changed by an external force.

Newton construct the concept of ‘force’ to describe how hard to accelerate an object with a given mass. Alternatively, force $\mathbf{F}$ is defined as the *rate of change of momentum* of an object. Mathematically,

$$\mathbf{F} = m\mathbf{a} \quad (1.16)$$

where m is the mass of the object, $\mathbf{a}$ the acceleration.

The unit of force is “Newton” (N). 1 Newton of force is required to push a 1 kg object to accelerate at 1 ms^{-2} .

Weight The weight of an object is not its own mass. Weight is defined as a force due to gravitational acceleration by the Newton second law,

$$\mathbf{F}_W = m\mathbf{g}. \quad (1.17)$$

where $\mathbf{g}$ is the gravitational acceleration with an average magnitude 9.8 ms^{-2} on the surface of earth and always pointing downward.

Newton’s third law

Action equals to reaction.

As force being a vector, in order to keep an object in *equilibrium* state of motion (i.e., inertial motion) a force with the same magnitude but in opposite direction must be applied.

Example 1.21. A 10 kg object is suspended by a string with a force $\mathbf{F}$. Since the system is in static equilibrium (motionless) the total force (resultant) with the weight $\mathbf{W}$ is zero:

$$\begin{aligned} \mathbf{F} + \mathbf{W} &= 0 \\ \mathbf{F} &= -\mathbf{W} \\ &= -10 \times (-9.8) \\ &= 98 \text{ N} \end{aligned}$$

The positive $\mathbf{F}$ implies upward direction.

1.6.2 Frictional force

Frictional force of rigid body

Frictional force is an opposite directional induced force between surface layers, with a magnitude proportional to the *normal reaction* of them.

Definition 1.8 (Mechanical friction between rigid bodies).

$$\mathbf{f} = \mu \mathbf{N}$$

where $\mathbf{f}$ is the induced frictional force, $\mathbf{N}$ the normal reaction, μ the coefficient of friction between the contact of the materials.

⚠. In physics, ‘normal’ often means the direction perpendicular to a surface.

There are two types of coefficient of friction for the same material; the coefficients of kinetic and static frictions. Static friction is always larger than the kinetic one, $f_s > f_k$.

- Static friction increases as the applied force increases. The magnitude of the forces are always equal with opposite direction, until the maximum force of static friction is surpassed.

	μ_k	μ_s
Rubber on dry concrete	0.7	1.0
Rubber on wet concrete	0.5	0.7
Wood on wood	0.3	0.5
Waxed wood on wet snow	0.1	0.14
Steel on steel (dry)	0.3	0.6
Steel on steel (oiled)	0.03	0.05
Teflon on steel	0.04	0.04
Bone lubricated with synovial fluid	0.015	0.016
Shoes on wood	0.7	0.9
Shoes on ice	0.05	0.1
Ice on ice	0.03	0.1
Steel on ice	0.02	0.4

Table 1.3: Coefficients of kinetic and static friction of materials.

- Kinetic friction is roughly a constant no matter how large external force is applied.

Example 1.22. Why is static friction stronger than kinetic friction?

Solution. Since static frictional force is directly proportional to the applied force. The maximum static frictional force should be equal to the applied force at the moment when an object starts to move, i.e.

$$F - \mu_s N = 0$$

where N is the normal reaction. For an object starting to move, the minimum applied force must be

$$F - \mu_k N \geq 0.$$

Comparing with these equations gives

$$\mu_s N \geq \mu_k N \quad \Rightarrow \quad \mu_s \geq \mu_k$$

as shown in Figure 1.6.

1.6.3 Fluid friction

From groups of particles in microscopic sizes like atoms, molecules, and plasma to the enormous huge sizes of groups of stars, clusters and galaxies all of them can be regarded as fluid under the right conditions.

Fluid frictional (resistant) force is approximately proportional to the square of the velocity and directly proportional to the cross-sectional area A of a **not small and not slow moving object**:

$$\begin{aligned} f &\propto Av^2 \\ f &= CAv^2 \end{aligned} \tag{1.18}$$

where C is the coefficient of fluid friction depending on the type of fluid and the shape of the object. It is often integrated into the density of the fluid.

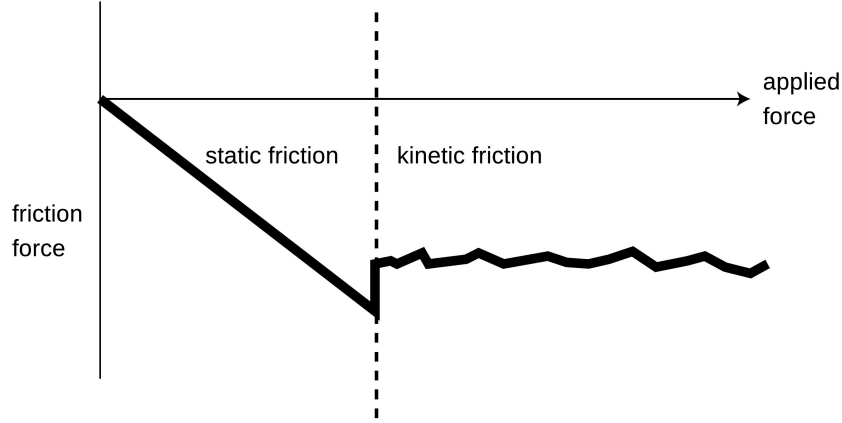


Figure 1.6: The absolute maximum value of static friction is slightly larger than kinetic friction at the instant that the body starts to move.

Motion through air

When an object moves through air the air molecules have to be pushed out of its way. The resulting reaction force pushes back against the body and retards the motion. This is the source of fluid friction in air.

Air resistance F_a is directly proportional to the transverse³ area A of an object where the air are pushed. F_a also increases as the increasing of the body's velocity v obeying $F_a \propto v^2$. Thus, as in (1.18)

$$F_a = CAv^2$$

where C is the coefficient of air resistance.

The resultant force of a freely falling object with mass m against air resistance is

$$\mathbf{F}_a + \mathbf{W} = m\mathbf{a}$$

where $\mathbf{W} = m\mathbf{g}$. Since $F_a \propto v^2$, it will end up equal to the weight of the falling object

$$\mathbf{W} + \mathbf{F}_a = 0.$$

Thus,

$$W = CAv_t^2$$

gives the *terminal velocity*

$$v_t = \sqrt{\frac{W}{CA}}. \quad (1.19)$$

Note. In the case of **small, round and not moving fast** object, we may apply the *Stokes' law* in which the drag force is

$$F_s = 6\pi r\eta v \quad (1.20)$$

where r is the radius of the object, η is the viscosity of the fluid, and v is the object's velocity.

As in this approximation the frictional force is just $F_s \propto v$, it simplifies some calculation especially to find the falling velocity of an object before reaching the terminal.

³The directions perpendicular to the direction of motion is transverse. The direction being the same as the direction of motion is called longitudinal.

Example 1.23. If the coefficient of air resistance of a person is $C = 0.88 \text{ kgm}^{-3}$, the terminal velocity of a falling person with mass 70 kg and an effective area of 0.2 m^2 is

$$v_t = \sqrt{\frac{W}{CA}} = \sqrt{\frac{70 \times 9.8}{0.88 \times 0.2}} = 62.4 \text{ m/sec.}$$

Terminal velocity is proportional to the square-root of the object dimension:

$$v_t \propto \sqrt{\frac{W}{A}} = \sqrt{\frac{L^3}{L^2}} = \sqrt{L}. \quad (1.21)$$

This result has interesting implications on the ability of animals to survive a fall. A speed of 8.6 m/sec is the terminal velocity of a 1 cm bug. Such a small creature can drop from any height without injury. Miners often encounter mice in deep coal mines but seldom rats. A simple calculation shows that a mouse can fall down a 100 m mine shaft without severe injury. However, such a fall will kill a rat.

Example 1.24. (1) Interpret the equation $v_t = \sqrt{mg/CA}$ in the case of $C = 0$.

(2) How would the terminal velocity of a 4 cm steel ball compare to that of a 1 cm ball?

Solution. (1) Clearly from the equation we can see that $v_t \rightarrow \infty$ if $C = 0$. This is the case when the object is freely falling in vacuum, i.e. no air resistance. Nevertheless, its not applicable in real world. Because

- The object will hit the ground before reaching a certain high velocity.
- Even if an object is falling along a very long hole through the ground to the centre of the earth, the gravitational acceleration will decrease gradually, and $g = 0$ at the earth's centre.
- The correct classical physics to describe motion is relativity. The theory of special relativity tell us that all material matter is unable to run faster or equal to the speed of light.

(2) If the objects density are the same, the terminal velocities depend on (1.21):

$$v_t \propto \sqrt{L}$$

where L is the linear dimension of the objects. Let v_4, v_1 be the terminal velocities of the balls. Therefore,

$$\frac{v_4}{v_1} = \frac{\sqrt{4L}}{\sqrt{L}} = 2.$$

$\therefore$ The larger ball falls faster than the smaller.

1.6.4 Stress and strain

Hook's law

An object lengthens slightly when you stretch it. The elongation ΔL of the it is directly proportional to the applied force:

$$\begin{aligned} F &\propto \Delta L \\ F &= k\Delta L \end{aligned} \quad (1.22)$$

where k is a constant depending on the shape and composition of the object and the direction of the force.

Stress and strain

Forces cause an object changing its shape. The deformation is directly proportional to the applied forces and inversely proportional to some areas of surfaces.

Tensile stress Forces pull on an object and cause its elongation, like the stretching of an elastic band.

Compressive stress Forces cause a compression of an object. It seems being the opposite situation of tensile stress.

Bulk stress An object is being squeezed from all sides, often happens in fluid. Fluid pressure is this particular case.

Shear stress Forces apply tangentially to opposite surfaces of a solid object to deform the object's geometry.

Strain S_t is the ratio of the *longitudinal elongation* to the original length:

$$S_t := \frac{\Delta L}{L}. \quad (1.23)$$

Stress S is the internal force per unit area acting on the material; it is defined by

$$S_s := \frac{F}{A}. \quad (1.24)$$

The ratio of stress to strain is a constant $Y = \frac{S_s}{S_t}$ known as *Young's modulus*. Thus,

$$Y = \frac{S_s}{S_t} = \frac{F/A}{\Delta L/L} \Rightarrow F = \frac{YA}{L} \Delta L. \quad (1.25)$$

Young's modulus Y is a constant depending on the material as well as A and L being also constants for a given object. Comparing to (1.22), identifying

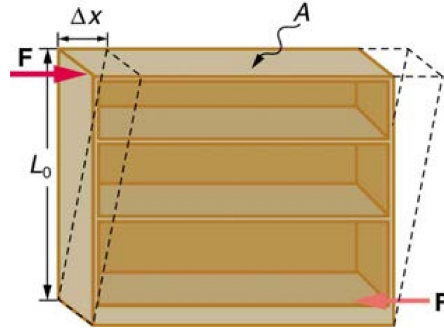
$$k = \frac{YA}{L}$$

is just the constant of Hooke's law.

Example 1.25. Consider a suspension cable for a gondola that includes an unsupported span of 3 km. Assuming the cable has a diameter of 5.6 cm and the maximum tension it can withstand is 3.0×10^6 N. Calculate the amount of stretch in the steel cable. (Young's modulus of steel 210×10^9 N/m²)

Solution. From (1.25) we have $\Delta L = \frac{F}{YA}L$, with the given F and the cross-sectional area $A = \pi r^2 = 2.8^2 \pi = 2.46 \times 10^{-3}$ m² to give

$$\Delta L = \frac{3 \times 10^6 \times 3000}{210 \times 10^9 \times 2.46 \times 10^{-3}} = 17.4 \text{ m}$$



Shear modulus: sideways stress

As $F \propto \frac{A\Delta x}{L_0}$, define the *shear deformation*

$$\Delta x = \frac{F}{SA} L_0$$

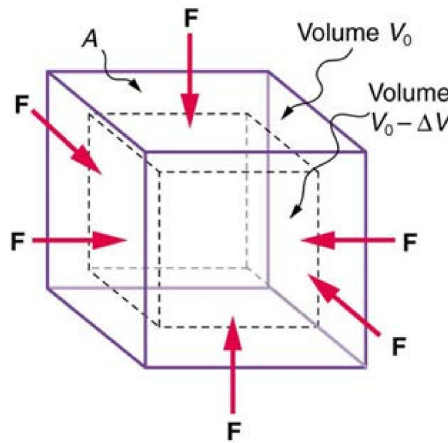
where S is the *shear modulus* and F is the force applied perpendicular to L_0 and parallel to the cross-sectional area A .

Bulk modulus: changes in volume

Similar to the shear modulus, the relationship of the change in volume to other physical quantities is given by

$$\Delta V = \frac{F}{BA} V_0$$

where B is the *bulk modulus*, V_0 is the original volume, and F/A is the force per unit area applied uniformly inward on all surfaces.



1.6.5 Projection of forces

It is convenient to resolve a single force into up to three independent forces to produce an equivalent effect. The magnitudes of the forces due to this projection are called *components*, with three *orthonormal* basis (unit vectors) pointing to three independent directions. In fact we have used this technique when we handled the kinematical projectile problems.

Example 1.26. A hockey puck having a mass of 0.3 kg slides on the horizontal, frictionless surface of an ice rink. Two forces act on the puck, as shown in Figure 1.7. The force $\mathbf{F}_1$ has a magnitude of 5 N, and the force $\mathbf{F}_2$ has a magnitude of 8 N. Determine both the magnitude and the direction of the puck's acceleration.

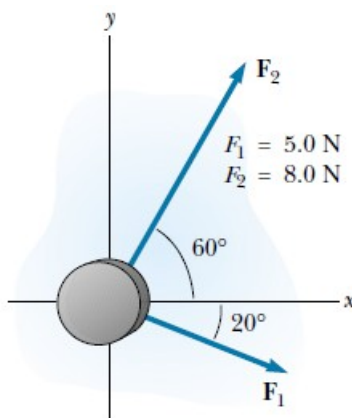


Figure 1.7: A hockey puck moving on a frictionless surface with acceleration.

Solution. The resultant force in the x direction is

$$F_x = F_1 \cos(-20) + F_2 \cos 60 = 8.7 \text{ N}.$$

In the y direction it is

$$F_y = F_1 \sin(-20) + F_2 \sin 60 = 5.2 \text{ N}.$$

Using Newton's second law, the x and y components of acceleration are

$$a_x = \frac{F_x}{m} = \frac{8.7}{0.3} = 29 \text{ ms}^{-2}, \quad a_y = \frac{F_y}{m} = \frac{5.2}{0.3} = 17 \text{ ms}^{-2}.$$

Thus, using the Pythagorean theorem the acceleration has a magnitude of

$$a = \sqrt{29^2 + 17^2} = 34 \text{ ms}^{-2}$$

and its direction relative to the positive x axis is

$$\theta = \tan^{-1} \left(\frac{a_y}{a_x} \right) = \tan^{-1} \left(\frac{17}{29} \right) = 30^\circ.$$

Example 1.27 (Standing broad jump). We assume that a jumper can generate with his feet a force equal to twice the body weight (Figure 1.8).

The horizontal and vertical components of the resultant force are:

$$\begin{aligned} F_r \cos 45^\circ &= 2W \cos \theta \\ F_r \sin 45^\circ &= 2W \sin \theta - W \end{aligned}$$

Now, we have two simultaneous equations with two unknowns. We can solve F_r and θ exactly to give

$$F_r = 1.16W \quad \text{and} \quad \theta = 65.8^\circ.$$

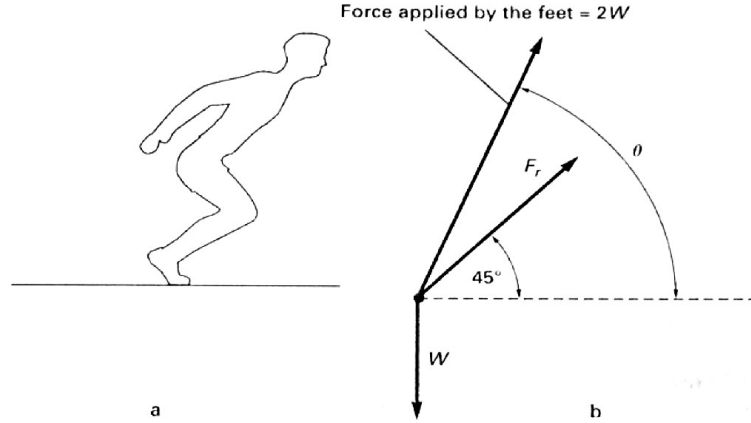


Figure 1.8: The standing broad jump and the associated forces.

The acceleration produced by the resultant force is

$$a = \frac{F_r}{m} = \frac{1.16W}{W/g} = 1.16g.$$

For the launching velocity v , we use $v^2 = u^2 + 2as$ with $u = 0$ and $s = 0.6$ m for the extent of the crouching position. Thus

$$v = 3.7 \text{ ms}^{-1}.$$

Using the projectile equation we have

$$R = \frac{v^2}{g} = \frac{13.7}{9.8} = 1.4 \text{ m}.$$

Note. The exerting force $2W$ and the resultant force F_r only happen during the person from crouching to extending the body **before** leaving off the ground. Analogous the previous examples, once the person launched there is no other source of acceleration except gravity.

1.7 Centripetal force

An object undergoing circular motion must be accelerating as it is changing the direction of its velocity. This centrally acceleration known as centripetal acceleration (radial acceleration) is given by $a_c = -\frac{v^2}{r}$ where v is the velocity of the object, directed along a tangent line to the curve at any instant. Since $v = r\dot{\theta}$, thus

$$a_c = -r\dot{\theta}^2.$$

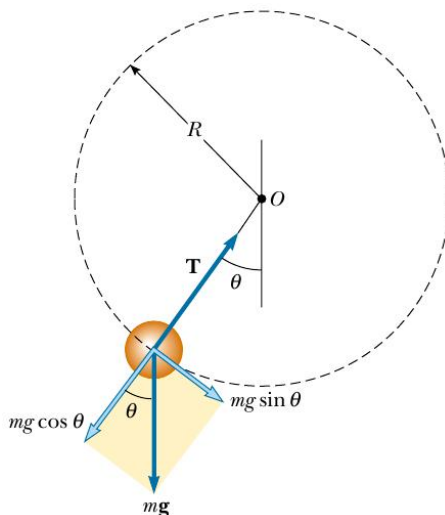
An acceleration must be produced by a force. According to Newton's second law, we get the magnitude of the *centripetal force*

$$F_c = -m\frac{v^2}{r} = -mr\dot{\theta}^2. \quad (1.26)$$

Note. An opposite directional force of the centripetal force known as the *centrifugal force* which seems to balance the system to avoid the circulating object flying away to maintain a circular motion is an example of the fictitious force (Coriolis' force) due to the inertial motion observed by a non-comoving observer.

Circular motion under gravity in a vertical plane

A small sphere of mass m is attached to the end of a cord of length R and whirls in vertical circle about a fixed point O .



Even though the motion is circular ($\therefore$ fixed radius) and is controlled to maintain a constant angular speed, the tension along the cord is varying depending on the angle θ .

Example 1.28. Determine the tension in the cord at any instant when the speed of the sphere is v and the cord makes an angle θ with the vertical.

Solution. We only need to consider the radial directional vectors. Thus,

$$T - mg \cos \theta = \frac{mv^2}{R} \quad \Rightarrow \quad T = m \left(\frac{v^2}{R} + g \cos \theta \right)$$

Note. A tension is a force along the length of a medium, especially a force carried by a flexible medium, such as a rope or cable.

Example 1.29 (Circular motion). A certain amount of force is needed to provide the acceleration of a circular motion. What if we are exerting a force perpendicular to the direction of motion in an attempt to make an object trace a circle of radius r , but the force isn't as big as mv^2/r ?

Solution. The object would move along the orbit of a parabola initially, then establish a circular motion with a radius r' , where

$$\frac{mv^2}{r'} < \frac{mv^2}{r}.$$

1.7.1 Forces on a curved path

Automobile rounds a curve

To calculate the maximum speed of an automobile rounding a curve without skidding on a flat ground, the centrifugal force F_c exerted on a moving car is

$$F_c = \frac{mv^2}{R} = \frac{Wv^2}{gR}.$$

For equilibrium (i.e. the car is able to keep in circular motion without skidding), the centrifugal force should be equal to the centripetal force, the latter is provided by the frictional force between the road and the tires. Thus,

$$\frac{Wv^2}{gR} = \mu W$$

where μ is the coefficient of friction between the tires and the road surface. Therefore,

$$v_{\max} = \sqrt{\mu g R}. \quad (1.27)$$

Safe speed on a curved path may be increased by banking the road along the curve as shown in Figure 1.9, in which the centrifugal force F_c could be compensated by part of the normal force N due to the inclination.

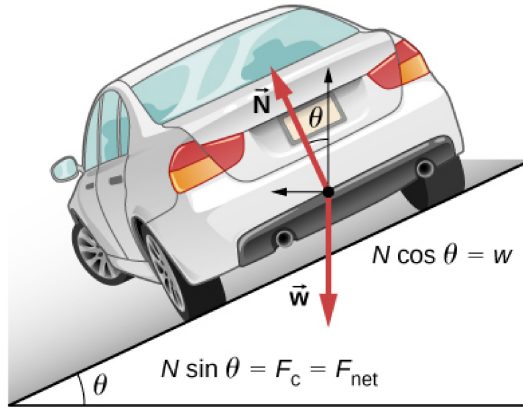


Figure 1.9: The car on this banked curve is moving away and turning to the left.

The vertical and horizontal components for equilibrium in this situation are

$$\begin{aligned} N \cos \theta &= W, \\ N \sin \theta &= \frac{Wv^2}{gR} \end{aligned}$$

where R is the **horizontal** radius of the road curvature. Eliminating N and W , the angle θ for the road bank is

$$\tan \theta = \frac{v^2}{gR}.$$

Centrifuges are used to hasten settling in fluids.

- Ultracentrifuges are the most advanced of these devices. Much the same forces are felt by people in a car, on an amusement ride, or in an airplane when rounding curves. The centripetal accelerations to which people are subjected are considerably less than any centrifuge can exert.
- Fighter pilots may experience accelerations as large as $5g$, or even $10g$, when turning their aircraft.
- g suits are required to keep fighter pilots from blacking out from loss of blood to the brain and to protect blood vessels in the legs from being stressed by too much blood. Astronauts are put into a large centrifuge training so that they can experience accelerations as large as those during a launch (up to $10g$).

Example 1.30. An ultracentrifuge spins at 50,000 revolutions per minute and the material being centrifuged is 5.0 cm from the pivot point. Calculate the centripetal acceleration and express it as a multiple of the acceleration of gravity (in g 's).

Solution. The speed in SI units is

$$v = \frac{s}{t} = \frac{50000 \times 2\pi r}{60} = 2.62 \times 10^2 \text{ ms}^{-1}.$$

The centripetal acceleration is then

$$a_c = \frac{v^2}{r} = \frac{(2.62 \times 10^2)^2}{0.05} = 1.37 \times 10^6 \text{ ms}^{-2}.$$

In multiples of g , this is

$$\frac{a_c}{g} = \frac{1.37 \times 10^6}{9.8} = 1.4 \times 10^5 \Rightarrow a_c = 1.4 \times 10^5 g.$$

This centrifuge creates an acceleration 140,000 times greater than gravity, meaning that the body being centrifuged feels a force 140,000 times its weight!

A runner on a curved track

Similar to the previous automobile problem, Figure 1.10(b) shows the vectors of the system. However, the person inclines an angle from the ground instead of banking the road in order to use gravity for centripetal force. In equilibrium, the horizontal and vertical components are

$$\begin{aligned} F_r \sin \theta &= F_{cp} = \frac{Wv^2}{gR}, \\ F_r \cos \theta &= W. \end{aligned}$$

Hence the angle θ toward the centre of rotation adjusted by the runner is

$$\tan \theta = \frac{v^2}{gR}.$$

For a speed of 6.7 m/sec on a 15 m radius track,

$$\tan \theta = \frac{(6.7)^2}{9.8 \times 15} = 0.305 \Rightarrow \theta = 17^\circ.$$

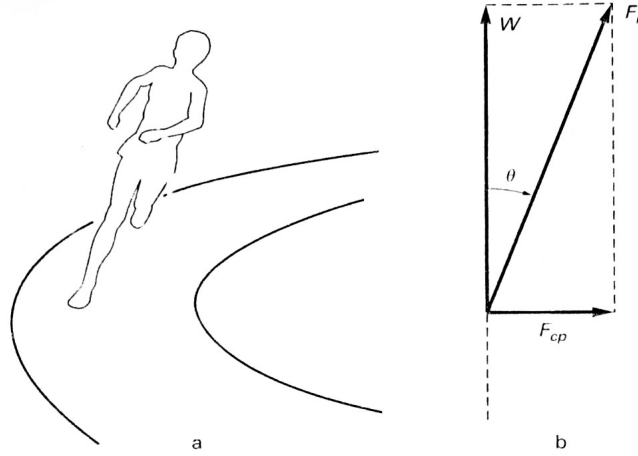


Figure 1.10: (a) Runner on a curved track. (b) Forces acting on the foot of the runner.

1.8 Gravity

1.8.1 Kepler's laws

Elliptical orbit law The planets orbit the sun in elliptical orbits with the sun at one focus.

Equal area law The line connecting a planet to the sun sweeps out equal areas in equal amounts of time (Figure 1.11), i.e.

$$\frac{dA}{dt} = \text{constant}.$$

Periods The time required for a planet to orbit the sun, called its period, is proportional to the long axis a of the ellipse raised to the $3/2$ power, i.e.,

$$T \propto a^{3/2}.$$

The constant of proportionality is the same for all the planets.

The equal area law implies that a planet in a **circular orbit** moves around the sun with constant speed. Since a small area of a sector is

$$\Delta A = \frac{r^2 \Delta \theta}{2},$$

Dividing both sides by Δt and taking the limit $\Delta t \rightarrow 0$ gives

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta A}{\Delta t} = \frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta}.$$

In particular, v is constant if the orbit is a circle; for r and $dA/dt = \text{constant}$.

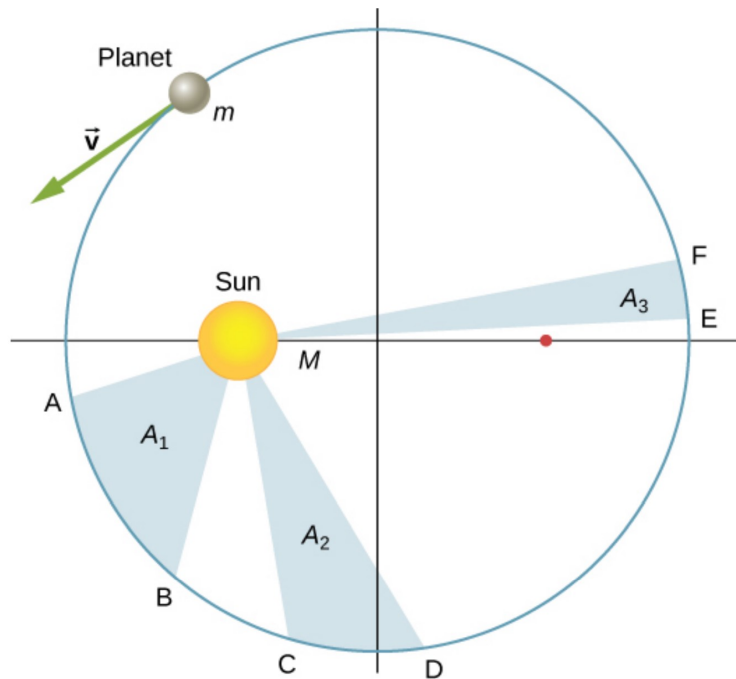


Figure 1.11: All the areas swept by the planet are equal in a fixed time interval.

1.8.2 Newton's law of gravity

Inverse square law

Gravitational force between two objects is inversely proportional to the square of the distance separation:

$$F \propto \frac{1}{r^2}.$$

Newton's law of gravity

Newton reasoned that the mass of an object should be proportional to gravitational attraction. Thus, he claimed that

$$\begin{aligned} F &\propto \frac{m_1 m_2}{r^2} \\ F &= \frac{G m_1 m_2}{r^2} \end{aligned} \quad (1.28)$$

where G is a proportionality constant called Newton's *gravitational constant*, with numerical value $6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$, which is one of the most important universal constants in nature.

Earth's gravity

Let M be the mass of the earth, m the mass of a test object (m is small comparing to M), r the radius of the earth. According to Newton's second law and his law of gravitation we have

$$F_I = m_I g \quad \text{and} \quad F_G = \frac{GMm_G}{r^2}$$

where we use F_I, F_G to represent the forces due to the *inertial mass* m_I and *gravitational mass* m_G . Why do we have two different masses for the mass m ? In general, the mass m is defined in two different ways, according to Newton:

- m_I is defined as the physical quantity of an object for how hard to make it to accelerate, i.e. how large a force needed to push it.
- m_G is defined by gravity, proportional to the gravitational attraction.

These two quantities in theory are not supposed to be equal. In general, $F_I \propto F_G$, i.e. $F_I = kF_G$. However, from various experiments, we found $k = 1$. Thus, **gravitational mass equals to inertial mass**. Obviously, it is not only for earth's gravity, Newton proposed that this rule is universal valid. Therefore, we have

$$\begin{aligned}\frac{GMm}{r^2} &= mg \\ g &= \frac{GM}{r^2}.\end{aligned}\tag{1.29}$$

Since G, M, r are constants, the gravitational acceleration g is also a constant, which is **independent** of the mass of the falling object (unless the object is too heavy or too large), i.e. Galileo principle of equivalence.

1.8.3 Inertial and non-inertial frames of references

We have defined inertial frame of reference. That is, we choose a frame (coordinate system) to measure the physical state of motion of an object as long as the frame is moving uniformly, i.e., no acceleration (uniform motion and motionless are relative).

It follows that *non-inertial system* sounds easy to be understood. It simply just a non-uniform moving system, i.e. $a \neq 0$. Nevertheless, sometimes it is easy to be confused by the definitions as often misled by daily experiences.

Example 1.31. Alice who is standing on the ground is watching a broken cable cable-car falling. Inside there, is the second person Bob. Who is in inertial system? Are both Alice and Bob in inertial systems?

Solution. • Consider Alice standing still on the ground. She is standing there under gravity (gravitational acceleration) that is the reason why she is able to be standing on the ground in the first place. Obviously, she is not in inertial system.

- The falling cable-car looks like happening in non-inertial system if we observe it on the ground. However, observers on ground are definitely not in inertial frames as mentioned. The person inside the falling cable-car is experiencing “weightless”, like what the astronauts experience in a spacecraft orbiting round the earth with the engine power-off. The cable-car person Bob does not see any free fall or gravity pull phenomenon if he releases any object. Because Bob himself, the cable-car and all objects inside the cable-car are actually free falling! According to Galileo's principle of equivalence (Einstein's as well), all freely falling objects are at the same gravitational acceleration. Therefore, Bob is in inertial system before the cable-car hits the ground.

We have the important find:

The existence of inertial system only happens while in free fall.

Most of the scientific theories are formulated with the assumption of the existence of an inertial reference system. In practice, experiments have been often being demonstrated in labs where obviously are not in inertial systems. We just simply ignore this gravitational factor. Accurate and valuable experimental results are often given in the labs of spacecrafts with weightless situation.

Note Relative motion only exists in initial system. Theoretically speaking, it is almost impossible to achieve this ideal situation. Imagine that you are inspecting a spinning football on which an ant is sitting on the surface, gravity is ignored. You cannot say that due to the ant's rotation in its perspective is equivalent to, in the ant's coordinate system you seem rotating with having real rotational physical effect. Because you are in the inertial system (ignore gravity), the rotating football in fact is in non-inertial system due to centripetal acceleration. In short, we see the ant rotating with centrifugal effect, however, the ant sees us rotating without any physical effect.

Example 1.32. A car is normally capable of an acceleration of 3 ms^{-2} . If it is towing a trailer with half as much mass as the car itself, what acceleration can it achieve?

Solution. Let m be mass of the car, a be the new acceleration of the whole system. The car manages to exert a force $F = m(3) = 3m$. The total mass of the car and the trailer is $m(1 + 1/2) = 3m/2$. Since the applied force is a constant. Thus

$$3m = \frac{3m}{2}a \quad \Rightarrow \quad a = 2 \text{ ms}^{-2}.$$

Chapter 2

Energy and conservation laws

2.1 Conservation of energy

2.1.1 Energy

All things—fields and matter can be transformed to energy. Gravity might be a sort of though probably would be more than that.

The total energy of a system is found by adding up contributions from the characteristics of an enclosed system such as motion of objects in it, heating of the objects, and the relative positions of objects those interact via forces. **The total energy of a closed system always remains constant.** Energy cannot be created or destroyed¹, but only transferred from one system to others. In Newtonian physics, mass is supposed to be invariant.

Mechanical energy

Definition 2.1. Mechanical energy is defined as the work done E required to bring an object moving along a path:

$$E = \int \mathbf{F} \cdot d\mathbf{s} \quad (2.1)$$

where $\mathbf{F}$ is the applied force, $\mathbf{s}$ is the displacement.

Using dimensional analysis,

$$[\text{Workdone}] = [F][s] = MLT^{-2}L = ML^2T^{-2}$$

That is, the unit of energy is $\text{kgm}^2\text{s}^{-2}$ or simply Nm (Newton-metre). In practice, we prefer the unit ‘Joule’ (J) which is defined by the requiring work to apply 1 N of force to bring an object to travel 1 m distance, i.e., $1\text{J} = 1\text{Nm}$.

Note. The infinitesimal displacement $d\mathbf{s}$ here indicates this vector quantity changing the direction along the path. A dot product for the definition of energy is a scalar quantity. In words,

$$\text{Work done} = \text{force} \cdot \text{distance travelled}$$

Since dot product is used, for a good efficiency to push or pull an object the best way is always to keep the applied force along the same direction of the path.

¹This is only valid in classical physics. Energy seems to be able to be created out from nothing in quantum mechanics, nevertheless, the energy content of the whole universe could be fixed.

Potential energy

Potential energy is often referred to the description of the potential of an object having tendency to do mechanical work at a certain height due to gravity. Therefore,

$$\text{Potential energy (PE)} = Wh = mgh \quad (2.2)$$

where m, h are the mass of the object and the height above a reference point.

Kinetic energy

To describe the energy of an object during motion, *kinetic energy* (KE) tells us about the energy at an instant with velocity v . KE can also be derived from the fundamental equation of work done, i.e., the product of the force and the distance traveled.

The required energy to increase an object to move from u to v is

$$\begin{aligned} \text{KE} &= \int \mathbf{F} \cdot d\mathbf{s} = \int m\mathbf{a} \cdot d\mathbf{s} = \int m \frac{d\mathbf{v}}{dt} \cdot d\mathbf{s} = \int_u^v m\mathbf{v}' d\mathbf{v}' \\ &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \end{aligned}$$

Setting $u = 0$, the instantaneous kinetic energy comparing to at rest is

$$\text{KE} = \frac{1}{2}mv^2 \quad (2.3)$$

2.1.2 Thermal energy

Heat energy, or technically called *thermal energy* is only meaningful defined as heat energy transferring from a hotter region to a colder region. We feel something hot because our body is colder than the object that we touch. As thermal energy is only valid in heat transfer, it is relative.

The typical processes of heat transfer are

- conduction: in solid
- convection: in fluid
- radiation: no medium needed, it is in fact an electromagnetic radiation.

Heat is kinetic energy

All matter is composed of molecules. The kinetic energy of a group of molecules is described by heat. Actually, we also need to consider the *distribution* of molecules in a given region at some physical state. A physical quantity to measure such a distribution of particles is known as *entropy*.

A physical quantity to relate the entropy and kinetic energy of molecules at a given condition is *temperature*, which may regarded as energy density, i.e., the average kinetic energy of a particle. Or in more accurate, the energy of a system per the numbers of *degrees of freedom* of the particles.

2.1.3 Power

From the definition of the unit of energy, Joule, you might wonder the “rate” to push (or pull) an object for 1 metre distance with the applied force 1 N. If the object is heavy, it will take a longer time to fulfil the task (ignoring all types of frictions). This is why the definition of *power* turns up.

Using a shorter time in able to complete the same work instead of using a longer time interval to complete we often call the former more powerful. Hence,

Definition 2.2. Power (P) is defined as the rate of change of energy, i.e.

$$P = \frac{dE}{dt}. \quad (2.4)$$

The unit of power is “Watt” (W), i.e. 1 Joule per second (Js^{-1}). A simple example to understand power is supposed that you normally to heat a glass of water take you 10 minutes to make it boiling, now you change to use a higher power kettle to heat the same amount of water to boiling in a shorter time interval.

Example 2.1. For an astronaut sealed inside a space suit, getting rid of body heat can be difficult. Suppose a 60 kg astronaut is performing vigorous physical activity, expending 200 W of power. If none of the heat can escape from her space suit, how long will it take before her body temperature rises by 6 °C, an amount sufficient to kill her? Assume that the amount of heat required to raise her body temperature by 1 °C is the same as it would be for an equal mass of water. (1 Joule is the amount of energy required to heat 0.24 g of water by 1 °C.)

Solution. We find the total generated energy first:

$$E = \frac{60 \times 6}{2.4 \times 10^{-4}} = 1,500,000 \text{ J}$$

The time taken is $1500000/200 = 7500$ second, or 125 minutes. That is, only 2 hours more or less, the generated heat would kill the astronaut.

2.1.4 Fundamental types of energy

There are only two types of energy—kinetic and potential, i.e., energy due to motion and the various internal potential energies.

- Kinetic energy
 - Energy for moving, include all forms of matter and field, examples: molecule, atom, electron, photon,² etc.
- Gravitational potential energy
 - Gravitational interaction is always attractive—the energy to pull all forms of matter and field together.
- Electrical and magnetic potential energy

²Photon has no mass! It cannot be described by the usual formula $\text{KE} = mv^2/2$.

- The interactions of electricity and magnetism are in the same origin just with in different manifestations. The unified theory is known as *electromagnetism*. The potential energy is the energy to attract or repulse electric charged matters to each other.
- Nuclear potential energy
 - The binding energy amongst protons and neutrons

Example 2.2. What is the form of energy of (1) sound, (2) chemical and (3) solar?

Solution. (1) Sound is the variation of air pressure due to the moving and vibrational air molecules. Sound wave is mechanical.

- (2) Chemical reaction is the breaking or formation of bondings between molecules or atoms. Chemical binding energy is electromagnetic.
- (3) Solar energy which is mainly produced by the nuclear reaction inside the sun is *nuclear fusion*. However, the energy from atomic bomb is *nuclear fission*. Both of them belong to nuclear energy. However, the form of energy transfer from the sun to the earth via *electromagnetic radiation* (light), the energy is in packet form called *photon*, and this discrete nature known as *quantum*. This will be discussed in the last chapter.

2.2 The transfer of mechanical energy

Work is the amount of energy transferred into or out of a system, not counting energy transferred by heat conduction.

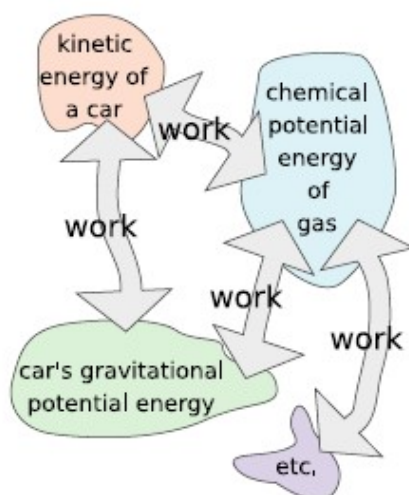


Figure 2.1: Work is a transfer of energy.

2.2.1 Metabolic rate

Animals do work by means of muscular movement. The energy required to perform the work is obtained from the chemical energy in the food eaten by the animal.

Only a small fraction of the energy consumed by the muscles is converted to work. For example, in bicycling at a rate of one leg extension per second, the efficiency of the muscles is 20%. In other words only one fifth of the chemical energy consumed by the muscle is converted to work. The rest is dissipated as heat.

The energy consumed per unit time during a given activity is called the *metabolic rate*.

Example 2.3. A 70 kg person jumping up 60 cm for 10 minutes at a rate of one jump per second. How much energy does he consume?

Solution. The external mechanical work performed by the leg muscles in each jump is

$$\text{Weight} \times \text{height of jump} = 70 \times 9.8 \times 0.6 = 411 \text{ J.}$$

The total muscle work during the 10 minutes of jumping is

$$411 \times 600 \text{ jumps} = 24.7 \times 10^4 \text{ J.}$$

If we assume a muscle efficiency of 20%, then in the act of jumping the body consumes

$$24.7 \times 10^4 \times \frac{1}{20\%} = 1.23 \times 10^6 \text{ J.}$$

2.2.2 Gravitational potential

As gravitational potential energy V is the ‘potential’ to release a certain amount of mechanical energy, we have

$$V = - \int_{r_1}^{r_2} \mathbf{F} \cdot d\mathbf{r} \quad (2.5)$$

where $d\mathbf{r}$ is the infinitesimal distance. In most cases since the increasing of displacement $\mathbf{r}$ always decreases the potential V a negative sign has been added. The reverse, i.e. the derivative of potential (energy) w.r.t. displacement is the force given by

$$\mathbf{F} = - \frac{dV}{d\mathbf{r}}.$$

Example 2.4. We would like to find the energy to bring an object with mass m from a height r_1 to r_2 (both are measured from the centre of the earth). The gravitational force is given by

$$F = - \frac{GMm}{r^2}.$$

Integrating F with the boundary condition gives the potential

$$V = - \int_{r_1}^{r_2} \frac{GMm}{r^2} dr = GMm \left(\frac{1}{r_2} - \frac{1}{r_1} \right).$$

Supposing an object is brought from a lower position to a higher position on earth, since $r_2 > r_1$, above equation gives a negative value of V . The usual meaning of the negative sign is:

- positive energy implies giving energy,
- negative energy means taking energy.

Since gravitational interaction is always attractive, energy is always needed to bring an object to a higher position.

Example 2.5. Find the energy required to bring an object with mass m to leave the earth.

We simply put $r_1 = r$ the earth radius, and put $r_2 \rightarrow \infty$ into above equation, we get

$$V = -\frac{GMm}{r}.$$

Since G, M, r are constant. The energy needed is directly proportional to the mass of the object.

Escape velocity

Escape velocity v is the velocity required to bring an object leaving from the effect of earth gravity (or other planets). To calculate the velocity for launching a rocket, the kinetic energy must be equal or larger than the gravitational potential

$$\frac{GMm}{r} = \frac{1}{2}mv^2 \quad \Rightarrow \quad v = \sqrt{\frac{2GM}{r}}$$

where we have ignored the minus sign for convenience. G, M, r are constants again, the escape velocity is independent of the mass of the rocket.

Gravity in a spherical cloud

Suppose that space probe of mass m is dropped into an unexplored spherical cloud of gas and dust, and accelerates toward the centre of the cloud under the influence of the cloud's gravity. Measurements of its velocity allow its potential energy, U , to be determined as a function of the distance r from the cloud's centre. The mass in the cloud is distributed in a spherically symmetry way, so its density, $\rho(r)$, depends only on r and not on the angular coordinates. The relation of the potential energy $U(r)$ to the density function $\rho(r)$ is given by

$$\rho(r) = \frac{1}{4\pi Gmr^2} \frac{d}{dr} \left(r^2 \frac{dU}{dr} \right). \quad (2.6)$$

To derive the equation consider the density of the cloud $\rho(r)$ depending on r only and the distribution being spherically symmetric, the mass of the thin shell of the spherical cloud is $4\pi r^2 \rho(r) dr$ where dr is the infinitesimal thickness of the shell of the space probe traveled in a small time interval. Thus, the gravitational attractive force between the probe and the centre of the cloud is

$$\frac{Gm}{r^2} \int_0^r 4\pi r^2 \rho(r) dr.$$

This is equal to the derivative of the potential with respect to r :

$$\frac{Gm}{r^2} \int_0^r 4\pi r^2 \rho(r) dr = \frac{dU}{dr}.$$

There is no negative sign for the potential because of the probe going inward. Differentiating above equation gives

$$4\pi r^2 \rho(r) Gm = \frac{d}{dr} \left(r^2 \frac{dU}{dr} \right)$$

$$\rho(r) = \frac{1}{4\pi Gmr^2} \frac{d}{dr} \left(r^2 \frac{dU}{dr} \right).$$

2.3 Linear momentum

2.3.1 Momentum

Momentum is a physical observable for describing motion. In Newtonian physics, momentum $\mathbf{p}$ is defined by the product of the mass of object and its velocity, i.e.

$$\mathbf{p} = m\mathbf{v}. \quad (2.7)$$

Since mass is a scalar, momentum $\mathbf{p}$ is a vector resulting pointing to the same direction as the velocity.

The fundamental important concept of momentum is the *conservation law of momentum*.

Definition 2.3. The total momentum of an enclosed system is always constant. Mathematically,

$$\mathbf{p}_1 + \mathbf{p}_2 + \dots + \mathbf{p}_n = \text{constant}$$

where $\mathbf{p}_1, \mathbf{p}_2, \dots$ are the individual momenta of the particles.

Newton's second law

The second law can be re-expressed in terms of the momentum:

$$\mathbf{F} = m\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{m\Delta\mathbf{v}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\mathbf{p}}{\Delta t} = \frac{d\mathbf{p}}{dt}.$$

That is, the applied force is equal to the **rate of change of the momentum**. In fact, this definition is more accurate and universal than the common definition “force equals to mass times acceleration.”

The derivative of the total linear momentum w.r.t. time in an enclosed system is always zero gives the vanishing of the resultant force:

$$\frac{d}{dt}(\mathbf{p}_1 + \dots + \mathbf{p}_n) = \mathbf{F} = 0$$

In other words, momentum is conserved in a enclosed system.

The differential of an impulse $\mathbf{J}$ is defined as the product of an applied force $\mathbf{F}$ over an infinitesimal time interval dt , such that

$$d\mathbf{J} = \mathbf{F}(t)dt.$$

The RHS is exact as it is defined to be written as a differential $d\mathbf{J}$. Taking the integral over a time interval we get the impulse

$$\mathbf{J} = \int_{t_i}^{t_f} \mathbf{F}(t)dt.$$

By Newton's second law, the integral becomes

$$\mathbf{J} = \mathbf{p}(t_f) - \mathbf{p}(t_i) = \Delta \mathbf{p}.$$

It is simply the change of momentum over the time interval. We do not need to know the detail of the changing momentum as using the fundamental theorem of calculus like the previous conservative force example. The boundary values define the impulse.

2.3.2 Impulse

Recall the definition of the average value of a function

$$\langle f(x) \rangle = \frac{1}{b-a} \int_a^b f(x) dx.$$

It implies that the average value $\langle f(x) \rangle$ is the height of the rectangle which gives the area by the product $\langle f(x) \rangle (b-a)$ equal to the area $\int_a^b f(x) dx$ under the curve. Similarly, define the average force

$$\langle \mathbf{F} \rangle = \frac{1}{\Delta t} \int_{t_i}^{t_f} \mathbf{F}(t) dt. \quad (2.8)$$

Thus, an impulse can be written as the product of average force and the time interval

$$\mathbf{J} = \langle \mathbf{F} \rangle \Delta t. \quad (2.9)$$

In other words, it is the change of momentum.

Impulsive forces

As a force is the rate of change of momentum, the average force is

$$\mathbf{F} = \frac{m\mathbf{v}_f - m\mathbf{v}_i}{\Delta t}$$

where the time interval Δt is often for the duration of collision or impact.

Because an impulsive force can be decreased by increasing Δt , an example of by using an air bag to increase the time duration during a car collision is shown in Figure 2.2.

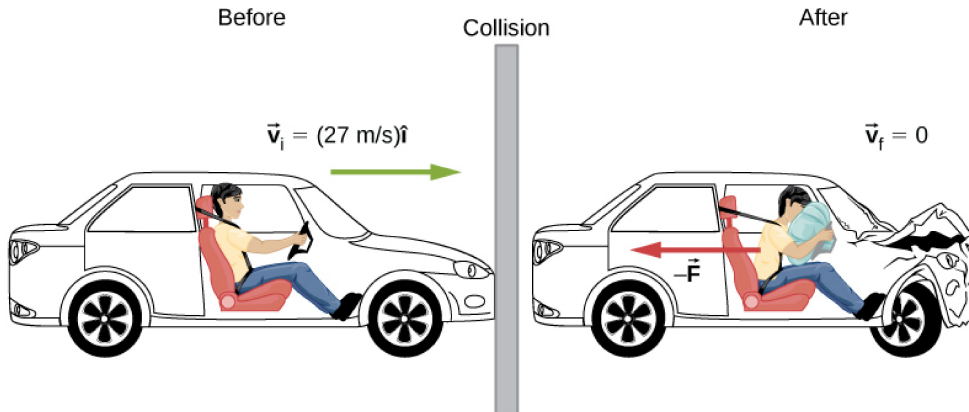


Figure 2.2: Inflating collision protective device.

2.3.3 Collisions

As momentum must be conserved, the total momentum of an enclosed system before and after collisions are the same.

One-to-Many The kinetic energy of a system after an *explosion* increases. Thus kinetic energy is not conserved $K_f > K_i$.

Many-to-One Kinetic energy decreases after the collision $0 < K_f < K_i$. This is a typical example of *inelastic* collision. If the kinetic energy completely lost $K_f = 0$, it is said to be *perfectly inelastic*.

Many-to-Many If two or more objects collide, and bounce off each other, the total kinetic energy of the system is conserved $K_f = K_i$. Such collision is *elastic*.

We may resolve the momenta into 2- or 3-dimensional coordinate systems if needed. Treat all the independent components separately, then using the Pythagorean theorem to combine them up.

Centre of mass

The total momentum of a system can be written as if a single object's momentum

$$M\mathbf{v} = \sum_i^n m_i \mathbf{v}_i$$

where m_i are the individual masses of the objects, M is the total mass. Write the velocities as the derivatives of the position vectors

$$M \frac{d\mathbf{r}}{dt} = \sum_i m_i \frac{d\mathbf{r}_i}{dt} \Rightarrow \frac{d}{dt} \mathbf{r} = \frac{d}{dt} \sum_i \frac{m_i}{M} \mathbf{r}_i$$

The position of the centre of mass of a system is defined by

$$\mathbf{r} = \sum_i \frac{m_i}{M} \mathbf{r}_i \quad (2.10)$$

as the expected value of $\mathbf{r}_i$.

For a continuous object, the summation in (2.10) becomes an integral. The position vector of the centre of mass is

$$\mathbf{r}_{CM} = \frac{1}{M} \int \mathbf{r} dm. \quad (2.11)$$

The integral is also called the *first moment*.

Rocket propulsion

The law of conservation of momentum can also explain the increase acceleration of launching a rocket even the force generated by the propeller is constant.

- A rocket are continuously ejecting burned fuel gases, which have both mass and velocity, and therefore some momentum. Thus, the rocket's momentum changes by this same amount.
- As the mass of the rocket continuously decreases, even though the force on the rocket is constant, due to $\mathbf{F} = m\mathbf{a}$ the resulting acceleration continuously increases.

Using the conservation of momentum, the change of momentum between the initial momentum of the whole rocket to the momenta of the rocket and the expelled burned fuel at a later time is zero

$$[(m - dm_g)(v + dv) + dm_g(v - u)] - mv = 0$$

where m and m_g are the masses of the rocket and the burned fuel gas, v the velocity of the rocket and u the constant velocity of the ejected burned gas **relative to the rocket**, hence $v - u$ implies relative to the ground. Expanding the equation gives

$$mdv = dm_g dv + dm_g u.$$

Since we have defined dm_g positive, the change of it implies the decrease of the mass of the rocket; $dm_g = -dm$.

The product of two infinitesimal changes is extremely small, we may just safely ignore it. Therefore

$$mdv = -udm.$$

As gas is constantly burned, u is a constant. Integrating the expression over the appropriate intervals gives

$$\int_{v_i}^v dv = -u \int_{m_i}^m \frac{1}{m} dm \quad \Rightarrow \quad v - v_i = u \ln \left(\frac{m_i}{m} \right) \quad (2.12)$$

This equation shows how the change of velocity of a rocket relates to the change of the rocket's mass. It gives the decreasing of the mass function of the rocket

$$m(v) = m_i \exp \frac{-(v - v_i)}{u}.$$

Rocket in a gravitational field

The previous analysis is under the assumption that a rocket is propelled in vacuum without gravity. Consider to launch a rocket on earth but still ignoring air resistance, we should have the change of momentum of the rocket equal to the impulse induced by gravity $d\mathbf{p} = d\mathbf{J}$, i.e.,

$$(m - dm_g)(v + dv) + dm_g(v - u) - mv = -mgdt.$$

As previous, $mdv + dm u = -mgdt$ in which we have put $dm_g = -dm$. After integration we obtain

$$v - v_i = u \ln \left(\frac{m_i}{m} \right) - g\Delta t \quad (2.13)$$

Unlike (2.12), the change of velocity here depends on the change of time Δt . The rocket will run faster if more mass of the burned fuel gas is expelled at a shorter time interval.

2.4 Angular momentum

Angular momentum is a dynamical physical effect due to rotation closely relating to the *torque* of a system.

2.4.1 Torque

The effectiveness of a force in producing a rotation is called a torque or a *moment of a force*. Torque is defined by $\tau = \ell_{\perp} F$.

Definition 2.4. In vector form, torque is the result of a cross product by a displacement vector and a force

$$\boldsymbol{\tau} = \boldsymbol{\ell} \times \mathbf{F}. \quad (2.14)$$

As a result of cross product (use the right hand rule), **torque is a vector** normal to the plane of the displacement $\boldsymbol{\ell}$ and the applied force $\mathbf{F}$.

Statics

Statics is a body or a system in the state of *equilibrium*, such as buildings, bridges, or patients in traction. When a body is in equilibrium, its acceleration is zero.

Conditions for equilibrium:

- Resultant force is zero
- No rotation; the net torque must be zero or that the sum of the clockwise torques must equal to the sum of the counter-clockwise torques.

That is,

$$\mathbf{F} = 0 \quad \text{and} \quad \boldsymbol{\tau}_{\text{cw}} = \boldsymbol{\tau}_{\text{ccw}}. \quad (2.15)$$

Example 2.6. (a) Calculate the force that the biceps muscle³ exerts to hold up the forearm and the book as shown in Figure 2.3.

- (b) Compare the force exerted by the biceps muscle with the weight of the forearm and book.

Solution. (a) For static equilibrium, $\tau_{\text{cw}} = \tau_{\text{ccw}}$, that is,

$$0.16w_a + 0.38w_b = 0.04F_B.$$

Solving for F_B

$$\begin{aligned} F_B &= \frac{0.16m_a g + 0.38m_b g}{0.04} \\ &= \frac{0.16 \times 2.5 \times 9.8 + 0.38 \times 4 \times 9.8}{0.04} \\ &= 470 \text{ N}. \end{aligned}$$

³Muscles can only contract, so they occur in pairs. In the arm, the biceps muscle is a flexor that closes the limb. The triceps muscle is an extensor that opens the limb.

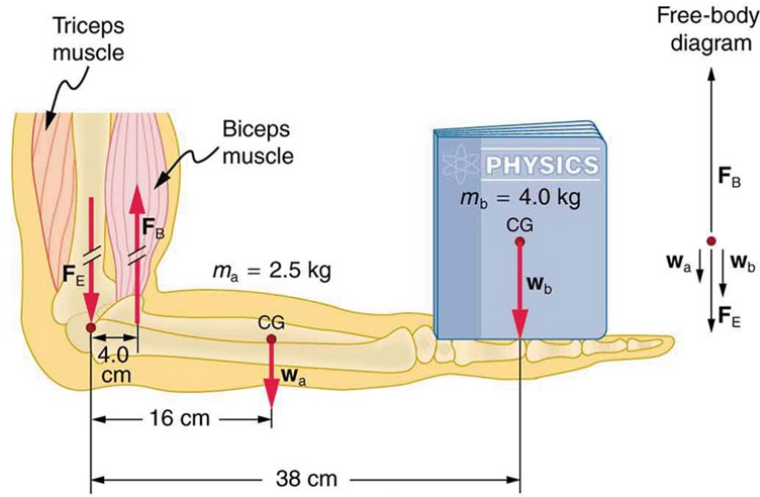


Figure 2.3: The biceps muscle is exerting a force much greater than the weight supported.

(b) The combined weight of the forearm and its load is

$$\begin{aligned} w &= w_a + w_b \\ &= m_a g + m_b g \\ &= 63.7 \text{ N.} \end{aligned}$$

The biceps exerts its force on the arm at a point close to the elbow, making it much less effective in producing a rotation than forces exerted farther from the elbow. The biceps is said to be at a *mechanical disadvantage*, as are most of the muscles in the body.

Example 2.7. Show that for a body made up of many equal mass, the equation for the centre of mass becomes a simple average of all the positions of the masses.

Solution. Imagine suspending the body where the suspended point O is the pivot position (no matter where the one of the masses or simply the attachment is). Let Δm be an infinitesimal mass element of the body. Taking moment about the pivot point O , we have

$$m\mathbf{r} \times \mathbf{g} = (\Delta m\mathbf{r}_1 + \Delta m\mathbf{r}_2 + \dots + \Delta m\mathbf{r}_n) \times \mathbf{g}$$

where $\mathbf{r}_1, \dots, \mathbf{r}_n$ are the distances from O to the mass elements, $\mathbf{r}$ is the distance from O to the centre of mass, m is the mass of the body. Thus,

$$\begin{aligned} \mathbf{r} &= \frac{\Delta m}{m}(\mathbf{r}_1 + \mathbf{r}_2 + \dots + \mathbf{r}_n) \\ &= \frac{1}{n}(\mathbf{r}_1 + \mathbf{r}_2 + \dots + \mathbf{r}_n). \end{aligned}$$

We have used the fact that $n\Delta m = m$ where n is the number of mass elements. The position of the centre of mass is just the average of all the positions of the mass elements.

2.4.2 Conservation of angular momentum

Similar to the conservation of momentum for linear motion of an object, there is a conserved quantity for rotation called *angular momentum* which is defined as

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \quad (2.16)$$

where $\mathbf{L}$ is the angular momentum of the rotating object about the axis of the pivot point, $\mathbf{r}$ is the displacement from the axis to the object, $\mathbf{p}$ is the momentum of the object.

Since $p = mv$, the magnitude of angular momentum written in terms of the object's mass is

$$L = mvr.$$

In an enclosed system, the angular momentum is a constant

$$L = \text{constant}.$$

Recall the uniform circular motion, $v = r\dot{\theta}$, angular momentum can be written in terms of the angular velocity

$$L = mr^2\dot{\theta}.$$

From (2.14), torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ and force is the rate of change of momentum

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times \frac{d\mathbf{p}}{dt}.$$

Consider the rate of change of angular momentum

$$\begin{aligned} \frac{d}{dt}(\mathbf{r} \times \mathbf{p}) &= \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt} \\ &= \mathbf{v} \times \mathbf{p} + \mathbf{r} \times \mathbf{F} \\ &= \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times \mathbf{F} \\ &= 0 + \mathbf{r} \times \mathbf{F} \\ &= \boldsymbol{\tau} \end{aligned}$$

Thus,

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{p}) = \frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}. \quad (2.17)$$

Therefore, torque is the rate of change of angular momentum. Comparing with the equations

$$\begin{aligned} \mathbf{F} &= \frac{d\mathbf{p}}{dt} && \text{(linear motion)} \\ \boldsymbol{\tau} &= \frac{d\mathbf{L}}{dt} && \text{(rotation)} \end{aligned}$$

we see that the mathematical form and the physical meaning of a torque looks like the definition of a force, giving the correspondences between linear motion and rotation.

Precession of a gyroscope

A spinning top can be explained by the relationships among the angular momentum, the torque and the weight of it (Figure 2.4). The torque created by the $\mathbf{r} \times M\mathbf{g}$ changes the direction of the angular momentum $\mathbf{L}$ according to $d\mathbf{L} = \boldsymbol{\tau}dt$, but not its magnitude.

The same principle can be applied as in Figure 2.5.

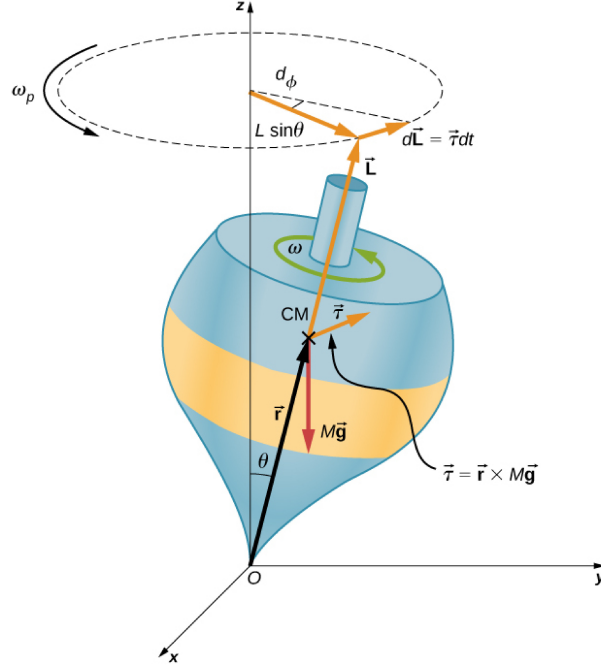


Figure 2.4: The rate of change of the angular momentum is balance by the torque.

2.4.3 Rotation of a rigid body

To find the angular momentum of a rotating rigid body as shown in Figure 2.6, we may consider the angular momentum of a small element of mass Δm at P rotating about the pivot point O with tangential velocity $\mathbf{v}$. Then add all those small angular momenta up to construct for the whole angular momentum of the rigid body.

Write the angular momentum in term of $\dot{\theta}$ instead of v as $\dot{\theta}$ is the same for the whole body,

$$r\Delta mv = r\Delta mr\dot{\theta} = \Delta mr^2\dot{\theta}.$$

The angular momentum of the body is given by

$$L = \sum_i^n \Delta m_i r_i^2 \dot{\theta}$$

where n is the total number of the mass elements.

Analogously, using the same method, the kinetic energy of a small rotating mass is

$$\frac{1}{2}\Delta mv^2 = \frac{1}{2}\Delta mr^2\dot{\theta}^2.$$

Thus, the kinetic energy of the rigid body due to rotating is given by

$$\text{KE}_R = \frac{1}{2} \sum_i^n \Delta m_i r_i^2 \dot{\theta}^2.$$

Both examples share a common term. If we let $I = \sum_i^n \Delta m_i r_i^2$, the angular momentum and rotational kinetic energy can be written in the neat way:

$$L = I\dot{\theta}, \quad \text{KE}_R = \frac{1}{2}I\dot{\theta}^2$$

The physical quantity I is called the *moment of inertia*.

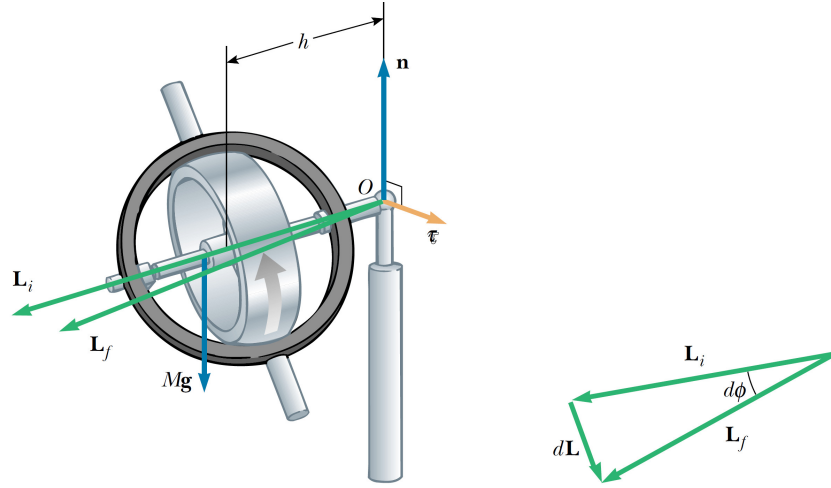


Figure 2.5: This gyroscope is better in practice since the friction at the pivot point is lesser.

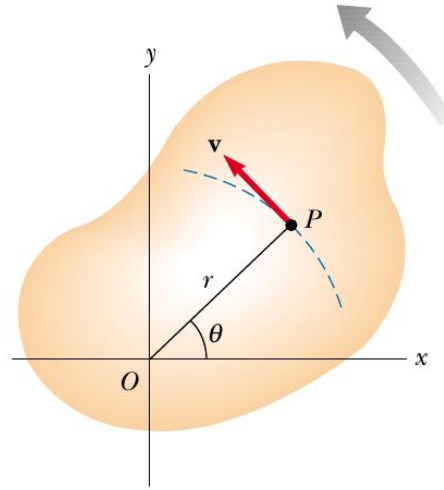


Figure 2.6: Rotation of a rigid body about the pivot point O .

Note. Different mass elements move in different tangential velocities. This is why we have written them in terms of the angular velocity which is independent of the positions of the mass elements.

The correspondences of linear and rotational motions

We may find the other expressions for rotational motion in the forms analogous to the forms of linear motion. Table 2.1 lists the correspondences of the physical quantities of linear and rotational motions.

Some physical quantities for rotation constructed by using the correspondences:

- Momentum is the product of mass and velocity, the rotational analogue — angular momentum is the product of moment of inertia and angular velocity:

$$L = I\dot{\theta}$$

- Force is the product of mass and acceleration. In rotation we have torque, it is the

Linear motion	Rotation
m mass	I moment of inertia
v velocity	$\dot{\theta}$ angular velocity
a acceleration	$\ddot{\theta}$ angular acceleration
s displacement	θ angle
p momentum	L angular momentum
F force	τ torque

Table 2.1: The correspondences of linear and rotational motions.

product of moment of inertia and angular acceleration:

$$\tau = I\ddot{\theta}$$

- Energy is defined as the product of force and distance travelled. In rotation, we have the product of torque and displaced angle:

$$E = \tau\theta$$

- Similarly, define the rotational kinetic energy:

$$\text{KE}_R = \frac{1}{2}I\dot{\theta}^2$$

It seems that there is no “rotational potential energy.” Potential energy is defined via the equation $\mathbf{F} = -\frac{dV}{d\mathbf{r}}$ where the force always lying along the displacement, and energy is a scalar. Pure rotation does not involve gravity at all. It makes sense that we do not have this type of equation.

Kinematic equations for rotation

Even without mass and force we can write similar forms of kinematic equations for rotation if $\ddot{\theta} = \text{const.}$

$$\begin{aligned}\dot{\theta}_f &= \dot{\theta}_i + \ddot{\theta}t, \\ \Delta\theta &= \dot{\theta}_i t + \frac{1}{2}\ddot{\theta}t^2, \\ \dot{\theta}_f^2 &= \dot{\theta}_i^2 + 2\ddot{\theta}\Delta\theta.\end{aligned}$$

where $\Delta\theta = \theta_f - \theta_i$. This set of “suvat-equations” like equations is good for both point particle and rigid body as the basic variable θ is independent of the radius of rotation.

2.4.4 Moment of inertia

Moment of inertia in rotational motion is analogous to a mass in translational motion. The moment of inertia I of an element of mass Δm located a distance r from the centre of rotation is defined by

$$\Delta I = \Delta m r^2. \quad (2.18)$$

As derived from the previous section, moment of inertia of a rigid body is defined by

$$I = \sum_i^n \Delta m_i r_i^2.$$

When $n \rightarrow \infty$, we get

$$I = \int r^2 dm. \quad (2.19)$$

Example 2.8. Two particles with mass m each and the separation d rotating in the xy -plane about the z -axis. (a) Calculate the moment of inertia of the system. (b) If the angular speed is ω , find the rotational kinetic energy.

Solution. (a) Because the distance of each particle is $d/2$ from the z -axis, the moment of inertia about the axis is

$$I = \sum_i m_i r_i^2 = m \left(\frac{d}{2}\right)^2 + m \left(\frac{d}{2}\right)^2 = \frac{1}{2}md^2.$$

(b) Using above result, the rotational kinetic energy is

$$\text{KE}_R = \frac{1}{2}I\omega^2 = \frac{1}{2} \frac{1}{2}md^2\omega^2 = \frac{1}{4}md^2\omega^2.$$

Example 2.9 (Moment of inertia of a thin rod with length l). If we take one end of a thin rod as the pivot point, then let ρ be the mass per unit length (density), r be the distance of the mass element to the pivot. The moment of inertia is

$$I = \int_0^l r^2 dm = \int_0^l r^2 \rho dr = \left[\rho \frac{r^3}{3} \right]_0^l = \frac{1}{3}ml^2.$$

The moments of inertia of the hoop and the solid cylinder shown in Figure 2.8 can be reduced from the hollow cylinder.

- When $R_1 = R_2$, the hollow cylinder becomes the hoop.
- When $R_1 = 0$, the hollow cylinder becomes the solid cylinder.

If we re-scale $M \rightarrow M/2$, $\ell \rightarrow \ell/2$ in the case of the long thin rod rotating about one of the ends in Figure 2.8 and double the number of rod, it becomes the long thin rod with rotation axis through the centre

$$I_{CM} = \frac{1}{3} \frac{M}{2} \left(\frac{\ell}{2}\right)^2 \times 2 = \frac{1}{12}M\ell^2$$

Parallel-axis theorem

If we know the moment of inertia of a rigid body rotating about its centre of mass, the moment of inertia of the body rotating about any parallel axis to the previous CM is the sum of the I_{CM} and md^2 , where m and d are the mass of the body and the displacement between the parallel axes. That is,

$$I_{\text{parallel-axis}} = I_{CM} + md^2. \quad (2.20)$$

Example 2.10. Apply the theorem for the previous examples relating the I_{CM} of a long thin rod rotating about the centre axis to the I about one end, we have

$$\frac{1}{12}M\ell^2 + M \left(\frac{\ell}{2}\right)^2 = \frac{1}{3}M\ell^2.$$

2.5 Pendulum

2.5.1 Simple pendulum

A simple pendulum consists of a small mass attaching to a suspended light cord with negligible mass allowing to perform a small vertical angle to and fro motion.

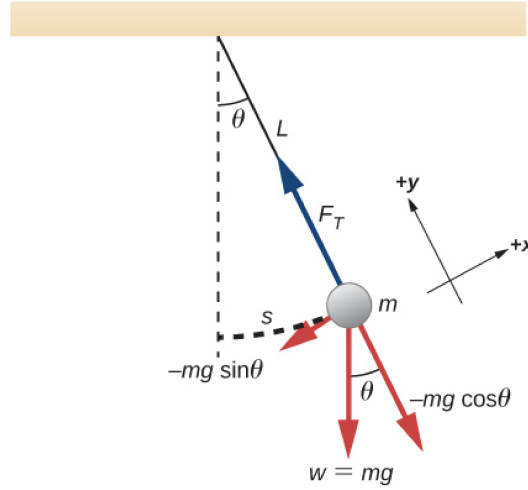


Figure 2.7: A simple pendulum

The motion of a simple pendulum is a *simple harmonic motion* as the tangential acceleration due to gravity is proportional to the horizontal displacement at a small angle θ but in opposite direction. The angular frequency and the period of the pendulum are (will be derived in the later chapter)

$$\omega = \sqrt{\frac{g}{\ell}}, \quad T = 2\pi\sqrt{\frac{\ell}{g}}. \quad (2.21)$$

2.5.2 Physical pendulum

Physical pendulum is like simple pendulum able to do a simple harmonic motion under gravity but more realistic; as it is a rigid body that we often find it in a grandfather clock and some pendulum related devices. The angular frequency and the period are

$$\omega = \sqrt{\frac{mg\ell}{I}}, \quad T = 2\pi\sqrt{\frac{I}{mg\ell}}. \quad (2.22)$$

In term of the weight, the period of oscillation can also be written as

$$T = 2\pi\sqrt{\frac{I}{W\ell}}. \quad (2.23)$$

Speed of walking and running

A human leg may be regarded as physical pendulum with a moment of inertia of a thin rod pivoted at one end. Given the moment of inertia of leg about the knee is $I = ml^2/3$,

where l is the length of the leg from the knee to the heel. Using (2.23) the period of oscillation is given by

$$T = 2\pi\sqrt{\frac{2}{3}\frac{l}{g}} \quad (2.24)$$

where we have put the distance $\ell = l/2$, being approximated distance from the knee to the centre of mass of the leg. Similar to a simple pendulum, the period in this case is independent of the mass of the leg.

Example 2.11. For a 90 cm long leg, the period is 1.6 sec.

Remark 2.1. In a most effortless walk, the legs swing at their natural frequency, the time for one step is $T/2$. Walking faster or slower requires additional muscular exertion and is more tiring. Similar considerations apply to the swinging of the arms.

The speed of walking v is proportional to

$$v \propto \frac{l}{T} \Rightarrow v \propto \frac{l}{\sqrt{l}} = \sqrt{l} \quad (2.25)$$

substituted by (2.24).

Remark 2.2. The natural walk of a small animal is slower than that of a large animal.

Speed of running

We assume:

- the length of the leg muscles is proportional to the length of the leg l ,
- the area of the leg muscles is proportional to l^2 ,
- the mass of the leg is proportional to l^3 .

The maximum force that a muscle can produce $F_{\max}$ is proportional to the area of the muscle. The maximum torque $\tau_{\max}$ is supposed to be

$$\tau_{\max} = F_{\max}l \propto l^3,$$

and the period of this pendulum is

$$T = 2\pi\sqrt{\frac{I}{\tau_{\max}}}.$$

Therefore, the period of oscillation in this case would be

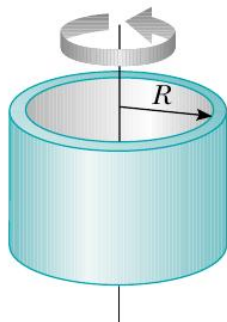
$$T \propto \sqrt{\frac{l^3 \times l^2}{l^3}} = l,$$

i.e., the period T is directly proportional to the length of leg l . Thus, this gives

$$v_{\max} \propto \frac{l}{T} \propto \frac{l}{l} = 1.$$

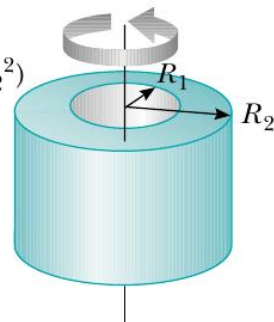
$\Rightarrow$ The maximum velocity is approximately **independent** of the length of legs. Thus, two people with different lengths of legs seem to be running at the same speed!

Hoop or
cylindrical shell
 $I_{\text{CM}} = MR^2$

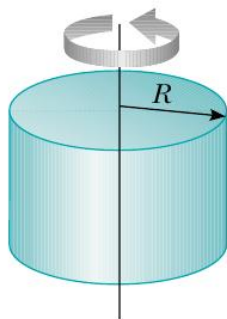


Hollow cylinder

$$I_{\text{CM}} = \frac{1}{2} M(R_1^2 + R_2^2)$$

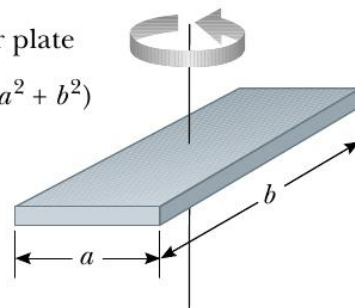


Solid cylinder
or disk
 $I_{\text{CM}} = \frac{1}{2} MR^2$

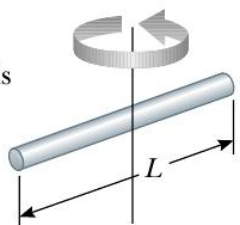


Rectangular plate

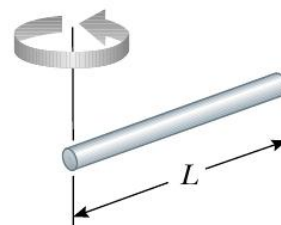
$$I_{\text{CM}} = \frac{1}{12} M(a^2 + b^2)$$



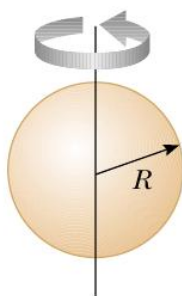
Long thin rod
with rotation axis
through center
 $I_{\text{CM}} = \frac{1}{12} ML^2$



Long thin
rod with
rotation axis
through end
 $I = \frac{1}{3} ML^2$



Solid sphere
 $I_{\text{CM}} = \frac{2}{5} MR^2$



Thin spherical
shell
 $I_{\text{CM}} = \frac{2}{3} MR^2$

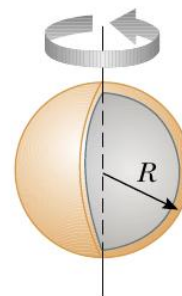


Figure 2.8: Moments of inertia of homogeneous rigid bodies with different geometries.

Chapter 3

Fluid and thermodynamics

3.1 Fluids

Solids and fluids transmit forces differently. When a force is applied to one section of a solid, this force is transmitted to the other parts of the solid with its direction unchanged. Because of a fluid's ability to flow, it transmits a force uniformly in all directions. Therefore, the pressure at any point in a fluid at rest is the same in all directions. The force exerted by a fluid at rest on any area is perpendicular to the area (Figure 3.1).

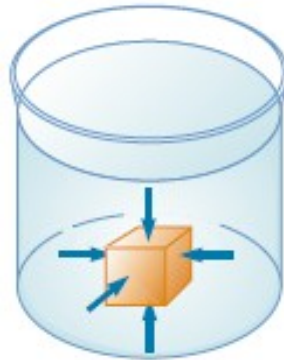


Figure 3.1: Pressure is defined by the acting force per unit area.

3.1.1 Pressure

The definition of pressure is usually reserved for fluid pressure. Similar to the definition of stress of rigid body, pressure is a physical quantity to describe how large the force is on a given surface area.

Definition 3.1. Pressure is the stress of a bulk deformation defined by the force per unit area

$$P = \frac{F}{A} \quad (3.1)$$

where F is the applied force to the normal direction of a surface area A .

Similarly, the pressure over an infinitesimal small area is defined as

$$P = \frac{dF}{dA}.$$

If ρ is the density of fluid, the difference of pressure between two positions 1 and 2 of an immersed object due to gravity is given by

$$P_2 - P_1 = \rho g l \quad (3.2)$$

where l is the vertical distance between the two positions.

Example 3.1. Please show equation (3.2).

Solution. Let h be the vertical distance from the top of the object to the surface of the fluid, i.e. the depth. Let A_1, A_2 be the horizontal cross sectional areas of the top and bottom of the object. Thus, we have

$$\begin{aligned} P_2 - P_1 &= \frac{F_2}{A_2} - \frac{F_1}{A_1} \\ &= \frac{\rho g(h+l)A_2}{A_2} - \frac{\rho g h A_1}{A_1} \\ &= \rho g l. \end{aligned}$$

Therefore, the difference of the pressures is independent of the shape of the object.

We have implicitly used the formula of fluid pressure expressed in terms of the depth of fluid due to gravity:

$$P = \rho g h \quad (3.3)$$

where h is the depth of the measuring position.

The unit of pressure is N/m². For convenience, we define the pascal: 1 Pa = 1 N/m². Another useful unit of pressure is *torr*:

$$\begin{aligned} 1 \text{ torr} &= 1 \text{ mm Hg} \\ &= 13.5 \text{ mm water} \\ &= 1.32 \times 10^{-3} \text{ atm} \\ &= 1.93 \times 10^{-2} \text{ psi} \\ &= 1.33 \times 10^2 \text{ Pa.} \end{aligned}$$

Example 3.2. Water is filled to a height H behind a dam of width w . Determine the resultant force exerted by the water on the dam.

Solution. The resultant force is the total force at normal direction acting on the dam. Since the force is not linearly gradually increasing with the depth, we let the depth **from the bottom** be y . The pressure at the depth y is

$$P = \rho g(H - y)$$

The infinitesimal force acting on the dam is

$$dF = \rho g(H - y)dA = \rho g(H - y)w dy.$$

Integrating the equation over the height of water gives the total resultant force

$$\begin{aligned} F &= \int_0^H \rho g(H - y)w dy \\ &= \rho g w \left(Hy - \frac{y^2}{2} \right)_0^H \\ &= \frac{1}{2} \rho g w H^2 \end{aligned}$$

Buoyancy

An object floating or immersed into a fluid experiences an upward force is called the *buoyancy* as a result of the different pressures along the vertical displacement under gravity.

Archimedes' principle Archimedes' principle states that a body partially or wholly submerged in a fluid is buoyed upward by a force that is equal in magnitude to the weight of the displaced fluid.

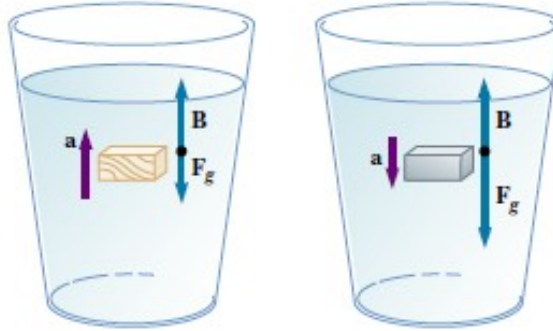


Figure 3.2: Because the density of the object in the left figure is smaller than water, the object is going upward. In the right figure, the object density is larger than water, hence it sinks.

Viscosity with buoyancy in free fall

Consider a round and small object freely falling via a tank of oil with the drag force $b\mathbf{v}$ and the buoyancy $\mathbf{B}$, the equation of motion is

$$mg - B - bv = m \frac{dv}{dt}.$$

Integrating the equation over the intervals gives

$$\int_0^t dt = \int_0^v \frac{m}{mg - B - bv} dv \Rightarrow t = -\frac{m}{b} \ln \left(\frac{mg - B - bv}{mg - B} \right)$$

Simplifying we get

$$v = \frac{mg - B}{b} (1 - e^{-bt/m}).$$

When $t \rightarrow \infty$, $v = (mg - B)/b$. It is exactly equal to the terminal velocity v_t by putting $dv/dt = 0$ into the first equation. The drag force $b\mathbf{v}$ is an example of the application of Stokes' law (1.20).

Pascal's principle

According to (3.1), pressure is the applied force per unit area. If P is a constant (different positions at the same attitude) we have

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

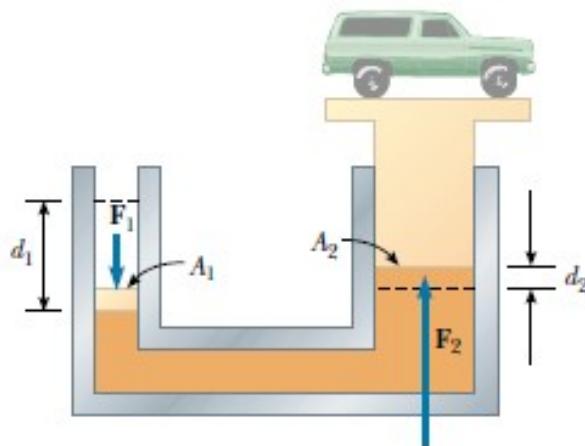


Figure 3.3: The pressures are assumed to be equal on the both sides.

giving a balance equation for different applied forces on different normal areas (Figure 3.3). If there is an object (the car) placed on the top of the A_2 , only a small applied force (less than the weight of the car) on the A_1 can lift the object up. The ratio A_2/A_1 is similar to the *mechanical advantage* of a lever. The principle which is similar to the system of a lever by using the principle of static equilibrium of torques

$$\ell_1 \times \mathbf{F}_1 = \ell_2 \times \mathbf{F}_2$$

can be used for lifting a heavy object.

Example 3.3. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 5 cm. This pressure is transmitted by a liquid to a piston that has a radius of 15 cm. What force must the compressed air exert to lift a car weighing 13,300 N? What air pressure produces this force?

Solution. Because the pressure exerted by the compressed air is transmitted undiminished throughout the liquid, we have (as in Figure 3.3)

$$\begin{aligned} F_1 &= \left(\frac{A_1}{A_2} \right) F_2 = \frac{\pi \times (5 \times 10^{-2})^2}{\pi \times (15 \times 10^{-2})^2} \times 1.33 \times 10^4 \\ &= 1.48 \times 10^3 \text{ N.} \end{aligned}$$

The air pressure that produces this force is

$$P = \frac{F_1}{A_1} = \frac{1.48 \times 10^3}{\pi \times (5 \times 10^{-2})^2} = 1.88 \times 10^5 \text{ Pa.}$$

This pressure is approximately twice atmospheric pressure.

Note. The work done of input is equal to the work done of output, i.e. $F_1 d_1 = F_2 d_2$, in accordance with the law of conservation of energy.

Blood pressure

The contraction of the heart chambers is triggered by electrical pulses that are applied simultaneously both to the left and to the right halves of the heart. First the atria contract, forcing the blood into the ventricles; then the ventricles contract, forcing the blood out of the heart. Because of the pumping action of the heart, blood enters the arteries in spurts or pulses. The maximum pressure deriving the blood at the peak of the pulse is called the *systolic pressure*. The lowest blood pressure between the pulses is called the *diastolic pressure*. In a young healthy individual the systolic pressure is about 120 torr and the diastolic pressure is about 80 torr. Therefore the average pressure of the pulsating blood at heart level is 100 torr.

As the blood flows through the circulatory system, its initial energy, provided by the pumping action of the heart, is dissipated by two loss mechanism: losses associated with the expansion and contraction of the arterial walls and *viscous friction* associated with the blood flow. Due to these energy losses, the initial pressure fluctuations are smoothed out as the blood flows away from the heart, and the average pressure drops. By the time the blood reaches the capillaries, the flow is smooth and the blood pressure is only about 30 torr. The pressure drops still lower in the veins and is close to zero just before returning to the heart. In this final stage of the flow, the movement of blood through the veins is aided by the contraction of muscles that squeeze the blood toward the heart. One-way flow is assured by unidirectional valves in the veins.

3.1.2 Surface tension

The molecules constituting a liquid exert attractive forces on each other. A molecule in the interior of the liquid is surrounded by an equal number neighbouring molecules in all directions. Therefore, the net resultant inter-molecular force on an interior molecule is zero (Figure 3.5). However, the situation is different near the surface of the liquid. Because

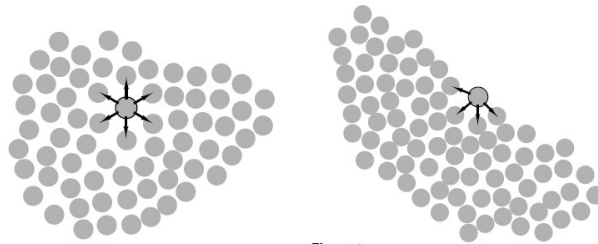


Figure 3.4: The inter-molecular force is canceled in the left diagram. In the right diagram, an unbalanced inter-molecular force arises at the boundary.

there are no molecules above the surface, a molecule here is pulled predominantly in one direction, toward the interior of the surface (Figure 3.5). This causes the surface of a liquid to contract and behave somewhat like a stretched membrane. This contracting tendency results in a *surface tension* that resists an increase in the free surface of the liquid.

Definition 3.2. Surface tension (intrinsic) γ is defined by

$$\gamma := \frac{F}{L} \quad (3.4)$$

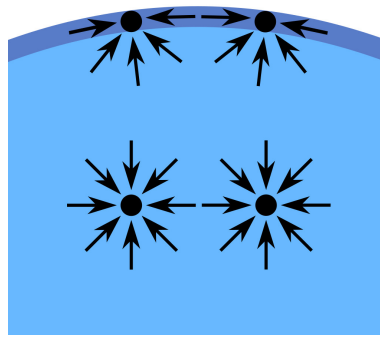


Figure 3.5: Molecules are pulled predominantly in the directions toward the interior of the surface.

where F is the pulling force due to inter-molecular force with direction perpendicular to L , which is the length of the movable side of an object.

If a surface has two movable sides, we may define

$$\gamma := \frac{F}{2L}.$$

Each of the sides contributes equally to the force; thus the factor $1/2$.

How liquid wets container

When a liquid is contained in a vessel, the surface molecules near the wall are attracted to the wall. This attractive force is called *adhesion*. At the same time, however, these molecules are also subject to the attractive *cohesive* force exerted by the liquid, which pulls the molecules in the opposite direction.

If the adhesive force is greater than the cohesive force, the liquid wets the container wall, and the liquid surface near the wall is curved upward. If the opposite is the case, the liquid surface is curved downward.

As an example two glass vessels contain water and mercury, respectively. At the edge, compare with the strength of interactions between media:

curved upward glass–water > water–water

curved downward glass–mercury < mercury–mercury

Capillary effect

In Figure 3.6, the adhesion between the red material and the liquid is greater than the cohesion of the liquid itself. If the tube is narrow the adhesive force is stronger enough against the weight of the liquid to rise it up. The adhesion between the blue material and the liquid is weaker than the cohesion, hence the liquid is depressed.

For calculating the height h of capillary rise, the weight W of the red column of the supported liquid is

$$W = \pi R^2 h \rho g$$

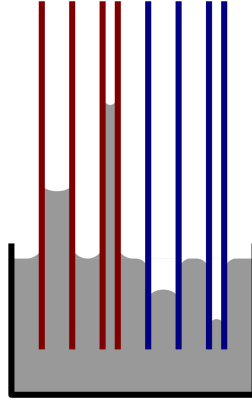


Figure 3.6: The adhesion between the red material and the liquid is greater than the cohesion of the liquid itself.

where R is the radius of the column and ρ is the density of the liquid. The maximum force F_m due to the surface tension along the periphery of the liquid is

$$F_m = 2\pi R\gamma.$$

The upward component of this force supporting the weight of the column of liquid is

$$2\pi R\gamma \cos \theta = \pi R^2 h \rho g$$

where θ is the angle between the curving direction and the vertical. Thus, the height of the column is

$$h = \frac{2\gamma \cos \theta}{R\rho g}.$$

In contrary, the depressed liquid in the blue tube will give a negative height. These effects are called *capillary action*.

Droplets

Another consequence of surface tension is the tendency of liquid to assume a half spherical shape (hemisphere). This tendency is most clearly observed in a liquid outside a container. Such an uncontained liquid forms into a hemisphere that can be noted in the shape of raindrops (Figure 3.7). The pressure inside the spherical liquid drop is higher than the pressure outside. The excess pressure ΔP in a liquid sphere of radius R is

$$\Delta P = \frac{2\gamma}{R} \tag{3.5}$$

Example 3.4. Show equation (3.7).

Solution. Because the liquid drop looks like a hemisphere on a surface (as an example like in Figure 3.7), the radius of the contact area between the droplet and table is R . The force due to the surface tension at the boundary (the circumference of the base circle) of



Figure 3.7: A drop of water on a leaf.

must equal to the force due to the interior pressure ΔP of the droplet, since pressure is non-directional. Thus,

$$2\pi R\gamma = \Delta P\pi R^2$$

$$\Delta P = \frac{2\gamma}{R}$$

3.2 Bernoulli's equation

3.2.1 Flow

The flow of a fluid is said to be *steady*, or *laminar*, if each particle of the fluid follows a smooth path, such that the paths of different particles never cross each other. A *flow* is

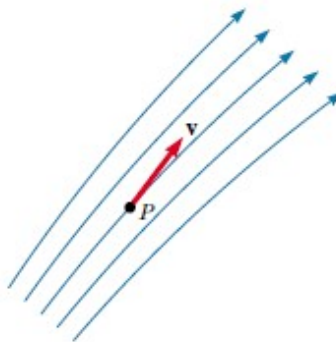


Figure 3.8: A particle in laminar flow follows a streamline, and at each point along its path the particle's velocity is tangent to the streamline.

defined as the rate of change of fluid volume with respect to time

$$Q = \frac{dV}{dt}$$

In steady flow, Q is constant. It follows the vanishing of the second derivative of $V(t)$.

If a flow moves above a certain critical speed, fluid flow becomes *turbulent*, which is irregular characterized by small whirlpool-like regions. The dynamics of the particles in turbulent flow are extremely difficult to predict.

Friction exists between fluid particles. *Viscosity* is used in the description of fluid flow to characterize the degree of internal friction in the fluid. The *viscous force* is associated with the resistance that two adjacent layers of fluid have to moving relative to each other as shown in Figure 3.9. Viscosity causes part of the kinetic energy of a fluid to be converted to internal energy.

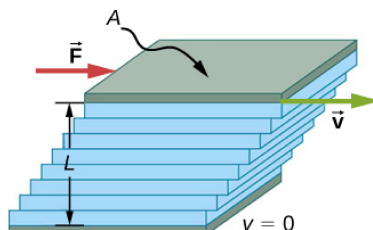


Figure 3.9: The viscous force $\mathbf{F}$ is associated with the resistance that two adjacent layers of fluid have to moving relative to each other.

Ideal fluid

The assumptions of an *ideal fluid*:

Nonviscous The internal friction is neglected. An object moving through the fluid experiences no viscous force.

Steady The flow remains constant.

Incompressible The density of an incompressible fluid is constant.

Irrotational The fluid has no angular momentum about any point.

Streamlines and the equation of continuity

The path taken by a fluid particle under steady flow is called *streamline*.

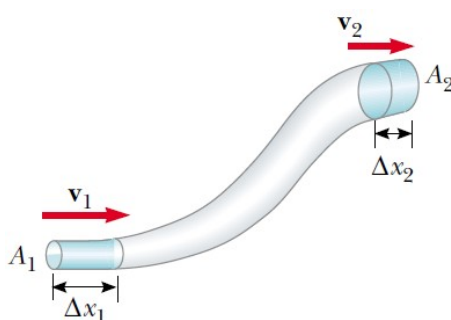


Figure 3.10: A fluid moving with steady flow through a pipe of varying cross-sectional area.

Consider an ideal fluid flowing through a pipe of nonuniform size (Figure 3.10). The particles in the fluid move along streamlines in steady flow. In a time t , the fluid at the bottom end of the pipe moves a distance $\Delta x_1 = v_1 t$. If A_1 is the cross-sectional area in this region, then the mass of fluid contained in the left shaded region is

$$m_1 = \rho A_1 \Delta x_1 = \rho A_1 v_1 t,$$

where ρ is the density of the ideal fluid. Similarly, at the upper end of the pipe we have

$$m_2 = \rho A_2 v_2 t.$$

For the conservation of mass and steady flow $m_1 = m_2$, we obtain

$$A_1 v_1 = A_2 v_2 = \text{constant}.$$

This expression is called the *equation of continuity*.

Bernoulli's equation

Any idea fluid moving in steady flow obey the *Bernoulli's equation*:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant} \quad (3.6)$$

where P is the pressure at the region, ρ fluid density and h is the height difference between two different regions (Figure 3.11). Equation (3.6) can be explicitly expressed as a relationship of the total pressures between two different regions

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2.$$

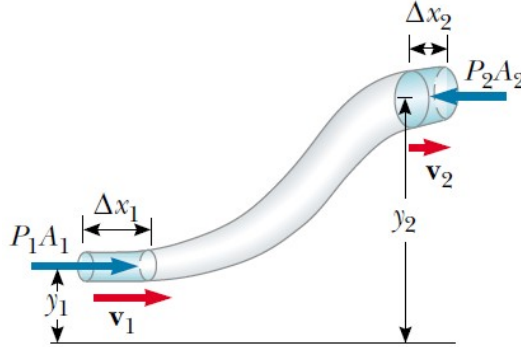


Figure 3.11: A fluid in laminar flow through a constricted pipe.

Example 3.5. Prove the Bernoulli's equation (3.6).

Proof. Multiplying the volume element ΔV of the liquid passing through the pipe at Δt on the left hand side of (3.6) gives

$$\begin{aligned} & P\Delta V + \frac{1}{2}\rho v^2\Delta V + \rho\Delta Vgh \\ &= F\Delta x + \frac{1}{2}\Delta mv^2 + \Delta mgh \\ &= \text{total energy of the region in an interval } \Delta t \\ &= \text{constant} \end{aligned}$$

where Δx is the horizontal displacement of the fluid. The first term is the work required to move the ΔV volume of liquid, second term is the kinetic energy of the lump of liquid, the third term is the potential energy of it. Law of conservation of energy directly implies the result. $\square$

If a steady flow fluid is passing through a horizontal pipe (Figure 3.12), we have the simple relation between pressure and velocity of the fluid:

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2. \quad (3.7)$$

Because the fluid is assumed being incompressible, v_2 must be larger than v_1 . Therefore,

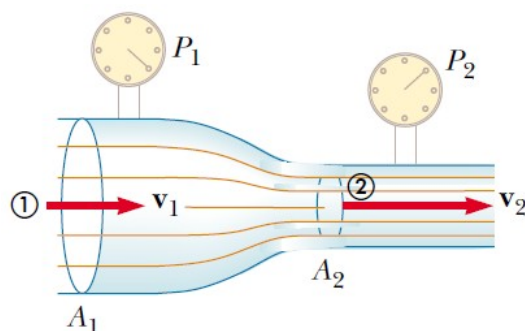


Figure 3.12: A steady flow fluid is passing through a non-uniform sectional area pipe.

P_2 is smaller than P_1 .

Example 3.6. Consider a very crowded classroom in which students are squeezed together. As soon as a door is opened and students begin to exit, the squeezing (pressure) is least near the door, where the motion (flow) is greatest.

Example 3.7 (Airplane wings). By the aid of Figure 3.13, explain why aeroplane can move upward.

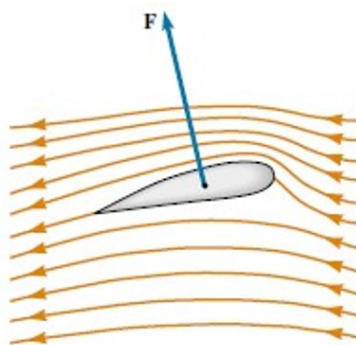


Figure 3.13: The cross section of an aeroplane wing. The red lines represent the direction of air flow.

Solution. The plane is running horizontally to the right hand side indicated from the figure. Taking the plane as the reference, air flowing to the direction opposite to the plane with $-v$ speed of the plane. Because the fluid is steady, the air particles traveling on the top of the plane's wing is greater than the lower part. It implies that the velocity of the air particles on the upper side is larger than the lower part. According to Bernoulli's equation (3.7), larger velocity implies lower pressure at the same region. The pressure difference between the upper and lower part of the wing results the lifting of the plane.

Example 3.8 (Spinning ball). Please describe the movement of the spinning ball in Figure 3.14

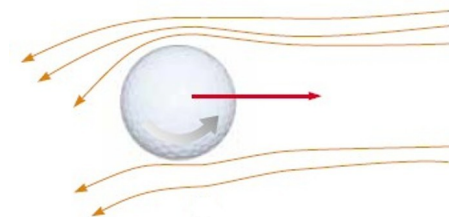


Figure 3.14: A spinning ball is moving horizontally to the right through air.

Solution. Suppose that the spinning ball is moving in speed v to the right direction with spinning speed u as shown in Figure 3.14. It follows that the speed of the air molecules on the upper side is $-(v + u)$, and $-v + u$ at the lower part of the ball. Since $|-(v + u)| > |-v + u|$, the speed of the air molecules on the upper part of the ball is greater than the lower part. The same argument is applied as in the previous problem, the upper part pressure of the ball is lower than the bottom. Therefore, the pressure difference resulting the ball is pushed upward. Consequently, combining the horizontal and vertical components of the movement, the path of the moving ball appears a curve bending upward shown in the figure.

Arteriosclerosis and blood flow

Arteriosclerosis is the most common of cardiovascular diseases. In arteriosclerosis, the arterial wall becomes thickened, and the artery is narrowed by deposits called *plaque*. This condition may seriously impair the functioning of the circulatory system. A 50% narrowing (*stenosis*) of the arterial area is considered moderate. Sixty to seventy percent is considered severe, and a narrowing above 80% is deemed critical. one problem caused by stenosis is made clear by Bernoulli's equation.

The blood flow through the region of constriction is speeded up. If, for example, the radius of the artery is narrowed by a factor of 3, the cross-sectional area decreases by a factor of 9, which results in a nine-fold increase in velocity. In the constriction, the kinetic energy increases by 9^2 , or 81. The increased kinetic energy is at the expense of the blood pressure; that is, in order to maintain the flow rate at the higher velocity, the potential energy due to pressure is converted to kinetic energy. As a result, the blood pressure in the constricted region drops. For example, if in the unobstructed artery the flow velocity is 50 cm/sec, then in the constricted region, where the area is reduced by a factor of 9, the velocity is 450 cm/sec. Correspondingly, the pressure is decreased by about 80 torr. Because of the low pressure inside the artery, the external pressure may actually close off the artery and block the flow of blood. When such a blockage occurs in the coronary artery, which supplies blood to the heart muscle, the heart stops functioning.

Stenosis above 80% is considered critical because at this point the blood flow usually becomes turbulent with inherently larger energy dissipation than is associated with laminar flow. As a result, the pressure drop in the situation presented earlier is even larger than calculated using Bernoulli's equation. Further, turbulent flow can damage the circulatory system because parts of the flow are directed toward the artery wall rather than parallel to it, as in laminar flow. The blood impinging on the arterial wall may dislodge some of the plaque deposit which downstream may clog a narrower part of the artery. If such clogging occurs in a cervical artery, blood flow to some part of the brain is interrupted causing an *ischemic stroke*.

Example 3.9. Compute the average velocity of the blood in the aorta of radius 1 cm if the flow rate is 5 liter/min.

Solution. Let v be the velocity of the blood, and r the radius of the aorta. The flow rate is given by

$$\begin{aligned}\text{flow rate} &= vr^2\pi \\ 5 \times \frac{1000}{60} &= v \times 1^2\pi \\ v &= 26.5 \text{ cm/sec.}\end{aligned}$$

We have changed the unit from litre/min to cm^3/sec . The average velocity of the blood is 26.5 cm/sec.

3.3 Thermodynamics

3.3.1 Thermal properties

Temperature

It is convenient to use a linear metrical scale to measure how hot of an object is by increasing the heat energy. Water is chosen as a reference substance that we define the scale from its melting point to boiling point by 100 units, each unit is called *degree Celsius*, or in short $^{\circ}\text{C}$. This method of revealing thermal energy density is called *temperature*, and it should be noted that the temperature is independent of the total heat energy of the system.

A simple example to show this is to compare a 0.5 litre volume of 80°C water with a 1 litre 80°C water. They have the same temperature but with different total energy content. The **thermal energy density** of a system defined in this way is called *temperature*.

The unit of temperature is *Kelvin* (K). It is nice the unit Celsius compatible to Kelvin with the same scale as $1 \text{ K} = 1^{\circ}\text{C}$.

Fahrenheit (F) is another scale of temperature mainly used in United States. Nevertheless, it is not SI and does not match very well to the unit *Kelvin* (K). The relationship between the Celsius and Fahrenheit temperature scales is

$$T_{\text{F}} = \frac{9}{5}T_{\text{C}} + 32^{\circ}\text{F}. \quad (3.8)$$

Note. Temperature is independent of the total heat energy of a system.

Absolute zero temperature

The pressure of a box of gas decreases as decreasing the temperature in an ideal situation. The temperatures of different type of gases tend to reaching to a fixed lowest value if we are trying to decrease their pressures towards zero as shown in Figure 3.15. Thus, the hypothetical *absolute zero temperature* of all substances by definition and observation is -273.15°C corresponding to zero pressure. We define this corresponding lowest temperature as our new origin of the scale, i.e., $0 \text{ K} = -273.15^{\circ}\text{C}$.

Why is the pressure zero at the absolute zero temperature?

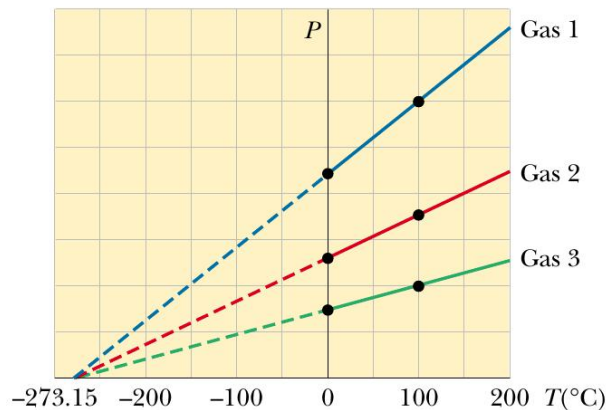


Figure 3.15: Pressure versus temperature for three dilute gases.

Thermal expansion

In most cases, substances expand volumes while being heated. The increment of the length of a rigid body by heat is given by

$$\Delta L = \alpha L \Delta T$$

where α is the coefficient of linear expansion, L the original length, and T the change of temperature. Written in the form of derivative we have

$$\frac{dL}{dT} = \alpha L. \quad (3.9)$$

The equation of thermal expansion in two dimensions is derived from the one dimension equation. Multiplying L on both sides of (3.9) gives

$$L \frac{dL}{dT} = \alpha L^2 \quad \Rightarrow \quad \frac{1}{2} \frac{dL^2}{dT} = \alpha L^2$$

Thus,

$$\frac{dA}{dT} = 2\alpha A \quad (3.10)$$

where $A = L^2$ is the area. Performing the same way, the expansion equation for three-dimension is

$$\frac{dV}{dT} = 3\alpha V \quad (3.11)$$

with $V = L^3$.

Heat capacity and specific heat

The *heat capacity* C of a particular sample of a substance is defined as the amount of energy needed to raise the temperature of that sample by 1°C . We see that if the *heat transfer* Q produces a change of temperature ΔT of a substance, then

$$Q = C \Delta T. \quad (3.12)$$

The *specific heat capacity* c of a substance is the heat capacity per unit mass. Thus, if energy Q transferred by heat to a mass m of a substance changes the temperature of the sample by ΔT , the specific heat of the substance is

$$c = \frac{Q}{m\Delta T}. \quad (3.13)$$

Quantitatively, the specific heat capacity is the amount of heat energy required to raise the temperature of 1 g of a substance by 1 degree.

Example 3.10. Our body is composed of 75% water and 25% proteins, with specific heat capacities 1 and 0.4, respectively. The small amount of fat and minerals are safely to be ignored. The specific heat of our body is approximately to be

$$0.75 \times 1 + 0.25 \times 0.4 = 0.85$$

Specific heat capacity can also be written in terms of derivatives as by varying the related physical quantities. If the pressure of an enclosed system is kept constant, the specific heat capacity in the form of derivative is given by

$$c_P = \frac{1}{m} \left(\frac{dQ}{dT} \right)_P.$$

Similarly, if the volume is kept constant it is

$$c_V = \frac{1}{m} \left(\frac{dQ}{dT} \right)_V.$$

The subscripts P and V are just labels for the related non-changing quantities. We also have the same versions for the heat capacities by removing the $1/m$ term.

Latent heat

Latent heat is the quantity of thermal energy required to **transform** a substance from one form (phase) to another form. It is also called *phase transition*.

Define the heat Q absorbed or released in a phase change of mass m by latent heat

$$Q = mL \quad (3.14)$$

where L is the latent heat of fusion or vaporization.

Example 3.11.

$$\text{ice} \longrightarrow \text{water}, \quad \text{water} \longrightarrow \text{steam}$$

Note. The temperature of a substance **does not** change during the phase transition, the only change is the form of substance (e.g. Figure 3.16):

$$100^\circ\text{C water} \longrightarrow 100^\circ\text{C water vapour}$$

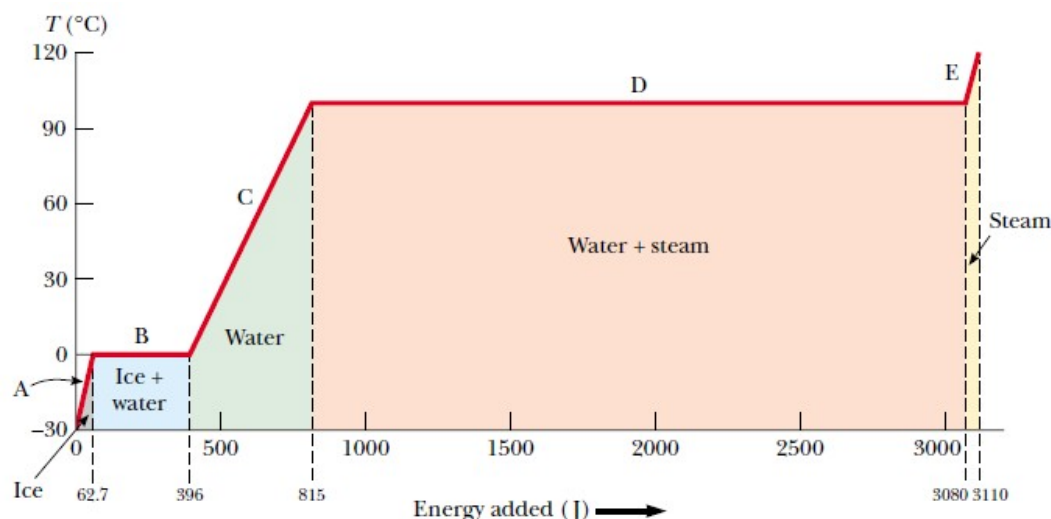


Figure 3.16: Latent heat energy is required to change the phase of water by keeping constant temperature.

3.4 Kinetic theory of matter

One of Einstein's paper in 1905 shows that liquid and gas molecules are in **random motion** continuously in an enclosed system. If a small piece of metal plate (e.g.) is suspended we can observe small movement of fluctuation or vibration of the object continuously. The random motion which are due to the bombardments of the gas molecules on the plate is called *Brownian motion*. The discovery was announced by Brown, however Einstein was the first to explain the phenomenon.

Note. What Brown has actually **done** is pouring pollen into a tank of water to observe the random motion of the grains.

3.4.1 Heat transfer

Molecules can move and vibrate, depending on the states of matters:

Solid with restricted movement

Liquid with less restricted movement

Gas with almost free and random movement

These motions induce kinetic energy known as *thermal energy*.

We have stated that macroscopically temperature is the energy content of a system per unit volume, or the energy density. In kinetic theory of matter we identify that temperature measures the average kinetic energy per particle

$$T \propto \frac{1}{2} m \langle v^2 \rangle. \quad (3.15)$$

It should be stressed that the definition of the temperature of a particle is equal to (proportional here) the average kinetic energy of the particle. But for example the temperature

of a system with ten moving particles at the same speed does not in general equal to taking the average of the temperatures of the individual particles! Unless they move in the same direction!

Energy is said to be transferred when higher energetic molecules collide on some lower kinetic energy molecules. We feel hot if we contact something whose *internal energy* is higher than the surface (skin) of our body.

The unit of thermal energy (heat) is also common measured in *calories*. One calorie (cal) is the amount of heat required to raise the temperature of 1 g of water by 1 °C. The formal definition of calorie is defined as the heat required to raise the temperature of 1 g of water from 14.5 °C to 15.5 °C. In terms of Joule, one calorie is equal to 4.184 J. Heat is often measured in the unit of kilo-calorie, abbreviated as Cal or kcal. 1 Cal = 1000 cal.

Conduction

The process of heat transfer via a **solid** substance is called *conduction*.

The molecules in solid are moving restrictly, the kinetic energy of moving molecules are mainly contributed by vibration. Thus, the efficiency of heat transfer is low due to the movement limitation of the molecules and atoms.

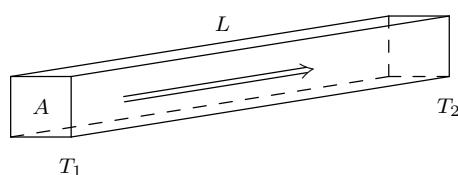
Insulator The substance has poor conduction properties.

Conductor The substance is effective to transfer thermal energy. Most of conductors are metals, there are *free electrons* on the outermost orbits of the electron shells, able to move freely amongs atoms inside even solid metal.

The *heat flow* passes through a block of material, the amount of heat conducted per second is given by

$$P_c = \frac{K_c A}{L} (T_1 - T_2) \quad (3.16)$$

where A is the cross section area, L the length, $T_1 - T_2$ the temperature difference, and K_c the *coefficient of thermal conductivity*.



Convection

Convection is the process of heat transfer in fluids (gases and liquids). Unlike solid the volume of the substance is not fixed, thus

- Hotter fluid implies less dense,
- Cooler fluid implies denser.

Due to the different densities, a convection current is generated due to **gravity**.

Imagine boiling a glass of water, the hotter water molecules at the bottom become less dense, the upper cooler molecules will go down (comparably heavy per unit volume) to replace the hotter molecules. The hot molecules will go upward. Thus, we have the *convection current*:

Fluid flows from the denser region into the less dense region.

Fluid flowing from a denser region into the less dense region is not only found in convection. It is also the mechanism of *diffusion* but without considering gravity.

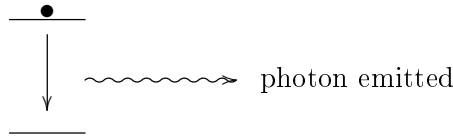
Heat transferred per unit time P_c of convection is given by

$$P_c = K_c A(T_1 - T_2) \quad (3.17)$$

where K_c is the coefficient of convection.

Radiation

Thermal radiation is *electromagnetic radiation* (EM). Electron from a higher energy state (shell) jumping to a lower energy state emits a photon.



Photon is the particle description of EM radiation carrying energy. EM radiation in classical physics is explained as the consequence of the process of **acceleration** (deceleration) of electric charge.

Thermal radiation is a sort of EM radiation with a longer wavelength (than red light). The accelerating motion of positive charged nuclei and negative charged electrons in matter **emit** thermal radiation.

- The acceleration of electrons are much larger than the heavier nuclei.
 - radiation is mainly generated by moving electrons.
- EM wave carries energy
 - material is continuously losing internal energy for converting to thermal radiation.

On the opposite, matter can absorb EM radiation resulting the **frequency** of electrons vibration increase, hence temperature increases. The amount of radiation absorbed by a material depends on its composition.

The rate of emission of radiant energy P_r by a unit area of a body at temperature T is given by

$$P_r = e\sigma T^4 \quad (3.18)$$

where $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ is the *Stefan-Boltzmann constant* e is the *emissivity* of the surface ranging from 0 to 1, depending on T and the nature of the surface.

Because σ and e are the same for absorption and emission, the net loss of radiant energy from a body is

$$P_r = e\sigma(T_1^4 - T_2^4)$$

where T_1, T_2 are the temperatures of the body and environment.

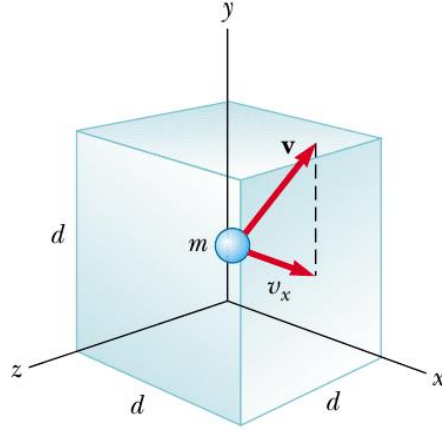


Figure 3.17: A gas molecule moving in an arbitrary direction with velocity $\mathbf{v}$, the component of x direction is v_x .

3.4.2 Ideal gases

Consider a cubical box with length d containing N number of *ideal gas* molecules. A moving gas molecule hitting a surface induces force due to the change of momentum during impact and bounce over the short time interval. Suppose it moving in an arbitrary direction with velocity $\mathbf{v}$, the component in x direction is v_x . The change of momentum of the gas molecule in that direction is

$$\Delta p_x = -mv_x - (mv_x) = -2mv_x.$$

The impulsive force along that direction in a given time interval is

$$F_1 = \frac{-2mv_x}{\Delta t} = \frac{-2mv_x}{2d/v_x} = \frac{-mv_x^2}{d}$$

where the force F_1 with a label '1' represents the first single molecule that we are considering. We simply write the force acting on the molecule by the wall in the same way without the negative sign according to Newton's third law $F_1 = \frac{mv_x^2}{d}$. Thus, the total force F exerted by the wall normal to the x direction is

$$F = \frac{m}{d}(v_{x1}^2 + v_{x2}^2 + \dots).$$

Write the total force in terms of the **average** velocity square

$$F = \frac{Nm}{d}\overline{v_x^2}$$

where N is the total number of gas molecules in the box. Since $v^2 = v_x^2 + v_y^2 + v_z^2$, the average value of v^2 for all the molecules in the box is

$$\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}.$$

The motion of molecules is completely random, the magnitudes of the three independent components should roughly the same

$$\overline{v^2} = 3\overline{v_x^2}.$$

Therefore, the total force exerted on the wall is

$$F = \frac{N}{3} \left(\frac{m\overline{v^2}}{d} \right).$$

Since we are interested in the total pressure exerted on the wall, thus

$$P = \frac{F}{A} = \frac{F}{d^2} = \frac{1}{3} \left(\frac{N}{d^3} m\overline{v^2} \right) = \frac{1}{3} \left(\frac{N}{V} \right) m\overline{v^2} = \frac{2}{3} \left(\frac{N}{V} \right) \left(\frac{1}{2} m\overline{v^2} \right).$$

Or, we may write the expression in the form of equation of state for ideal gas. Let, for a single molecule,

$$\frac{1}{2} m\overline{v^2} = \frac{3}{2} k_B T$$

where $k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$ is called the *Boltzmann constant*. For N particles contained in an enclosure with volume V , the equation of state for an ideal gas is

$$PV = NkT. \quad (3.19)$$

In most cases the number of particles N is fixed. In practice we have

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad (3.20)$$

When temperature is kept to be constant, we have the convenient form of *Boyle's law*

$$P_1 V_1 = P_2 V_2.$$

When pressure is constant, we get the *Charles's law*

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}.$$

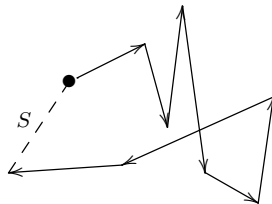
3.4.3 Diffusion

Random walk

Random walk (Figure 3.18) is a description of a molecule moving in fluid in which it may encounter collisions with other molecules as a result the path of it will be totally **random** after collisions.

Let L be the average distance travel (*mean free path*) of a molecule between collisions, the displacement after N collisions from the starting point statistically would be

$$S = L\sqrt{N}. \quad (3.21)$$



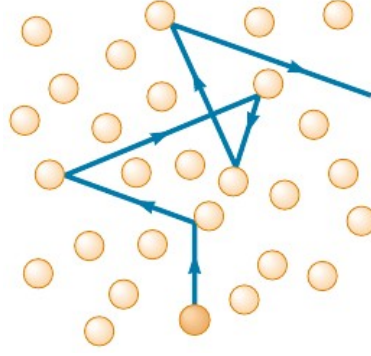


Figure 3.18: A molecule moving through a fluid collides with other molecules in a random fashion.

Diffusion

Diffusion is a process that particles in a high concentration region move to a region of lower concentration due to **random walk**. Examples like a concentrated coloured droplet is spreading around a sink of water, oxygen and nutrients diffuse into cells for the elimination of the waste products.

As particles move randomly, more particles doing random walk in a higher density region than in a lower density region, the resultant movement is from the higher concentration region to the lower.

Using (3.18) the number of collisions of a particle diffuses through a distance S being

$$N = \frac{S^2}{L^2}$$

and the total distance travel is NL . The time taken for a particle with velocity v is given by

$$t = \frac{\text{total distance}}{v} = \frac{NL}{v} = \frac{S^2}{Lv}.$$

Note. Gases are less denser than liquid. Generally the mean free path of a gas molecule is greater and the diffusion time is shorter than a liquid molecule.

Example 3.12. Diffusion of water molecules:

- Mean free path of a diffusing molecule is $\sim 10^{-8}$ cm.
- Velocity is $\sim 10^4$ cm/sec at room temperature.

The time required to diffuse through a distance of 1 cm is

$$t = \frac{S^2}{Lv} = \frac{1^2}{10^{-8} \times 10^4} = 10^4 \text{ sec} = 2.8 \text{ hours}.$$

For a typical size of a tissue cell 10^{-3} cm, the time required is only 10^{-2} second. This short time is good enough to transport water for our bodies.

Flux of diffusion

Flux is defined by the number of transporting particles per unit area per unit time. The flux of diffusion is given by

$$J = \frac{v_D \Delta C}{2}.$$

- ΔC are the difference of concentrations (densities) between two regions.
- v_D is diffusion velocity which is the displacement by diffusion divided by the diffusion time, i.e., $v_D = s/t$.
- The factor 2 accounts for the molecules diffusing both toward and away from the two different concentration regions.

Given two different concentration regions with displacement Δx , the diffusion velocity is $v_D = Lv/\Delta x$. We get

$$J = \frac{Lv\Delta C}{2\Delta x}.$$

The diffusion coefficient is defined by $D = \frac{Lv}{2}$.

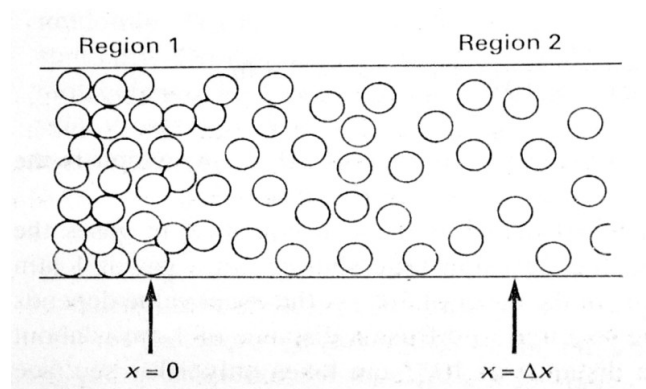


Figure 3.19: Molecules in the denser region 1 diffuse to the region 2.

In the limit, $\frac{\Delta C}{\Delta x} \rightarrow \frac{dC}{dx}$, the flux is the product of diffusion coefficient and the *gradient of concentration*

$$J = D \frac{dC}{dx}. \quad (3.22)$$

The equation of flux of diffusion can be written as

$$J = \frac{D}{\Delta x} (C_1 - C_2).$$

For example, D of salt (NaCl) is $1.09 \times 10^{-5} \text{ cm}^2/\text{sec}$. The diffusion coefficients for biologically important molecules are in the range from 10^{-7} to $10^{-6} \text{ cm}^2/\text{sec}$.

Note. It should be noted that although molecules **mainly** diffuse from denser to less dense region, there still exists diffusion from less dense to denser region.

Diffusion through membranes

A membrane consists of many pores (holes) in microscopic scale.

- Molecules larger than the pores can not pass through,
- molecules smaller than the pores can pass through the membrane but the efficiency of diffusion will be slowed down due to the decrease of cross-sectional area of the channel.

The net flux of molecules J flowing through a membrane with permeability P is given by

$$J = P(C_1 - C_2)$$

Many kinds of membrane in living systems only permit water molecules to pass through, such one-way passage of water is called *osmosis*.

3.5 Laws of thermodynamics

Zeroth law of thermodynamics

Body A thermally contacts to body B with temperature T_A and T_B , the whole system C (say) will reach an equilibrium state with temperature T_C .

As the law appears so natural and quite matches to our daily experiences scientists placed it to the first position of the laws of thermodynamics. Since the first law has been developed before this law, that is why it is called the ‘zeroth law’.

3.5.1 Work and heat in thermodynamic processes

Consider a gas contained in a cylinder fitted with a movable piston (Figure 3.20). At equilibrium, the gas occupies a volume V and exerts a uniform pressure P on the cylinder’s walls and on the piston. Assume that the gas expands *quasi-statically*, that is, slowly enough to allow the system to remain essentially in thermal equilibrium at all times. As the piston moves up a distance dy , the work done by the gas on the piston is

$$dW = Fdy = PAdy = PdV.$$

Because the gas expands, dV is positive, and so the **work done by the gas is positive**. In thermodynamics, positive work represents a transfer of energy out of the system. The total work done by the gas as its volume changes from V_i to V_f is given by

$$W = \int_{V_i}^{V_f} PdV$$

Clearly, work done by the gas is the area under the pressure function as shown in Figure 3.21. Therefore, the work done by a system depends on the initial and final states and on the path followed by the system between these states.

The energy transfer by heat Q into or out of a system also depends on the process. Consider the situations depicted in Figure 3.22. In (a), an *energy reservoir* is a source of energy that is considered to be so great that a finite transfer of energy from the reservoir does not change its temperature. Because the piston is moving upward, the gas is doing

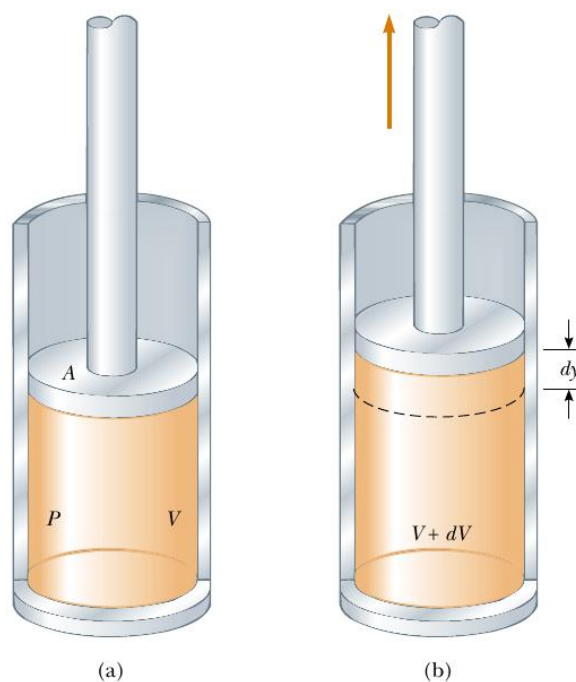


Figure 3.20: Gas at a pressure P does work on a moving piston as the system expands from V to $V + dV$.

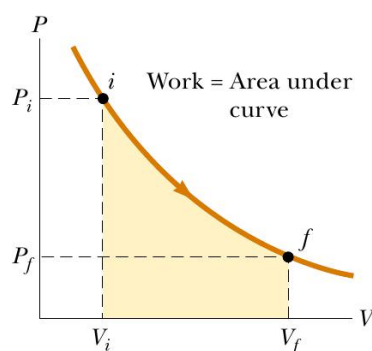


Figure 3.21: A gas expands quasi-statically (slowly) from state i to state f .

work on the piston. However, in (b), when the membrane is broken, the gas expands rapidly into the vacuum until it occupies a volume V_f and is at a pressure P_f . In this case, the gas **does no work** because there is no movable piston on which the gas applies a force. The initial and final states of the ideal gas in both cases are identical, but the paths are different. In the first case, the gas does work on the piston, and energy is transferred slowly to the gas. In the second case, no energy is transferred, and the value of the work done is work. Because heat and work depend on the path, neither quantity is determined solely by the end points of a thermodynamic process.

3.5.2 First law of thermodynamics

The total energy is conserved.

Quantitatively, suppose that a system undergoes a change from an initial state to a final state. During this change, energy transfer by heat Q to the system occurs, and work

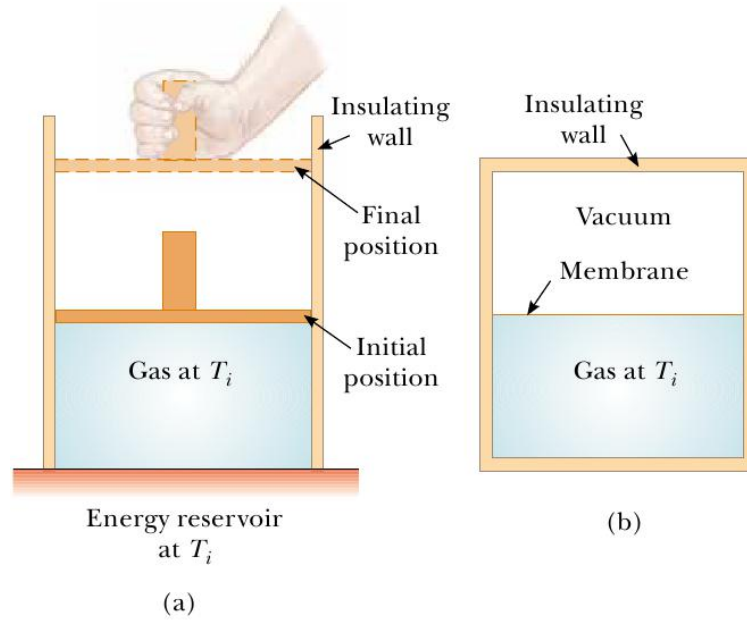


Figure 3.22: (a) A gas at T expands slowly while absorbing energy from a reservoir in order to maintain a constant temperature. (b) A gas expands rapidly into an evacuated region after a membrane is broken.

W is done **by** the system. As an example, suppose that the system is a gas in which the pressure and volume change from P_i and V_i to P_f and V_f . If the quantity $Q - W$ is measured for various paths connecting the initial and final equilibrium states, we find that it is the same for all paths connecting the two states. We conclude that the quantity $Q - W$ is determined completely by the initial and final states of the system, and we call this quantity the change in the internal energy of the system. Although Q and W both depend on the path, the quantity $Q - W$ is **independent** of the path. If the internal energy is U , then the change in internal energy ΔU can be expressed as

$$\Delta U = Q - W.$$

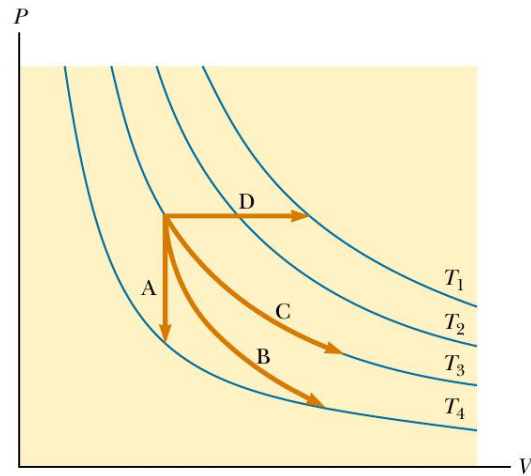
Adiabatic process

An *adiabatic process* is one during which no energy enters or leaves the system by heat, i.e., $Q = 0$. An adiabatic process can be achieved either by thermally insulating the system from its surroundings or by performing the process rapidly, so that there is little time for energy to transfer by heat. Applying the first law of thermodynamics to an adiabatic process,

$$\Delta U = -W.$$

From this result, we see that if a gas expands adiabatically such that W is positive, then ΔU is negative and the temperature of the gas decreases. Conversely, the temperature of a gas increases when the gas is compressed adiabatically. However, the process described in Figure 3.22(b) is unique and is called an *adiabatic free expansion*. In this special adiabatic process, both $Q = 0$ and $W = 0$. As a result, $\Delta U = 0$. Since the internal energy of an ideal gas depends only on its temperature, thus we expect no change in temperature during an adiabatic free expansion.

Example 3.13. Characterize the paths in the figure as isobaric, isovolumetric, isothermal, or adiabatic. Note that $Q = 0$ for path B.



Second law of thermodynamics

One of the most important physical laws; nature has a **tendency from order to disorder**. This basic rule is not only valid in physics, it seems fitting to any models of science.

However, from an engineering standpoint, perhaps the most important implication of the second law is the limited efficiency of heat engines. The second law states that a machine capable of continuously converting internal energy completely to other forms of energy in a cyclic process **cannot be constructed**.

Energy conservation of human body

The process is roughly via

take food $\rightarrow$ in the body $\rightarrow$ give heat and work

The energy of the enclosed system is conserved by the first law.

Let E_c be the chemical energy stored in the tissue of a body, E_t be the internal thermal energy of the body. If the body weight is not lost (E_c and E_t kept constant), over a given period of time the energy intake must be exactly equal to the sum of the work done and the heat lost by the body.

Example 3.14. Recall the process:

take food $\rightarrow$ inside the body $\rightarrow$ give heat and work

Why are the arrows pointing to in one way direction? Could we write the equation as

release food $\leftarrow$ inside the body $\leftarrow$ absorb heat and work ?

The first law of thermodynamics **only** tells us the energy conservation of energy, this problem may be explained by the topic of *arrow of time*.

Meaning of thermodynamics

Thermo- relates to the subject of heat energy. Then what is dynamics?

*Evolution of physical state under time translation; often appearing in terms of the rate of change of physical states with respect to **time**.*

There are many ways to define *time*. Examples:

Cosmological time It started at the moment of the birth of the universe (If it existed).

Psychological time It is the time that you feel fine and all right.

Biological time The “internal clock” of living organisms

3.6 Second law of thermodynamics

3.6.1 Heat engines

A *heat engine* is a device that converts internal energy (heat) to mechanical energy.

In a typical process by which a power plant produces electricity, coal or some other fuel is burned, and the high temperature gases produced are used to convert liquid water to steam. This steam is directed at the blades of a turbine, setting it into rotation. The mechanical energy associated with this rotation is used to drive an electric generator. Another heat engine; the internal combustion engine in an automobile; uses energy from a burning fuel to perform work that results in the motion of the automobile.

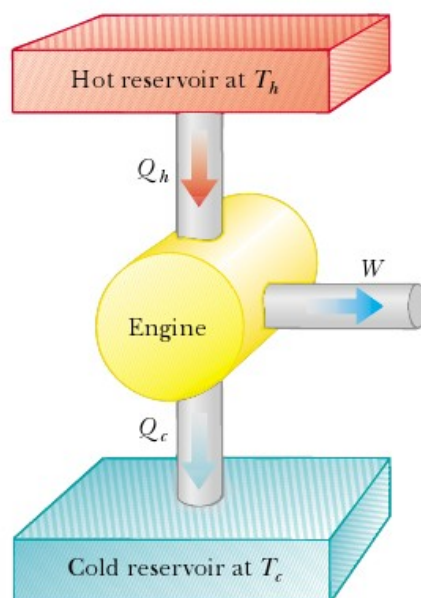


Figure 3.23: The engine absorbs energy Q_h from the hot reservoir, expels energy Q_c to the cold reservoir, and does work W .

A heat engine (Figure 3.23) carries some working substance through a cyclic process during which

1. the working substance absorbs energy from a high temperature energy reservoir,

2. work is done by the engine, and
3. energy is expelled by the engine to a lower temperature reservoir.

The engine absorbs a quantity of energy Q_h from the hot reservoir, does work W , and then gives up a quantity of energy Q_c to the cold reservoir. Because the working substance goes through a cycle, its initial and final internal energies are equal, and so $\Delta U = 0$. Hence, from the first law of thermodynamics, $\Delta U = Q - W$, and with no change in internal energy, the net work W done by a heat engine is equal to the net energy Q_{net} flowing through it. Therefore,

$$W = Q_h - Q_c.$$

The net work done in a cycle process is the area enclosed by the curve representing the process on a PV diagram (Figure 3.24).

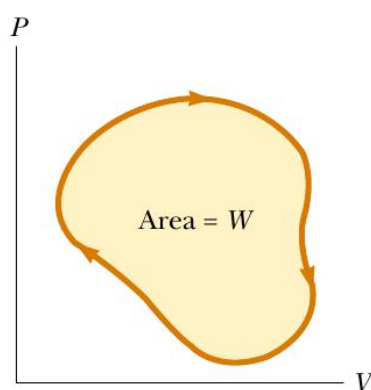


Figure 3.24:

Now, we would like to ask is how efficient of this heat engine? Can we get 100% efficiency, or in other words, can we transform **all** the heat energy to pure mechanical energy theoretically?

Thermal efficiency

The thermal efficiency e of a heat engine is defined as the ratio of the net work done by the engine during one cycle to the energy absorbed at the higher temperature during the cycle:

$$e = \frac{W}{Q_h} = \frac{Q_h - Q_c}{Q_h} = 1 - \frac{Q_c}{Q_h}.$$

For 100% efficiency, only if $Q_c = 0$, i.e., if no energy is expelled to the cold reservoir — impossible.

The Kelvin-Planck form of the **second law of thermodynamics** states the following:

It is impossible to construct a heat engine that, operating in a cycle, produces no effect other than the absorption of energy from a reservoir and the performance of an equal amount of work.

The Carnot engine

In 1824, Sadi Carnot showed that a heat engine operating in an ideal, reversible cycle — called a Carnot cycle (Figure 3.26) — between two energy reservoirs is the most efficient engine possible. Such an ideal engine established an upper limit on the efficiencies of all other engines. That is, the net work done by a working substance taken through the Carnot cycle is the greatest amount of work possible for a given amount of energy supplied to the substance at the upper temperature.

Figure 3.25 shows the PV diagram for the Carnot cycle. We see that the net work done, W equals the net energy received in on cycle, $Q_h - Q_c$. Note that $\Delta U = 0$ for the cycle.

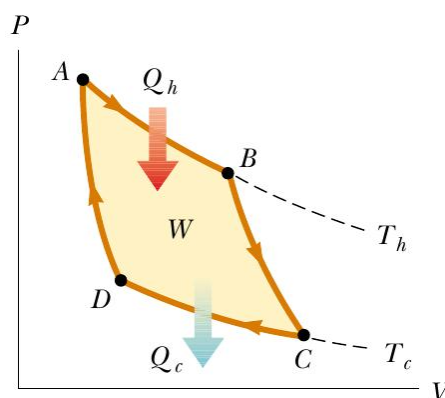


Figure 3.25:

We shall move to discuss other form of second law of thermodynamics. Let us think about this problem first:

Example 3.15. A glass will break into pieces by hitting on the ground if it falls from a table. The energy of this system is conserved, however. Parts of the energy of the glass will go into heat, sound and other forms of energy at the moment of breaking. Why is the reverse process not possible? In other words, why in nature these energies do not come back to the broken pieces to form a glass then to fly back from the ground to the top of the table?

Solution. The broken glasses (pieces) plus the emitted sound, heat etc. are more **disorder** than the unbroken glass.

3.6.2 Probability and entropy

Consider tossing three coins, surely the outcome of each coin to be either head (H) or tail (T). The combinations of the outcomes of the coins will be:

HHH HHT HTH THH TTT TTH THT HTT

If HTT is the particular outcome that we want, the probability to get it (HTT) is $1/8$. We probably have to do the experiment few times to get this result by chance.

Naïvely speaking, repeating to do this experiment requires **time**. More coins will cause more difficult to get the required arrangement of the coins by chance. This is to say, we would take **much more time** for the particular outcome.

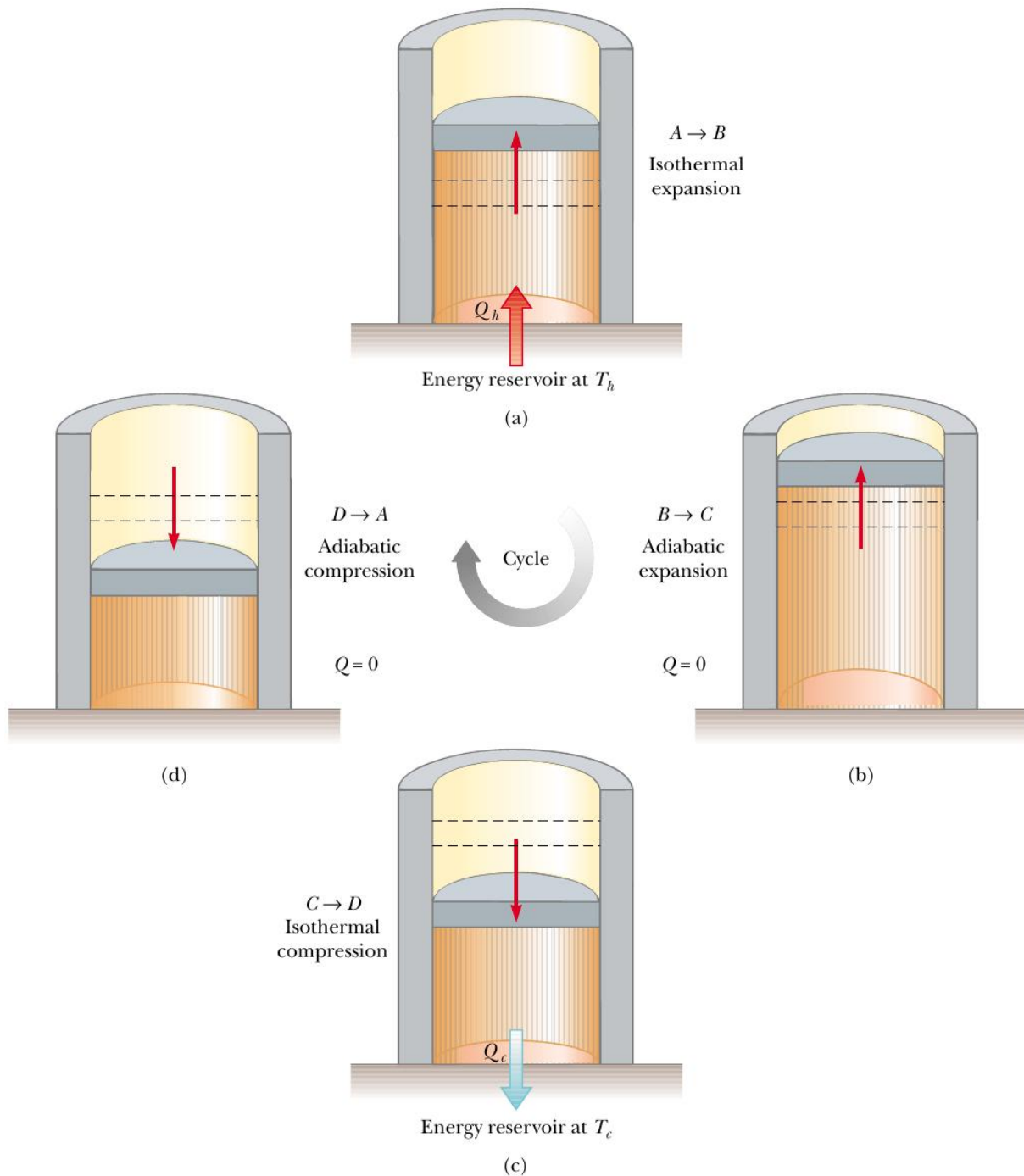


Figure 3.26: The Carnot cycle. In process $A \rightarrow B$, the gas expands isothermally while in contact with a reservoir at T_h . In process $B \rightarrow C$, the gas expands adiabatically ($Q = 0$). In process $C \rightarrow D$, the gas is compressed isothermally while in contact with a reservoir at $T_c < T_h$. In process $D \rightarrow A$, the gas is compressed adiabatically. The upward arrows on the piston indicate that weights are being removed during the expansions, and the downward arrows indicate that weights are being added during the compressions.

Example 3.16 (Food and energy). Recall the energy conservation law:

take food $\rightarrow$ in the body $\rightarrow$ give heat and work

Food Some particular arrangements of molecules by chemical bondings (energy).

Dissipated heat and work plus waste product Chaotic (random) thermal radiation, work done and different arrangements of molecules (than food) of the waste product.

The time taken for the reversing process by chance probably longer than **the life time of the universe**. It implies that in practice there is not possible to get the reverse process. This reason is also applied for the example of the broken glass.

The second law of thermodynamics can be expressed as:

The direction of spontaneous change in a system is from an arrangement of lesser probability to an arrangement of greater probability.

Entropy is a quantity to measure the degree of ordering.

Note. The entropy S_A of a system A is higher than the entropy S_B of a system B , if the system A is more disorder than the system B .

Entropy

The original definition of entropy does not look relating to anything of order or disorder. In statistical mechanics, the behaviour of a substance is described in terms of the statistical behaviour of its atoms and molecules. One of the main results is that isolated systems tend toward disorder and that entropy is a measure of this disordering.

In terms of thermodynamics, consider any infinitesimal process in which a system changes from one equilibrium state to another. If dQ is the amount of energy transferred by heat when the system follows a reversible path between the states, then the **change in entropy** dS is defined by

$$dS = \frac{dQ}{T}$$

where the temperature T is constant because the process is infinitesimal. Entropy, like the internal energy, is a *state function*. The change in entropy during a process depends only on the end points and therefore is independent of the actual path followed.

In addition, the entropy of the statistical thermodynamics is defined by

$$S = k_B \ln \Omega \tag{3.23}$$

where k_B as usual is the Boltzmann constant, Ω is the number of microscopic configurations that correspond to a thermodynamic system in a state specified by certain macroscopic variables.

Information

In information science, entropy is a measure of unpredictability of information content. Consider tossing a fair coin that the probability to obtain a head is equal to get a tail. We may say the entropy of the coin toss is as high as it could be because there is no way to predict the outcome of the tossing coin. We define one binary bit as

$$\log_2 2 = 1$$

where the base 2 stands for “yes” or “no”, “1” or “0” etc, the argument 2 is the number of configuration of the ensembles. In our tossing coin case the 2 is the head or tail outcomes. *Shannon entropy* is defined by

$$H = - \sum_i p_i \log_b p_i \quad (3.24)$$

where b usually is 2, e , or 10, p_i is the probability mass function.

3.6.3 Difference between heat and other forms of energy

The main feature that distinguishes heat from other forms of energy is the random nature of its manifestations. For example, when heat flows via conduction from one part of the material to another part, the flow occurs through the sequential increase in the internal energy along the material. This internal energy is in the form of random chaotic motion of atoms. Similarly, when heat is transferred by radiation, the propagating waves travel in random directions. The radiation is emitted over a wide wavelength range (mainly in the infrared range), and the phases of wave along the wave front are random. By comparison, other forms of energy are more ordered. Chemical energy, for example, is present by virtue of specific arrangements of atoms in a molecule. Potential energy is due to the well-defined position, or configuration, of an object.

While one form of energy can be converted to another, heat energy, because of its random nature, cannot be completely converted to other forms of energy. For example, consider a gas in a cylinder with a piston. Heat flows into the gas; this increases the kinetic energy of the gas molecules and, therefore, raises the internal energy of the gas. The molecules moving in the direction of the piston collide with the piston and exert a force on it. Under the influence of this force, the piston moves. In this way, heat is converted into work via *internal energy*.

The heat added to the gas causes the molecules in the cylinder to move in random directions, but only the molecules that move in the direction of the piston can exert a force on it. Therefore, the kinetic energy of only the molecules that move toward the piston can be converted into work. For the added heat to be completely converted into work, all the gas molecules would have to move in the direction of the piston motion. In a large ensemble of molecules, this is very unlikely.

In work, the energy is in an ordered motion; in heat, the energy is in random motion. Although some of the random thermal motion can be ordered again, the ordering of all the motion is very improbable. Because the probability of completely converting heat to work is vanishingly small, the second law of thermodynamics states categorically that it is impossible. As an example of the stated heat energy (Figure 3.23), heat can be partially converted to work as it flows from a region of higher temperature to a region of lower temperature. Heat can be completely converted into work only if the heat is rejected into

a reservoir at absolute zero temperature. However, absolute zero temperature cannot be attained both on experiment and theory. Therefore, heat cannot be completely converted into work.

3.6.4 Thermodynamics and living system

Why do we need food? (again...)

The simple and straightforward answer of this question is that food provides us energy to maintain our life.

Example 3.17. If one does not do any work, the temperature of the body is kept constant by some equipment, and if there are also some methods to pause the mechanism of his/her body, then, does he/she still need food?

Solution. The answer is positive. Because our body is built as a highly ordered configuration system comparing to the ingredients (cells, molecule, atoms) of the structure falling apart. We take food to maintain the highly ordered configuration structure.

A single protein molecule in the body may consist of a million atoms bound together in an ordered sequence. Cells are more complex still. Their specialized functions within the body depend on a specific structure and location. From the second law of thermodynamics such a highly ordered system, left to itself, tends to become disordered, and once it is disordered, it ceases to function. Work must be done on the system continuously to prevent it from falling apart. For example, the blood circulating in veins and arteries is subject to friction, which changes kinetic energy to heat and slows the flow of blood. If a force were not applied to the blood, it would stop flowing in a few seconds. The concentration of minerals inside a cell differs from that in the surrounding environment. This represents an ordered arrangement. The natural tendency is toward an equalization with the environment. Work must be done to prevent the contents of the cell from leaking out. Finally, cells that die must be replaced, and if the animal is growing, new tissue must be manufactured. For such replacement and growth, new proteins and other cell constituents must be put together from smaller, relatively more random subcomponents. Thus, the process of life consists of building and maintaining ordered structures. In the face of the natural tendency toward disorder, this activity requires work.

The work necessary to maintain the ordered structures in the body is obtained from the chemical energy in food. Except for the energy utilized in external work done by the muscles, all the energy provided by food is ultimately converted into heat by friction and other dissipative processes in the body. Once the temperature of the body is at the desired level, all the heat generated by the body must leave through the various cooling mechanisms of the body. The heat must be dissipated because, unlike heat engine, the body does not have the ability to obtain work from heat energy. The body can obtain work only from chemical energy. Even if the body did have mechanisms for using heat to perform work, the amount of work it could obtain in this way would be small—the second law sets the limits.

The amount of disorder in a system can be expressed quantitatively by *entropy*. In all cases, the increase in the entropy in the surroundings produced by the living system is always greater than the decrease in entropy obtained in the living system itself. The total process of life, therefore, obeys the second law. Thus, living systems are perturbations in the flow toward disorder. They keep themselves ordered for a while at the expense of the

environment. This is a difficult task requiring the use of the most complex mechanisms found in nature. When these mechanisms fail, as they eventually must, the order falls apart, and the organism dies.

Solar energy

The usual understanding is:

The sun gives energy. Plants absorb solar energy and Earth's minerals to transform to food for living things. Animals eat plants and we eat plants and animals...and so on. In addition, the sun gives us heat and warm....

The mechanism of our body is not designed for absorbing solar energy to use directly. Perhaps we would die of starvation and skin cancer if we only take sun bath without taking food. That is why there is the *food chain*.

Example 3.18. From the first law of thermodynamics, energy is conserved. Why living systems and Earth have to absorb solar energy continuously?

Solution. The electromagnetic radiation (EM) from the sun is “cold”. Because

1. The EM radiation shinning on Earth are roughly having similar wavelengths due to the nuclear fusion reaction in the core of the sun.
2. Radiation directly shines on Earth, i.e. supposed being no scattering or diffusion.

Living systems and anything on Earth absorb the “ordered” thermal energy in order to balance the tendency of going disorder. Disordered thermal radiation is continuously being emitted to the surrounding from Earth.

Information

Imagine leaving a house without maintenance and cleaned up for a long time but you still live in there. As a result the house probably would be very dirty, untidy and messed up with everything. This is a typical example of order to disorder.

Now, we would like to change the room back into an order form. Clearly we need “energy” for doing work such as to put things (books, soft drink cans, cleaning the floor...) back to the right places as well as doing cleaning. Nevertheless, the most important factor is that we need *information* to understand how the work has to be done. That is, for examples, we need information to know where we should put the things to the right places, how to clean the stuff, and all the knowledge for doing these.

Note. Information is physical.

Information and entropy

The relation of information to entropy was discovered not long ago. Each different arrangement (the way of ordering) of a system reveals different set of information.

Example 3.19. Information of a computer software is encoded by a bunch of numbers of 1 and 0.

Example 3.20. The human DNA consists of about a billion molecular units in a well-determined sequence. This huge of information for example guides the assembly of the various proteins and enzymes required for the functioning of the organisms.

3.7 Heat and life

Living organisms are mostly directly affected by temperature. The tolerant range of metabolism is about 2 °C to 120 °C due to relying on water.

- The simplest of living organisms can function near this extreme range.
- Large-scale living systems (e.g. human) are restricted to a much narrower range.

Note. *Warm-blooded* animals have ability to maintain their internal body temperature enduring severe environment better than *cold-blooded* animals (e.g. reptiles).

3.7.1 Metabolic rate

A man weighing 70 kg lying quietly awake consumes about 70 Cal/hr which is called *basal metabolic rate*.

The amount of energy consumed by a person depends on his or her weight and build. *Metabolic rate* is defined by

$$\frac{\text{rate of energy consumed}}{\text{surface area of body}} \approx \text{constant}$$

The surface area of a person is approximately given by

$$\text{Area (m}^2\text{)} = 0.202 \times W^{0.425} \times H^{0.725}$$

where W and H are the weight and height of the person.

Activity	Metabolic rate (Cal/m ² -hr)
Sleeping	35
Lying awake	40
Sitting upright	50
Standing	60
Walking (3 mph)	140
Moderate physical work	150
Bicycling	250
Running	600
Shivering	250

Table 3.1: Metabolic rates for selected activities of human.

Example 3.21 (Energy from food). The oxidation of glucose sugar molecule is given by



Enzymes (catalysts) are required for doing those chemical reactions.

Activity (Cal/m ² -hr)	Rate (Cal/m ²)
8 hr sleeping (35)	280
8 hr moderate physical labor (150)	1200
4 hr reading, writing, TV watching (60)	240
1 hr heavy exercise (300)	300
3 hr dressing, eating (100)	300
Total expenditure	2320

Table 3.2: One day's metabolism: A person with surface area 1.7 m² can take 400 g of carbohydrates, 200 g of protein, and 171 g of fat to compensate this consumption.

Food	Weight	Protein	C _x H _y O _z	Fat	Energy
Whole milk	976	32	48	40	660
Egg, 1	50	6	0	12	75
Hamburger, 1	85	21	0	17	245
Carrots, 1 cup	150	1	10	0	45
Potato (baked)	100	2	22	0	100
Apple	130	0	18	0	70
Bread, rye	23	2	12	0	55
Doughnut	33	2	17	7	135

Table 3.3: Some common food composition measured in gram and kilocalorie.

Note. Our body can eliminate the excess of certain substances, e.g. water and salt. But **no** mechanism to eliminate an excess in caloric intake. The excess energy is stored in the form of tissue (fatty tissue).

Example 3.22 (Woman's pregnancy). The energy requirements are expected to increase during pregnancy due to the growth and metabolism of the fetus. We assume

- the weight gain of the fetus is uniform (in 270 days),
- the fetus weighs 3 kg at birth, each day it gains 11 g,
- 75% of tissue consists of water and inorganic minerals, only 2.75 g of the daily mass increase is due to organic materials (protein).

$$\text{Calories required} = \frac{2.75 \text{ g protein}}{\text{day}} \times \frac{4 \text{ Cal}}{\text{protein g}} = 11 \text{ Cal/day}$$

where the energy content of protein 4 Cal per gram is assumed. We must add the basal metabolic rate for this relatively small value:

$$0.13 \times 40 \times 24 + 11 = 125 + 11 = 136 \text{ Cal/day}$$

This small value suggests that a woman may not be necessary to increase her food intake during pregnancy.

3.7.2 Green house effect

Compare these cases:

- (1) Sunny without much polluted atmosphere
 - wavelengths of solar radiation are similar, due to nuclear fusion reaction
 - same radiation direction
- (2) Cloudy, and polluted atmosphere
 - solar radiation is diffused
 - range of wavelengths becomes large
 - radiation scattered to various directions

The entropy in case (1) (more ordered) is lower than the case (2) (less ordered). We feel in case (2) hotter than in case (1). Why?

✎. The wavelength of EM radiation always increases by scattering. Longer wavelength EM wave is more wave-like, and hence more non-directional. The frequency of scattered sun light is around in the infra-red spectral region.

Chapter 4

Wave

4.1 Oscillations

Oscillation is a kind of periodic motions. Define a frequency ν to be the inverse of the period T of a periodic motion

$$\nu = \frac{1}{T}. \quad (4.1)$$

The unit of frequency is s^{-1} , we often use Hz instead.

4.1.1 Simple harmonic motion

Simple harmonic motion is periodic motion, in which the acceleration of the system is proportional to the displacement and acts in the opposite direction of it. For example, a spring obeys Hooke's law

$$F = -kx$$

where the force is opposite to the direction of the displacement. The maximum absolute value of the displacement is the *amplitude*.

Since any period motion can be described by sine, cosine functions, or some combination of them, the variation of the amplitude of a spring attaching to a mass on a smoothness horizontal plane at any time t under a simple harmonic motion is

$$x(t) = A_0 \cos\left(\frac{2\pi}{T}t\right) = A_0 \cos(\omega t). \quad (4.2)$$

In fact, it does not matter if we use sine or cosine functions for the definition. Here using the cosine for the initial instant $t = 0$ we get the maximum amplitude A_0 .

We may translate a function to the right or left relative to the coordinate system. The equation for simple harmonic motion in general is

$$x(t) = A_0 \cos(\omega t - \phi) \quad (4.3)$$

where ϕ is the phase difference. In this example the function is shifted to the right. Differentiating (4.3) with respect to t gives the velocity

$$v(t) = \dot{x}(t) = A_0 \omega \sin(\omega t - \phi). \quad (4.4)$$

The second derivative of the wavefunction gives the acceleration

$$a(t) = \ddot{x}(t) = A_0 \omega^2 \cos(\omega t - \phi). \quad (4.5)$$

Oscillation of a mass on a spring

Oscillation is due to the conservation law of energy that the energy of the system is kept to be constant by periodically changing the kinetic energy to potential energy stored by the spring, and the vice versa.

Using Hooke's law and Newton's second law we get

$$m \frac{d^2x}{dt^2} = -kx \quad \Rightarrow \quad \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

Inserting the equations of the acceleration (4.5) and the simple harmonic motion (4.3) into above gives

$$-A_0\omega^2 \cos(\omega t - \phi) = -\frac{k}{m}A_0 \cos(\omega t - \phi).$$

Thus, we obtain the angular frequency

$$\omega = \sqrt{\frac{k}{m}}. \quad (4.6)$$

It follows the period

$$T = 2\pi\sqrt{\frac{m}{k}}. \quad (4.7)$$

4.1.2 Energy and the simple harmonic oscillator

Stretching the spring of the previous simple harmonic oscillator problem will store energy into it. Integrating the force $F = kx$ gives the potential energy stored in the deformation of the spring

$$U = \int_0^x kx dx = \frac{1}{2}kx^2. \quad (4.8)$$

With the kinetic energy K we have the total energy of the system

$$E = U + K = \frac{1}{2}kx^2 + \frac{1}{2}mv^2. \quad (4.9)$$

Plugging (4.3) and (4.4) into it gives

$$\begin{aligned} E &= \frac{1}{2}kA_0^2 \cos^2(\omega t - \phi) + \frac{1}{2}mA_0^2\omega^2 \sin^2(\omega t - \phi) \\ &= \frac{1}{2}kA_0^2 \cos^2(\omega t - \phi) + \frac{1}{2}mA_0^2 \left(\frac{k}{m}\right) \sin^2(\omega t - \phi) \\ &= \frac{1}{2}kA_0^2(\cos^2(\omega t - \phi) + \sin^2(\omega t - \phi)) \\ &= \frac{1}{2}kA_0^2. \end{aligned} \quad (4.10)$$

Because both of k and A_0 are constant, the total energy is conserved.

The positive definition of potential energy U in (4.8) gives a concave upward parabola which looks like a well (Figure 4.1). The potential well diagram pictorially quite suit for our understanding of the stability of the motion with the vertex for the stable equilibrium point. The magnitude of the velocity at any position is easily obtained by (4.9) and (4.10)

$$|v| = \sqrt{\frac{k}{m}(A_0^2 - x^2)}. \quad (4.11)$$

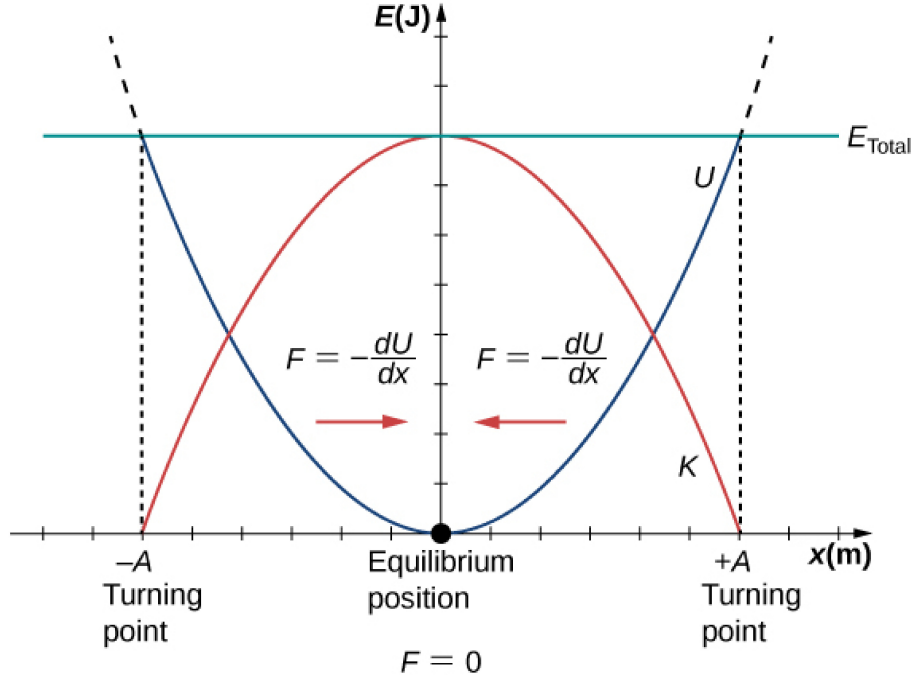


Figure 4.1: A graph of the kinetic energy K , potential energy U , and total energy E_{Total} of a simple harmonic oscillator.

4.1.3 Pendulums

Simple pendulum

A simple pendulum consists of a small mass attaching to a suspended light cord with negligible mass allowing to perform a small vertical angle to and fro motion (Figure 4.2).

The motion of a simple pendulum is the simple harmonic motion as the tangential acceleration due to gravity is proportional to the horizontal displacement at a small angle θ but in opposite direction. As shown in Figure 4.2 the torque of the system is

$$\begin{aligned}
 I \frac{d^2\theta}{dt^2} &= -\ell(mg \sin \theta) \\
 m\ell^2 \frac{d^2\theta}{dt^2} &= -\ell mg \sin \theta \\
 \frac{d^2\theta}{dt^2} &= -\frac{g}{\ell} \sin \theta \\
 \frac{d^2\theta}{dt^2} &\approx -\frac{g}{\ell} \theta
 \end{aligned}$$

Analogous of the angular frequency (4.6) of a vibrating spring, we identify

$$\omega = \sqrt{\frac{g}{\ell}} \quad (4.12)$$

and the period

$$T = 2\pi \sqrt{\frac{\ell}{g}}. \quad (4.13)$$

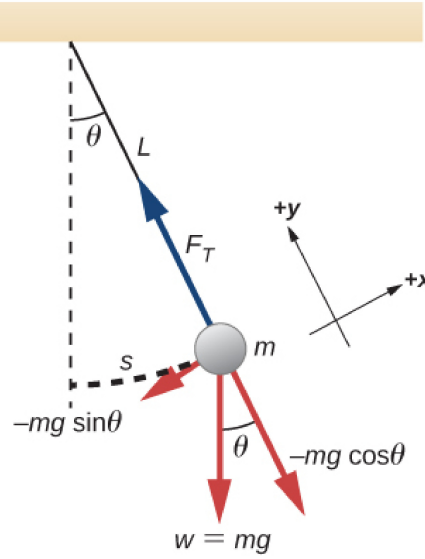


Figure 4.2: A simple pendulum allows the mass m doing a to and fro motion.

Physical pendulum

Physical pendulum is like simple pendulum able to do a simple harmonic motion under gravity but more realistic; as it is a rigid body that we often find it in a grandfather clock and some pendulum related devices.

For the angular acceleration equation using the same idea as in the previous simple pendulum does to find the torque equation except we are not able to write the moment of inertia I explicitly

$$I \frac{d^2\theta}{dt^2} \approx -\ell mg \theta \quad \Rightarrow \quad \frac{d^2\theta}{dt^2} = -\frac{mg\ell}{I} \theta.$$

Thus we have the angular frequency and the period

$$\omega = \sqrt{\frac{mg\ell}{I}}, \quad (4.14)$$

$$T = 2\pi \sqrt{\frac{I}{mg\ell}}. \quad (4.15)$$

Torsional pendulum

A torsional pendulum consists of a rigid body suspended by a light wire or spring as shown in Figure 4.3. When the body is twisted some small maximum angle Θ and released from rest, the body oscillates between $-\Theta \leq \theta \leq +\Theta$.

As the previous we use the torque equation for the angular acceleration:

$$I \frac{d^2\theta}{dt^2} = -\kappa \theta \quad \Rightarrow \quad \frac{d^2\theta}{dt^2} = -\frac{\kappa}{I} \theta$$

where κ is the torsion constant of the wire or string. Thus the angular frequency and the

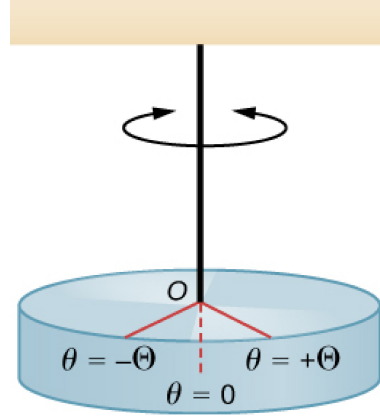


Figure 4.3: The rigid body oscillates between $-\Theta \leq \theta \leq +\Theta$.

period of a torsional pendulum are

$$\omega = \sqrt{\frac{\kappa}{I}}, \quad (4.16)$$

$$T = 2\pi \sqrt{\frac{I}{\kappa}}. \quad (4.17)$$

Although the name of torsional pendulum contains ‘pendulum’ its basic principle is closer to the simple harmonic oscillation of a one-dimensional vibrating string with an attaching mass. The set of equations of torsional pendulum does not contain gravity term g as the driving force (torque) coming from the twisted wire or string.

4.2 Wave motion

Any periodic motion of matter or energy can be represented by *wave*. Wave has the properties of spreading and repeating the motion with respect to time. Some common examples are vibrating string, rotating motor, pendulum, sound, electromagnetic wave, etc.

The propagation of wave relies on medium, except electromagnetic wave which does not.¹

Transverse wave A traveling wave that causes the particles of the disturbed medium to move perpendicular to the wave motion.

Longitudinal wave A traveling wave that causes the particles of the medium to move parallel to the direction of wave motion.

Example 4.1. The motion of water molecules on the surface of deep water in which a wave is propagating is a combination of transverse and longitudinal displacement, with the result that molecules at the surface move in nearly circular paths. Each molecule is displaced both horizontally and vertically from its equilibrium position (Figure 4.4).

Example 4.2. Why is a wave pulse traveling on a string regarded as a transverse wave?

¹EM wave in fact propagates on spacetime which in some sense might be regarded as medium as well. The exception actually is gravitational wave which is spacetime itself that the propagation is on nothing.

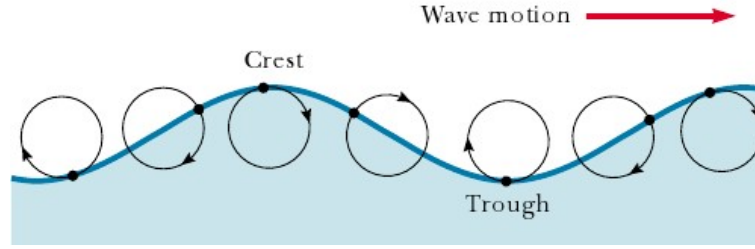


Figure 4.4: The motion of water molecules on the surface of deep water. (Credit: Serway & Jewett[5])

Solution. Particles (mark or label a position) of the string only move up and down vertically without any horizontal motion to the direction of propagation of the pulse. It is a transverse wave.

4.3 Sound wave

Sound is regarded as the manifestation of the disturbance of matter that carries energy from one place to other place without a transfer of mass. The propagation of sound wave relies on media. For example, air as the medium of sound wave in which the air molecules collide other adjacent molecules to propagate the mechanical energy away, due to the directional air pressure from the source. Sound wave is the **variation of air pressure**.

Sound is a mechanical wave produced by vibrating bodies. Energy is transferred, but not the air molecules themselves. A typical example of hearing is

mechanical vibrational source $\rightarrow$ medium (air) $\rightarrow$ receiver (ear)

Frequency ν is defined by the inverse of period, i.e. the number of oscillations per second,

$$\nu = \frac{1}{T}$$

where T is the period of the vibration. The unit of frequency is Hz (Hertz) as well as s^{-1} .

✎. In most cases, the magnitude square Ψ^2 of compression and rarefaction (air pressure) defines *intensity* I , such that

$$I \propto \Psi^2.$$

4.3.1 Speed of sound waves

The speed of **all mechanical waves** follows an expression of the general form

$$v = \sqrt{\frac{\text{elastic property}}{\text{inertial property}}}$$

where the inertial property of matter implies the density.

The speed of sound waves depends on the *compressibility* and the inertia of the medium. If a the medium has a *bulk modulus*² $\mathcal{B}$ and density ρ , the speed of sound

²Bulk modulus is defined by $\mathcal{B} = P/(V/V_0)$ where P is the pressure, V/V_0 is the deformation of volume known as the bulk strain.

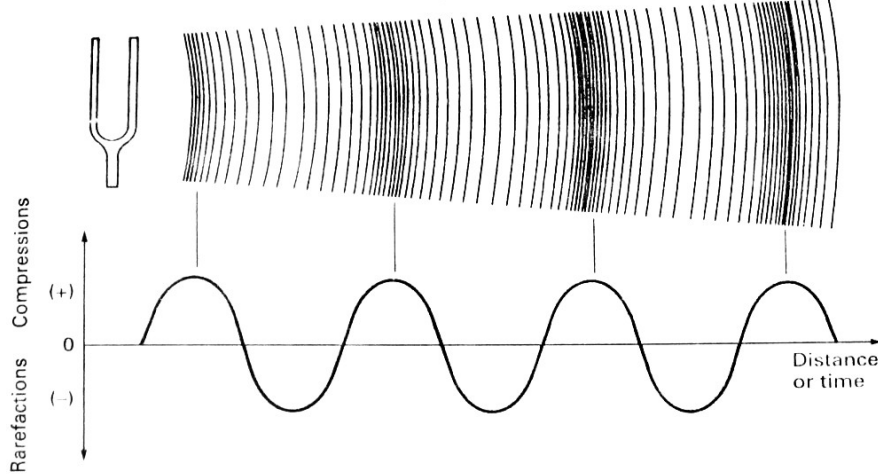


Figure 4.5: Define the compression and rarefaction of pressure as the upper and lower peaks of sinusoidal wave.

waves in the medium is

$$v = \sqrt{\frac{\mathcal{B}}{\rho}}. \quad (4.18)$$

On the other hand, the speed of transverse waves on a string is given by

$$v = \sqrt{\frac{T}{\mu}} \quad (4.19)$$

where T and μ are the tension (force) and the density of the string.

The speed of sound also depends on the temperature of the medium. For sound traveling through air, the relation between wave speed and medium temperature is

$$v = v_0 \sqrt{\frac{T}{T_0}} = v_0 \sqrt{1 + \frac{T_C}{273}} \quad (4.20)$$

where $v_0 = 331 \text{ ms}^{-1}$ is the speed of sound in air at 0°C and T_C is the temperature in degrees Celsius.

Example 4.3. The speed of sound in air at 20°C is approximated as

$$v = 331 \sqrt{1 + \frac{20}{273}} \approx 343 \text{ ms}^{-1}.$$

Example 4.4. Find the speed of sound in water, which has a bulk modulus of $2.1 \times 10^9 \text{ N/m}^2$ and a density of 10^3 kg/m^3 .

Solution. Using (4.18) we find that

$$v = \sqrt{\frac{\mathcal{B}}{\rho}} = \sqrt{\frac{2.1 \times 10^9}{10^3}} = 1.4 \text{ kms}^{-1}.$$

Sound waves travel in liquids slower than in solids because liquids are more compressible than solids.

4.4 Light wave

In classical physics, light is treated as wave. In fact, the frequencies of visible light are just in a narrow range of electromagnetic wave spectrum. Light is a transverse wave (Figure 4.6) but **not mechanical**.

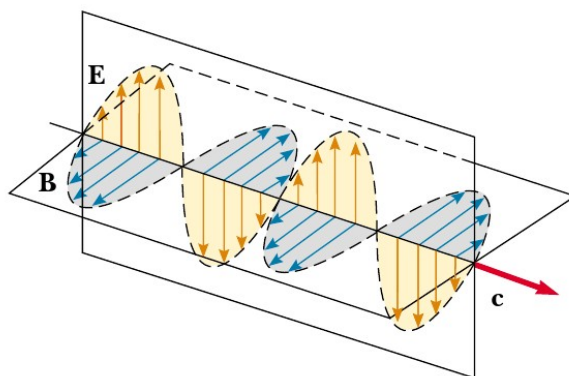


Figure 4.6: Light is transverse wave. (Credit: Serway & Jewett[5])

Light wave which is an electromagnetic radiation (wave) is often expressed in terms of the variation of electric (E) and magnetic (B) fields. In Figure 4.6, we see that both of the variations of E and B fields are perpendicular to the direction of propagation, and the fields themselves are also perpendicular to each other. The velocity of light is **always** equal to a constant value c , which is approximately $3 \times 10^8 \text{ ms}^{-1}$ in vacuum.

4.5 Wave motion

Sound and light are in totally different nature but they share many similar physical properties:

- reflection
- refraction
- interference
- diffraction
- diffusion (is the mixture of above properties)

4.5.1 Sinusoidal wave

Mathematically, a simple single tone sound wave may be described as a sine (or cosine) function. More complicated sound wave is the combination of the sinusoidal waves with the different *wavelengths* and the amplitudes.

Definition 4.1. A wavelength is the distance of the wave traveled in a cycle of vibration.

The relationship of wavelength λ , velocity of wave propagation v and the time taken for one complete cycle T is

$$\lambda = vT = \frac{v}{\nu} \quad (4.21)$$

where ν is the frequency of the wave. Speed of sound wave v depends on the media of the sound propagation. In air at 20 °C the speed is about $3.3 \times 10^2 \text{ ms}^{-1}$.

If a wave propagates to the right with a speed v , the wavefunction (i.e., amplitude function of the wave) $\Psi(x, t)$ at a later time t is (Figure 4.7)

$$\Psi(x, t) = \Psi_0 \sin \left(\frac{2\pi}{\lambda} (x - vt) \right). \quad (4.22)$$

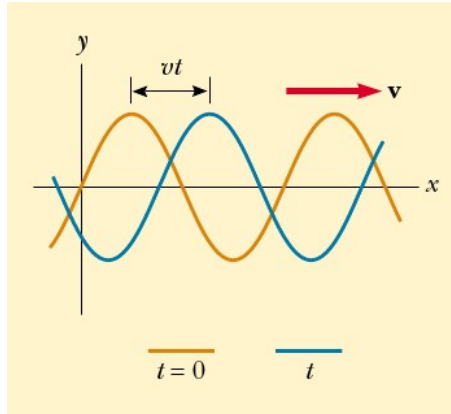


Figure 4.7: (Credit: Serway & Jewett[5])

Or it may be defined in terms of cosine function if one likes.

$$\Psi(x, t) = \Psi_0 \cos(kx - \omega t)$$

where

- Ψ_0 is a constant being the maximum amplitude.
- k is the *wave number* defined by $k = 2\pi/\lambda$.
- ω is the angular velocity (we often just call it frequency).

Choosing cosine instead of sine function is a matter of convenience. With using cosine and exponential functions, the wavefunction is written in the form

$$\Psi(x, t) = \frac{1}{2} \Psi_0 (e^{i(kx - \omega t)} + e^{-i(kx - \omega t)})$$

where $i = \sqrt{-1}$.

Example 4.5. The collision of air molecules due to sound wave induces variation of air pressure. The total pressure is

$$P = P_a + P_0 \sin 2\pi \nu t$$

where P_a is the ambient air pressure (at sea level at 0 °C is $1.01 \times 10^5 \text{ Pa} = 1.01 \times 10^6 \text{ dyn/cm}^2$), P_0 is the maximum pressure change due to the sound wave. The amount of energy transmitted per unit time in the normal direction of unit area is the intensity I given by

$$I = \frac{P_0^2}{2\rho v}$$

where ρ is the density of the medium.

4.5.2 Superposition

Fourier series

Fourier series is a mathematical method to write a wave as a summation of a series of the sinusoidal waves.

As an example, we can distinguish different sounds produced by different musical instruments with the same ‘base frequencies’ even though the details are different.

Complex musical waves compose of many *standing waves* with different frequencies by *superposition*, the lowest frequency is known as *fundamental*, the higher ones are *harmonics*. For a given note played by different instruments, they have the same fundamental frequency but the harmonic contents of the waves are different among instruments (See Figure 4.8).

Superposition of waves

As vectors, waves can be added up to form a resultant wave. The summation of waves is known as *superposition principle*.

Example 4.6. Adding two waves with the same amplitudes but in different wave numbers and frequencies give a new wave equation (Figure 4.9):

$$\begin{aligned}\Psi(x, t) &= \Psi_0[\cos(k_1x - \omega_1t) + \cos(k_2x - \omega_2t)] \\ &= 2\Psi_0 \cos(kx - \omega t) \cos(\Delta kx - \Delta\omega t)\end{aligned}$$

where

$$\begin{aligned}k &= \frac{1}{2}(k_1 + k_2), & \omega &= \frac{1}{2}(\omega_1 + \omega_2), \\ \Delta k &= \frac{1}{2}(k_1 - k_2), & \Delta\omega &= \frac{1}{2}(\omega_1 - \omega_2).\end{aligned}$$

The red color wave is described by the term $2\Psi_0 \cos(kx - \omega t)$ which may be regarded as the resultant wave with amplitude variation being modulated by the term $\cos(\Delta kx - \Delta\omega t)$ called *envelope*.

Wave packets

The superposition of waves forms an envelope of wave at some space and time. The pulse form of waves are called *wave packets* (Figure 4.10).

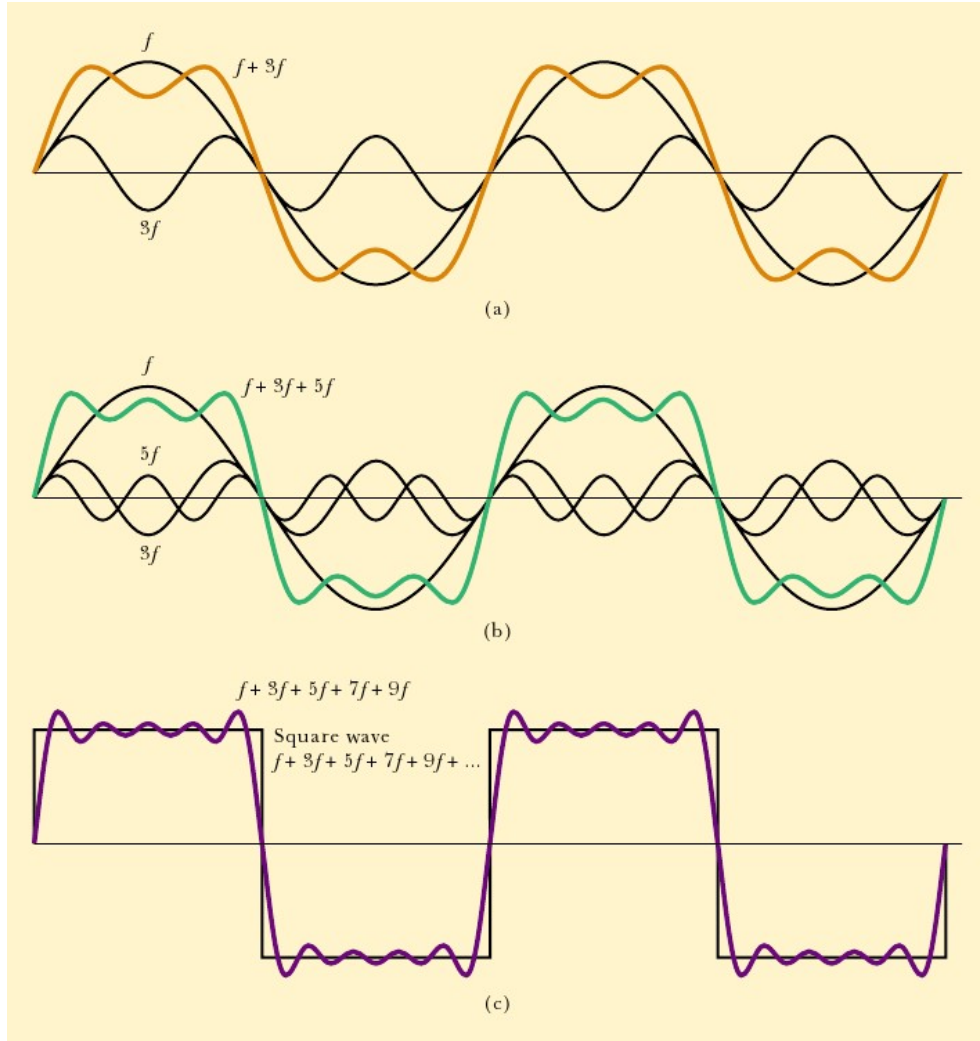


Figure 4.8: Fourier synthesis of a square wave, which is represented by the sum of odd multiples of the first harmonic, which has frequency ν . (a) Waves of frequency ν and 3ν are added. (b) One more odd harmonic of frequency 5ν is added. (c) The synthesis curve approaches the square wave when odd frequencies up to 9ν are added.

Example 4.7. By the superposition principle the addition of infinite waves in a range around k_0 is

$$\psi(x, t) = \int_{k_0 - \Delta k}^{k_0 + \Delta k} a \exp[i(kx - \omega t)] dk$$

where a is taken to be a constant amplitude for various k , and k_0 is the mean of the wave numbers. Without bothering the detail, we get

$$\psi(x, t) = a \exp[-i(\omega_0 t - k_0 x)] \frac{2 \sin[(\omega' t - x) \Delta k]}{\omega' t - x}.$$

Taking the real part of the equation, the graph looks like Figure 4.10.

The high intensity peak region of a wave packet could be regarded as a ‘particle’. If the wave are electromagnetic, this lump of energy packet could be interpreted as a photon.

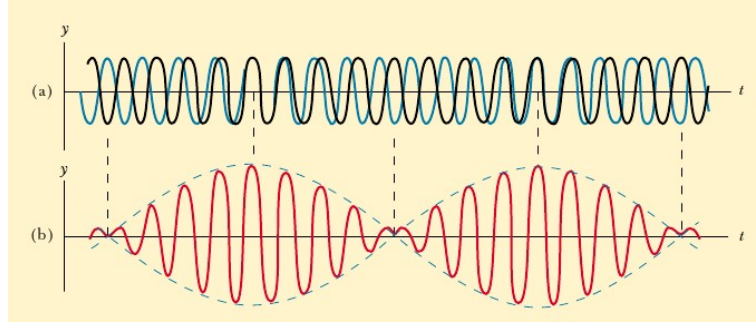


Figure 4.9: Superposition of two waves of the same amplitude. The envelope of the resultant wave is $\cos(\Delta kx - \Delta\omega t)$

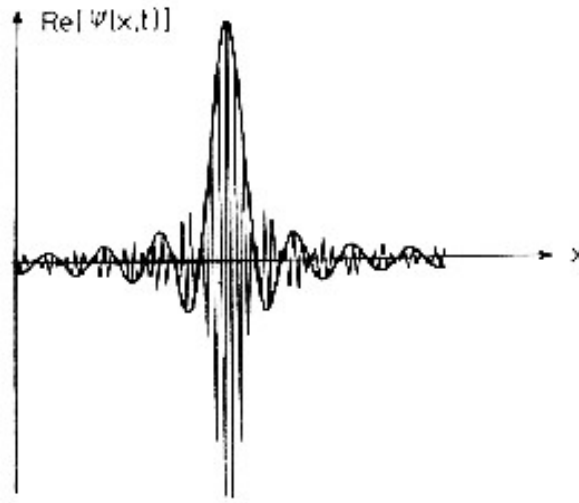


Figure 4.10: The rapid oscillations are described by $\cos(k_0x - \omega t)$ with t fixed. The envelope is given by $\sin[(\omega't - x)\Delta k]/(\omega't - x)$ with t fixed.

4.6 Wave phenomena

4.6.1 Reflection

When wave enters one medium from another, part of the wave is reflected at the interface, and part of it enters the medium. If the interface between the two media is smooth on the scale of the wavelength, the reflection is specular. If the surface has irregularities that are larger than the wavelength, the reflection is *diffuse*.

Figure 4.11 shows that the incident angle (against normal) is always equal to the reflected angle for a perfect smooth reflection.

4.6.2 Refraction

Similar to light, sound wave can be refracted through media. *Refraction* is a result of deflection of wave (e.g. light) caused by the boundary of two media while passing through them.

For light ray, the phenomenon is described by the *refractive index* which is defined by

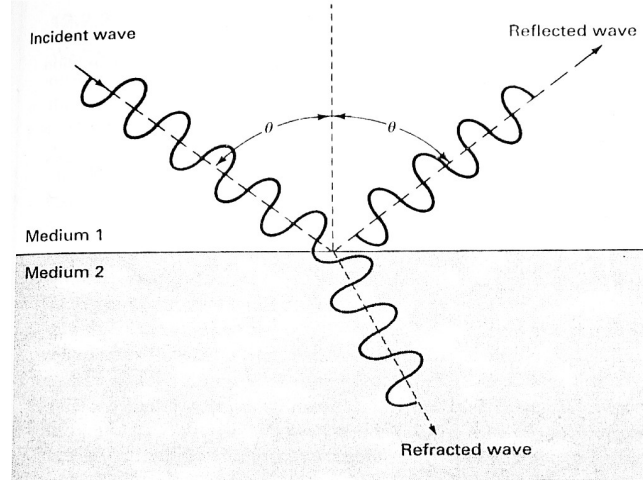


Figure 4.11: When wave is entering from a medium to a different medium, part of the wave is reflected, and the rest is refracted at the boundary of the media.

the speed of light travel through a medium relative to in vacuum. That is,

$$n = \frac{\text{Speed of light in vacuum } c}{\text{Speed of light in the medium } v}. \quad (4.23)$$

Note. The refracted angle (against normal) depends on the refractive indices of the two media.

For a sound wave, the ratio of the intensities of transmitted wave to incident wave is given by

$$\frac{I_t}{I_i} = \frac{4\rho_1 v_1 \rho_2 v_2}{(\rho_1 v_1 + \rho_2 v_2)^2}$$

where the subscripts 1 and 2 correspond to the media. If they represent air and water, only 1% of sound energy will enter into water in normal direction.

4.6.3 Interference

Wavefunction may be geometrically represented by vector. In quantum mechanics they are also called as wave vector or state vector. Addition of wave vectors follows the rule of addition of vectors to form a new wave vector. Mathematically,

$$|\Psi\rangle = |\psi_1\rangle + |\psi_2\rangle + |\psi_3\rangle + \dots$$

where $|\psi_1\rangle, |\psi_2\rangle, \dots$ are just some fancy way of writing vectors in quantum mechanics. This property is just superposition, as we have discussed the special case of adding two waves in the last section.

In Figure 4.12, when the amplitudes of the two waves projecting on some region of the screen happens being the same sign (positive, say), the amplitude of the resultant wave will be increased — *constructive interference*. If the wave amplitudes are one positive and the other negative, it is *destructive*. The distribution of wave amplitudes on the screen is called the *interference pattern*.

Phase (always relating to an angle) is the *path difference* between two overlapping waves.

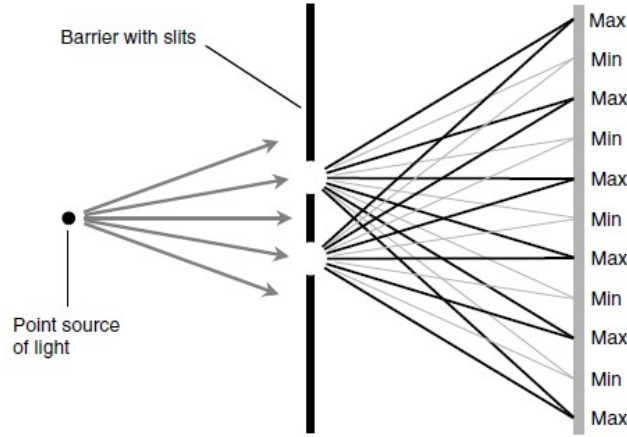


Figure 4.12: When the waves becoming from the slits are at the same phase, constructive interference occurs. On the contrary, opposite phases gives destructive interference.

Example 4.8. If two waves are traveling to the right and have the same frequency, wavelength, and amplitude but differ in phase, what is the resultant wave as a result of superposition? Interpret your result.

Solution. Let the wavefunctions of the waves be

$$\psi_1 = \psi_0 \sin(kx - \omega t) \quad \text{and} \quad \psi_2 = \psi_0 \sin(kx - \omega t + \phi),$$

respectively, where ϕ is the phase difference between the waves. The resultant wave is the summation of the waves by the superposition principle:

$$\begin{aligned} \psi &= \psi_1 + \psi_2 \\ &= \psi_0 [\sin(kx - \omega t) + \sin(kx - \omega t + \phi)] \\ &= 2\psi_0 \cos\left(\frac{\phi}{2}\right) \sin\left(kx - \omega t + \frac{\phi}{2}\right). \end{aligned}$$

The resultant wave function ψ is also sinusoidal having the same frequency and wavelength described by the sine function term. However, the amplitude is varying modulated by the term of cosine for different phase difference ϕ . The variation of the amplitude which is the envelope is also a wave.

When $\cos(\phi/2) = 1$, i.e. $\phi = 2n\pi$ where $n = 0, 1, 2, \dots$, the situation gives a constructive (in phase) interference, the resultant amplitude is doubled. Similarly, when $\phi = (2n + 1)\pi$ for $n = 0, 1, 2, \dots$, there is the destructive interference (out of phase). The example fully describes the Young's two-slit experiment in figure 4.12.

4.6.4 Standing wave

Consider that a stretched rope fixed by two ends (Figure 4.13) with only some fixed wave forms can be created. Such waves are called standing waves.

As an example, only some particular frequencies of sound wave are formed in a flute. These sort of frequencies are called *resonant frequencies*. The wavelength λ and the length L of a hollow pipe (flute) is related by

$$L = \frac{n\lambda}{2}.$$

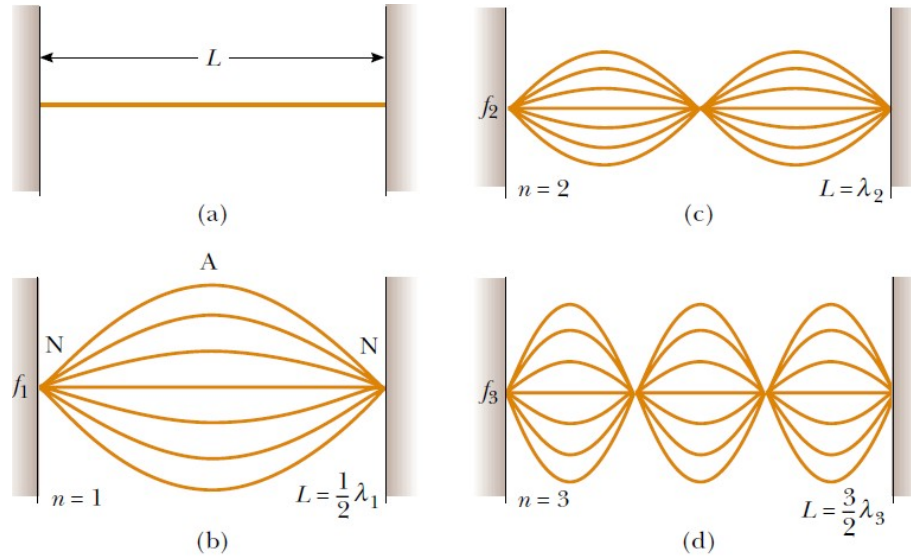


Figure 4.13: An example of standing wave: (a) A string of length L fixed at both ends. The normal modes of vibration form a harmonic series; (b) the fundamental, or first harmonic; (c) the second harmonic; (d) the third harmonic.

Resonances happen at $n = 1, 2, 3, \dots$

Figure 4.14 is the motion of air molecules in standing longitudinal waves in a pipe, along with schematic representations of the waves. The graphs with the letter A represent the largest displacement amplitudes, not the pressure amplitudes. N stands for the node where the amplitude is zero.

Example 4.9. Find the resultant wave of being created by the superposition of two transverse sinusoidal waves having the same amplitude, frequency, and wavelength but traveling in opposite directions in the same medium.

Solution.

$$\psi_1 = \psi_0 \sin(kx - \omega t), \quad \psi_2 = \psi_0 \sin(kx + \omega t)$$

where ψ_1 represents a wave traveling to the right and ψ_2 represents one traveling to the left, with the same amplitude ψ_0 . Adding these two functions according to superposition principle gives the resultant wave function ψ :

$$\begin{aligned} \psi &= \psi_1 + \psi_2 \\ &= \psi_0 \sin(kx - \omega t) + \psi_0 \sin(kx + \omega t) \\ &= (2\psi_0 \sin kx) \cos \omega t \end{aligned}$$

which is the wave function for a standing wave as there is no wave propagation that the wave varying with respect to time with changing the amplitude but no position translation.

4.6.5 Diffraction

Wave has a tendency of spreading when it propagates through a medium. This phenomenon is known as *diffraction*.

The degree of spreading depends on the relation of the wavelength and the dimension of the wave entering into the medium. A pattern of distribution similar to the interference

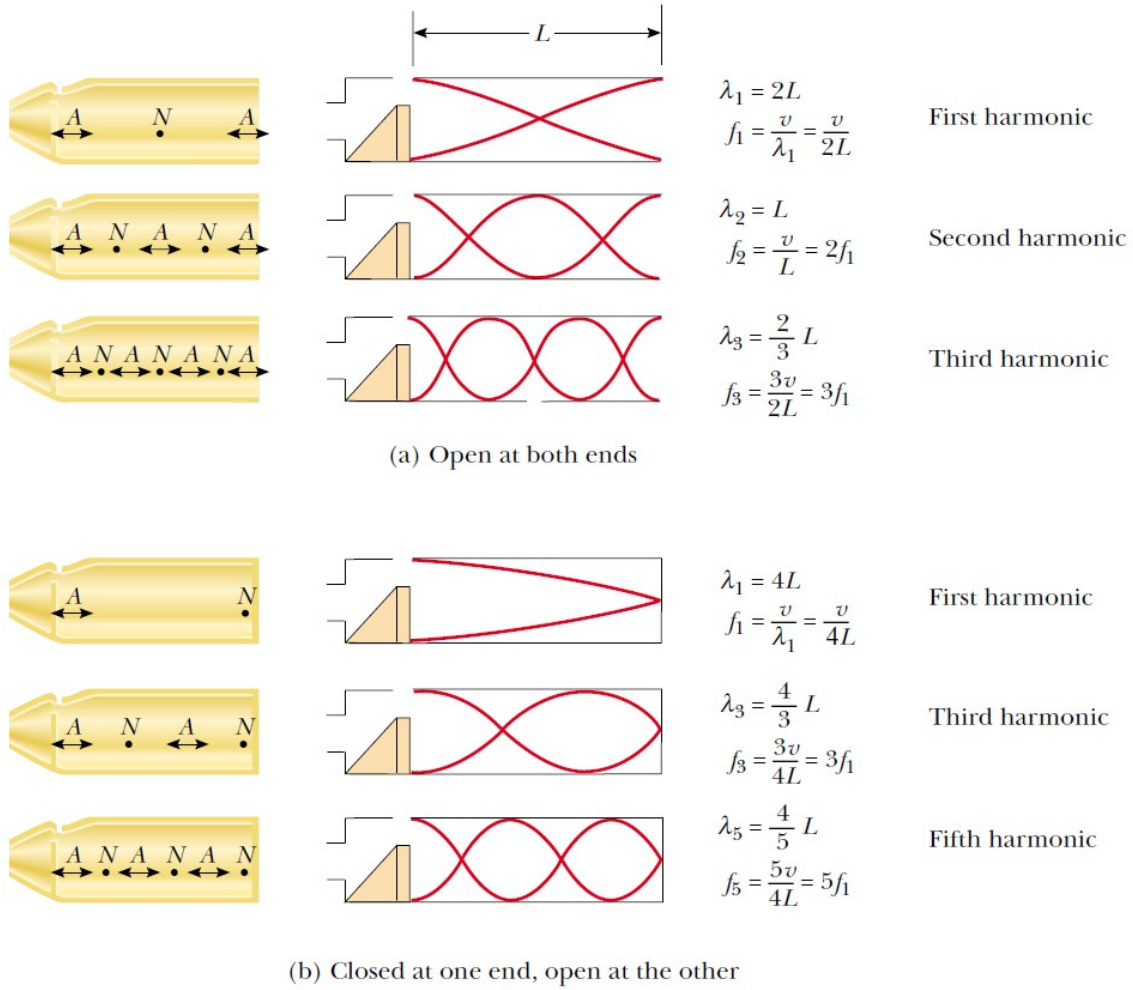


Figure 4.14: (a) In a pipe open at both ends, the harmonic series created consists of all integer multiples of the fundamental frequency: $f_1, 2f_1, 3f_1, \dots$ (b) In a pipe closed at one end and open at the other, the harmonic series created consists of only odd-integer multiples of the fundamental frequency: $f_1, 3f_1, 5f_1, \dots$

pattern of two-slit experiment will occur, the constructive and destructive phenomena arise when

$$d \sin \theta = \frac{1}{2}(n+1)\lambda \quad (4.24)$$

where d is the dimension of the hole, and $n = 0, 2, 4, 6, \dots$ for destructive, $n = 1, 3, 5, \dots$ for constructive. Equation (4.15) also applies for the two-slit experiment in Figure 4.12 as well, by replacing the d as the distance between the slits instead.

The wavelength of light is too small for the spreading to be observed in our daily experience. So we always see light rays as straight lines by any edges. Sound waves, on the other hand, are easy to be diffracted since the wavelengths are often larger than many openings such as doors and spacing between buildings in our daily experiences. That is quite common why we always hear the sound but do not see the source.

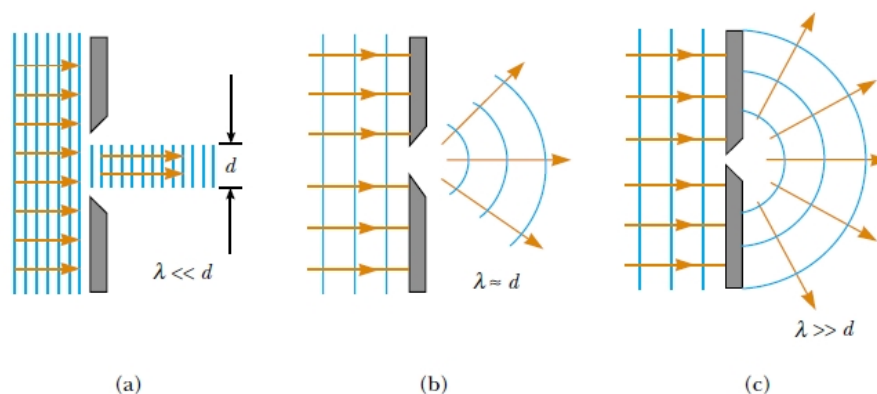


Figure 4.15: (a) The dimension of the opening is greater than the wavelength of the incident wave a lot $d > \lambda$, no diffraction occurs. (b) When $d \approx \lambda$, diffraction occurs. (c) More spherical wave emerges when $d < \lambda$.

4.7 Hearing and the ear

Hearing is produced by the response of the nerves in the ear to pressure variations in the sound wave.

Human ear is divided into three sections:

Outer ear Pinna, ear canal, tympanic membrane (eardrum)

Middle ear Ossicles – composed of a linkage of three bones: hammer, anvil, stirrup.

Inner ear Cochlea – a snail shell shape spiral cavity. There are three parallel ducts inside: vestibular canal, cochlear duct, tympanic canal.

4.7.1 Sound amplification of ear

The mechanical construction of the ear can amplify the sound pressure mainly in the middle ear:

1. The area of the eardrum is about 30 times larger than the oval window.
2. The ossicles act as a lever with a mechanical advantage of about 2.
3. In around 3,000 Hz, an increase in the pressure at the eardrum due to the resonance of the ear canal, the pressure is increased by 2.

Therefore, at around 3,000 Hz, the intensity at the oval window is amplified by a factor of about

$$(2 \times 30 \times 2)^2 = 120^2 = 14400$$

4.7.2 Frequency and pitch

The human ear is capable of detecting sound at frequencies between about 20 and 20,000 Hz, and most sensible between 200 and 4,000 Hz due to close to the resonant frequency of ear.

The sensation of *pitch* is related to the frequency. Pitch is a perceptual property that allows the ordering of sounds on a frequency related scale. Pitch increases as frequency increase, but not really directly proportional to each other.

Intensity and loudness

The range of intensities at 3,000 Hz is able to be responded around 10^{-12} W/m² to 1 W/m². These two extremes are called the *threshold of hearing* and the *threshold of pain*.

The ear responds sound intensity being logarithmic than linear. The unit of measuring the logarithmic intensity is decibel (dB) and is defined by

$$\text{Logarithmic intensity} = 10 \log \frac{\text{sound intensity in W/m}^2}{10^{-12} \text{ W/m}^2}. \quad (4.25)$$

Note. From the unit of intensity of sound wave we see that it is the power per unit area. Recall that the previous section when we discussed the flux of diffusion it is the number of particles passing through a region per unit area per unit time. Similarly, heat flow shares the same meaning as it is the energy flow per unit area per unit time. All of them are intensities in different aspects.

Example 4.10. If the intensity from a source is 4×10^{-3} W/m² at 12 m, what will the intensity be at 4 m?

Solution. Since intensity (strength) propagating from a point source obeys inverse square law, at 4 m distance the intensity is

$$4 \times 10^{-3} \times \frac{12^2}{4^2} = 3.6 \times 10^{-2} \text{ W/m}^2$$

Example 4.11. The intensity I of a sound wave is given by

$$I = \frac{\Delta P_{\max}^2}{2\rho v}$$

where $\Delta P_{\max}$ is the variation of the maximum pressure amplitude, $v = 343$ m/s the speed of sound waves in air, and $\rho = 1.2$ kg/m³ the density of air. Please determine the pressure amplitude associated with the limits of threshold of hearing and threshold of pain.

Solution. The intensity of threshold of hearing and threshold of pain are 10^{-12} W/m² and 1 W/m², respectively. We directly use the given formula to find the variations of air pressures. For the threshold of hearing,

$$\Delta P_{\max} = \sqrt{2 \times 1.2 \times 343 \times 10^{-12}} = 2.870 \times 10^{-4} \text{ Pa}.$$

The variation of air pressure for the threshold of pain is

$$\Delta P_{\max} = \sqrt{2 \times 1.2 \times 343 \times 1} = 28.70 \text{ Pa}.$$

It seems that our ears can detect a very small amount of variation of air pressure.

Ultrasonic

A sound wave with the frequency higher than the audible frequency of human is known as *ultrasonic*.

Since the degree of spreading of a wave will decrease if the wavelength becomes shorter. Ultrasound imaging is a special technique analogous to X-ray. The ultrasonic is capable to probe the interior of a body to form a visible image. Ultrasonic examinations are safer than X-rays and often can provide as much information.

4.8 Doppler effect

We can hear changing pitches (frequencies) when a sound source is moving away or approaching to us. The pitch becomes higher when the source is approaching, and the pitch will be lower if the source is going away. This phenomenon is called *Doppler effect*.

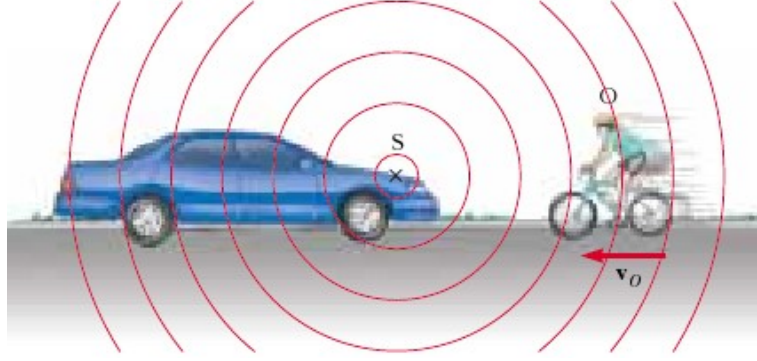


Figure 4.16: An observer O (the cyclist) moves with a speed v_o toward a stationary point source S, the horn of a parked car. The observer hears a frequency ν' that is greater than the source frequency.

The frequency of the sound ν' detected by the observer is given by

$$\nu' = \nu \frac{v}{v \pm v_s}$$

where ν is the frequency when the source is at rest, v_s is the speed of the source, and the “+” represents the source receding, “−” means approaching.

For the situation in Figure 4.16 — a stationary sound source and moving observer, the frequency of the sound ν' detected by the observer is

$$\nu' = \frac{v \pm v_o}{v} \nu.$$

‘+’ stands for the approaching observer as shown in Figure 4.16. Use ‘−’ if the observer is moving away.

Proof. For the moving sound source with stationary observer case, the frequency of the moving source observed at the observer’s frame is $\nu' = v/\lambda'$. Since the sound source is approaching, the wavelength appears to be shorter than that at the moving frame:

$$\lambda' = \lambda - s = \lambda - v_s T = \lambda - v_s \frac{1}{\nu}$$

where s is the displacement of the advanced wavefront at a time (period) T . Thus

$$\begin{aligned} \lambda' &= \frac{\lambda \nu - v_s}{\nu} = \frac{v - v_s}{\nu} \\ \nu' &= \nu \frac{v}{v - v_s}. \end{aligned}$$

The proof of the situation of leaving source is similar by replacing “−” with “+” instead. $\square$

In general, the frequency heard by an observer whenever there is relative motion between a source of sound waves and the observer is given by

$$\nu' = \left(\frac{v \pm v_o}{v \pm v_s} \right) \nu \quad (4.26)$$

where $v \pm v_o$ is the relative velocity of speed of sound and the observer. $(v \pm v_s)T$ is the changed wavelength due to the doppler shift where T is the period.

4.9 Shock waves

Figure 4.17 shows the situation when the speed v_S of a source **exceeds** the wave speed v . The circles represent spherical wave fronts emitted by the source at various times during its motion. At $t = 0$, the source is at S_0 , and at a later time t , the source is at S_n . In the time t , the wave front centered at S_0 reaches a radius of vt . In this same amount of time, the source travels a distance $v_S t$ to S_n . At the instant the source is at S_n , waves are just beginning to be generated at this location, and hence the wave front has zero radius at this point.

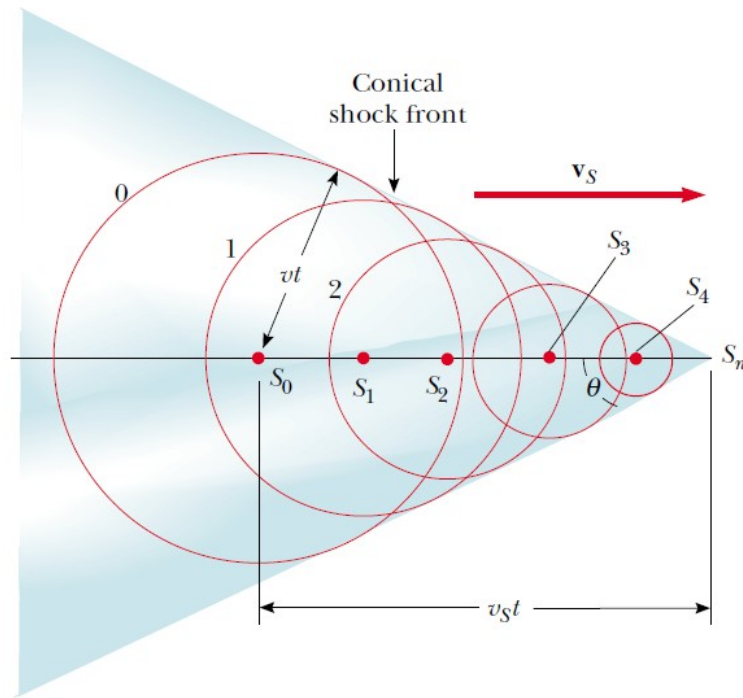


Figure 4.17: A shock wave produced when a source moves from S_0 to S_n with a speed v_S , which is greater than the wave speed v in the medium.

The tangent line drawn from S_n to the wave front centered on S_0 is tangent to all other wave fronts generated at intermediate times. Thus, we see that the envelope of these wave fronts is a cone whose apex half-angle θ is given by the *Mach number*

$$\frac{1}{\sin \theta} = \frac{v_S t}{vt} = \frac{v_S}{v}.$$

The conical wave front produced when $v_S > v$ (supersonic speeds) is known as a *shock wave*.



Figure 4.18: The V-shaped bow wave of a boat is formed because the boat speed is greater than the speed of the water waves.

Chapter 5

Optics

5.1 Spectrum of electromagnetic wave

Visible light is the electromagnetic (EM) radiation (wave) with wavelengths around 400 to 700nm (Figure 5.1).

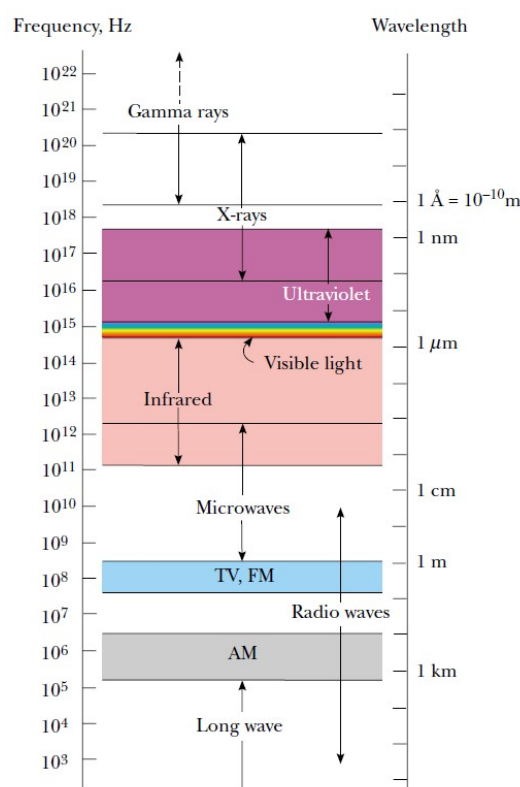


Figure 5.1: Spectrum of electromagnetic wave

Most of the electromagnetic radiation from the Sun that reaches the Earth's surface is in the region of visible light and ultraviolet. Plants use light to convert carbon dioxide and water into organic materials by *photosynthesis*. Animals have evolved light-sensitive organs which are their main source of information about the surroundings. About 70% of a person's sensory input is obtained through the eyes. On the contrary, some bacteria and insects can produce light through chemical reactions.

Photon

Matter possesses duality properties—wave and particle. EM radiation is considered as the propagation of a bunch of packets (quanta) of energy known as *photons*. Unlike other regular particles, photon has **no mass** and is assumed having no volume neither. The speed of propagation of a photon is always at $c = 3 \times 10^8 \text{ ms}^{-1}$ in vacuum.

Given a frequency ν , each photon has a **fixed** amount of energy E ,

$$E = h\nu = \hbar\omega \quad (5.1)$$

where

$$\hbar = \frac{h}{2\pi}, \quad \omega = 2\pi\nu$$

Both h and ‘h-bar’ $\hbar$ are commonly called the *Planck’s constant*, which has a value of $h = 6.63 \times 10^{-34} \text{ Js}$.

Note. Intensity (or strength) is defined as the square of amplitude in wave mechanics. Photon—the quantum version of light, the corresponding intensity is proportional to the number of photons.

- Intensity depends on the **number** of photons.
- Energy of a photon depends on the **frequency** of it (the wave packet, the envelope).

We may say the intensity of a beam of photons (e.g. Laser) but it is conceptually incorrect to talk about the intensity of a single photon.

5.2 Spherical mirrors

A spherical mirror has the shape of a section of a sphere. Two types of spherical mirror; concave and convex mirrors, depend on which sides are silvered.

5.2.1 Concave mirror

A *concave mirror* allows light to be reflected from the inner, concave surface as shown in Figure 5.2.

Light rays diverging from the point of radius of curvature are not always reflected to the same *focal point* if the rays diverge from the object are not at small angles (Figure 5.3). This kind of distortion is known as *spherical aberration*. The phenomenon also exists in lens system.

Note. Light rays which diverging from the object make a small angle with the *principal axis* are called *paraxial rays*.

Magnification

Image is formed by a spherical concave mirror if an object is placed at a suitable position in front of a mirror along the principle axis from the vertex. The magnification is given by

$$M = \frac{\text{dimension of image}}{\text{dimension of object}}.$$

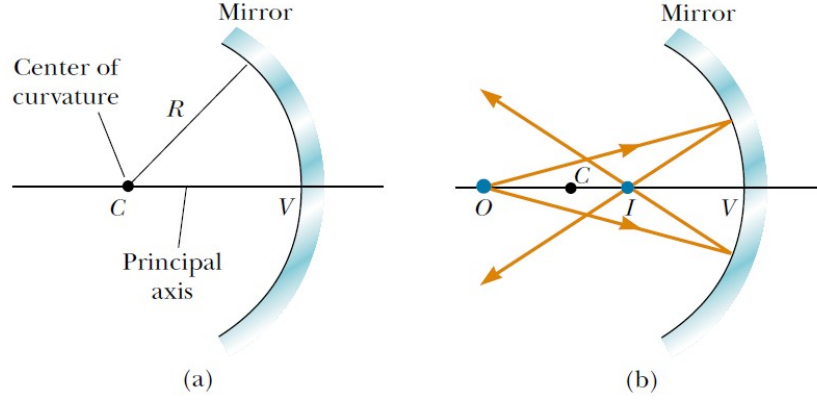


Figure 5.2: (a) A concave mirror of radius R . The centre of curvature C is located on the principal axis. (b) A point object placed at O in front of a concave spherical mirror of radius R , where O is any point on the principal axis farther than R from the mirror surface, forms a real image at I . If the rays diverge from O at small angles, they all reflect through the same image point.

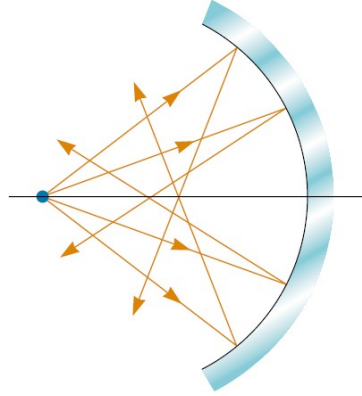


Figure 5.3: Spherical aberration of concave mirror

In the case of concave or convex mirrors, using the similar triangles method the magnification may be written as

$$M := -\frac{\text{image distance from the mirror}}{\text{object distance from the mirror}} = -\frac{q}{p} \quad (5.2)$$

where $p, q > 0$ (front side of the mirror) as shown in Figure 5.4. The negative sign appears in the definition as the triangles turning up side down against to each other. It follows negative value of M for an inverted image. In practice, a better ray diagram for locating the image is shown in Figure 5.7 in which the measurement of the angle of reflection is avoided.

Mirror equations

For paraxial rays, the *mirror equation* relates the object distance p , image distance q and the radius of curvature R :

$$\frac{1}{p} + \frac{1}{q} = \frac{2}{R}$$

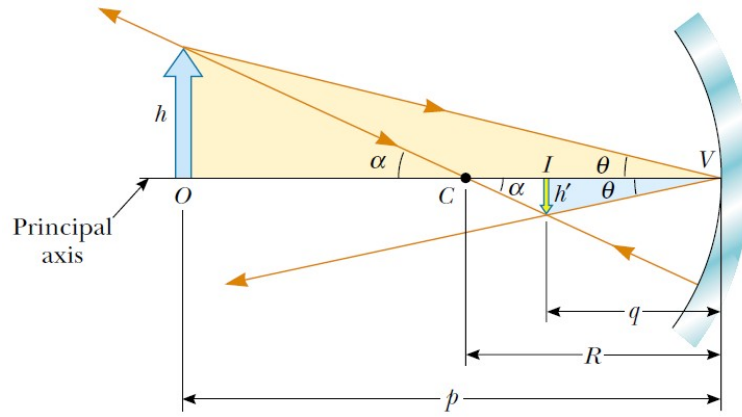


Figure 5.4: The image formed by a spherical concave mirror when the object O lies outside the centre of curvature C .

Define the *focal point* which is at the mid way of the radius of curvature and the mirror

$$f = \frac{R}{2}$$

where f is the *focal length*. Therefore,

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}. \quad (5.3)$$

All the parallel incident light rays (or light rays from infinity) are reflected to the focal point of the concave mirror as shown in Figure 5.5.

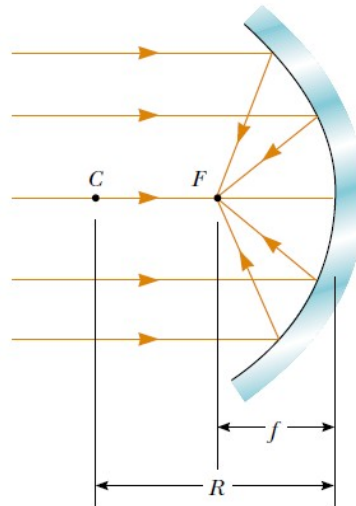


Figure 5.5: Light rays from a distant object reflect from a concave mirror through the focal point F .

5.2.2 Convex mirror

Figure 5.6 shows the formation of an image by a *convex mirror* also known as a *diverging mirror*; the rays from any point on an object diverge after reflection as if they were coming

from a point source behind the mirror. The image in Figure 5.6 is virtual because the reflected rays only appear to originate at the image point, as shown by the dashed lines. Furthermore, the image is always upright and smaller than the object.

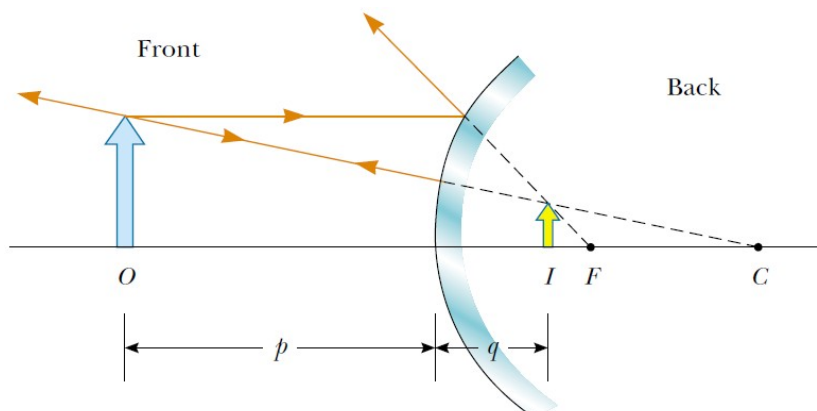


Figure 5.6: The image formed by the convex mirror is virtual and upright.

We may use the same set of equations for a convex mirror. The thing that we have to notice is the sign conventions.

Sign conventions for mirrors

- p is positive if object is in front of mirror (real object).
- p is negative if object is in back of mirror (virtual object).
- q is positive if image is in front of mirror (real image).
- q is negative if image is in back of mirror (virtual image).
- Both f and R are positive in front of mirror (concave).
- Both f and R are negative in back of mirror (convex).
- If M is positive, image is upright.
- If M is negative, image is inverted.

5.2.3 Ray diagrams for mirrors

The positions and sizes of images formed by mirrors can be conveniently determined with *ray diagrams*. As an example, consider Figure 5.7:

- Ray 1 is drawn from the top of the object parallel to the principal axis and is reflected through the focal point F .
- Ray 2 is drawn from the top of the object through the focal point and is reflected parallel to the principal axis.
- Ray 3 is drawn from the top of the object through the centre of curvature C and is reflected back on itself.

Since two rays are needed to position the image location, in practice it is convenient only to draw the ray 1 and 2.

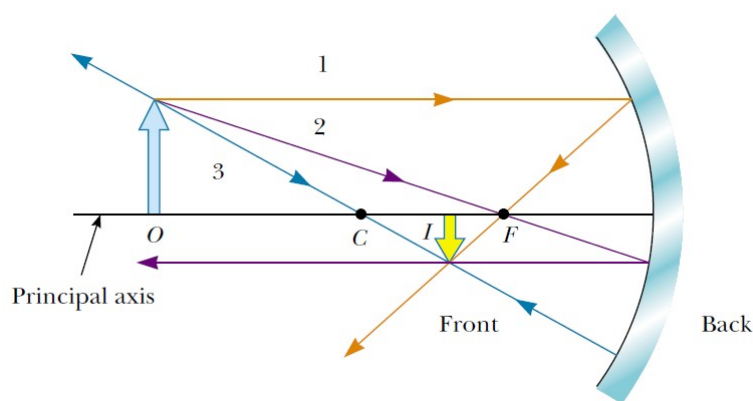


Figure 5.7: The object is located so that the centre of curvature lies between the object and a concave mirror surface, the image is real, inverted, and reduced in size.

5.3 Lens

Similar to mirrors, there are concave and convex lenses. But unlike a mirror reflects light rays which through lens are due to **refraction**.

5.3.1 Depth of field

A sharp image is always obtained with a small *aperture* size (Figure 5.8). The reasons are:

- The imperfection of lens (or the cornea of an eye); small aperture allows incoming light passing through around the centres of them.
- A sharp image always forms only on a 2-dimensional (plane) surface, small aperture decreases the formation of images in front of and behind the retina.

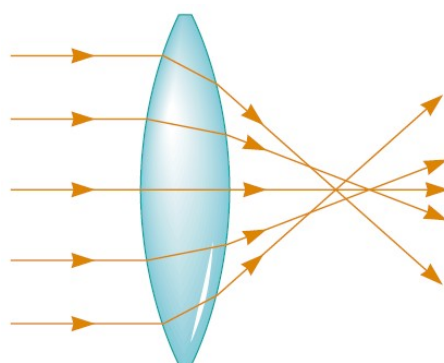


Figure 5.8: Spherical aberration – Light rays refracted by the outer region away from the centre of the lens always form an image closer to the lens.

The range of object distances over which a good image is formed for a given setting of the focus is called the *depth of field*. It is inversely proportional to the diameter of the aperture.

5.3.2 Lens makers' equation

The refraction of a lens depends on the refractive indices of the media. Taking the refractive index of air is 1 for convenient, the material of lens supposed being glass is n . We have the *lens makers' equation* for a **thin lens**

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (5.4)$$

where R_1, R_2 are the radii of the lens (Figure 5.9), f is the focal length when the object distance is assumed to be infinity.

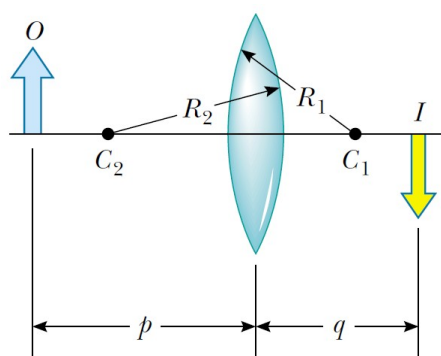


Figure 5.9:

The relationship among the focal length, image and object distances for a thin lens is given by the lens equation same as for a mirror

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}. \quad (5.5)$$

5.3.3 Ray diagrams for thin lenses

Three different cases of ray diagrams for convex and concave lenses are shown in Figures 5.10, 5.11 and 5.12.

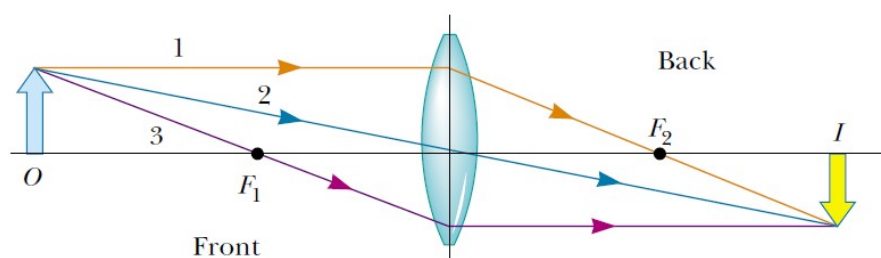


Figure 5.10: The object is in front of and outside the object focal point F_1 of a converging lens, the image is real, inverted, and on the back side of the lens.

Sign conventions for thin lenses

- p is positive if object is in front of lens (real object).

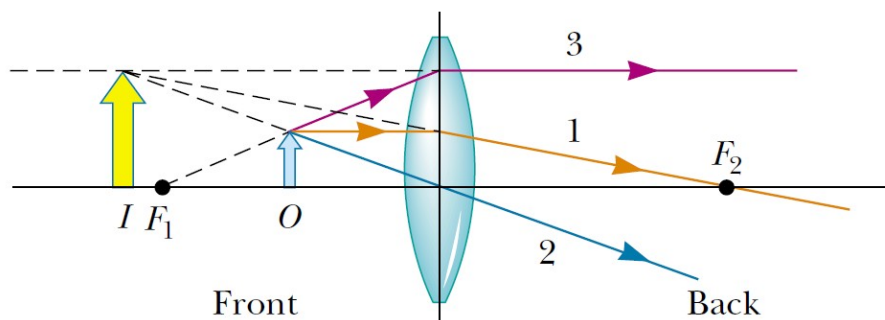


Figure 5.11: The object is between F_1 and a converging lens, the image is virtual, upright, larger than the object, and on the front side of the lens.

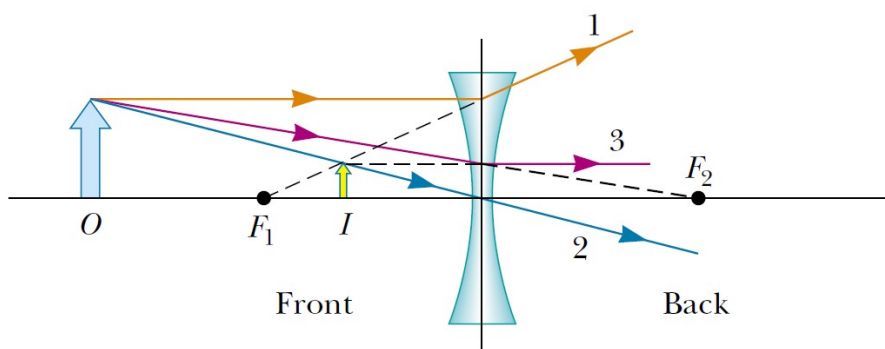


Figure 5.12: An object is anywhere in front of a diverging lens, the image is virtual, upright, smaller than the object, and on the front side of the lens.

- p is negative if object is in back of lens (virtual object).
- q is positive if image is in back of lens (real image).
- q is negative if image is in front of lens (virtual image).
- R_1 and R_2 are positive if centre of curvature is in back of lens.
- R_1 and R_2 are negative if centre of curvature is in front of lens.
- f is positive if the lens is converging.
- f is negative if the lens is diverging.

5.3.4 Combination of thin lenses

If two thin lenses are used to form an image, the system can be treated in the following manner. First, the image formed by the first lens is located as if the second lens were not present. Then a ray diagram is drawn for the second lens, with the image formed by the first lens now serving as the object for the second lens. The second image formed is the final image of the system. The overall magnification of a system of thin lenses equals the product of the magnifications of the separate lenses.

Two thin lenses in contact

Consider an example of a system of two lenses in contact with focal lengths f_1 and f_2 . If p is the object distance for the combination, application of the thin-lens equation to the first lens gives

$$\frac{1}{p} + \frac{1}{q_1} = \frac{1}{f_1}$$

where q_1 is the image distance for the first lens. Pretending this image as the object for the second lens, the object distance for the second lens must be $-q_1$. Thus, for the second lens,

$$\frac{1}{-q_1} + \frac{1}{q} = \frac{1}{f_2}$$

where q is the final image distance from the second lens. Adding these equations eliminates q_1 and gives

$$\begin{aligned} \frac{1}{p} + \frac{1}{q} &= \frac{1}{f_1} + \frac{1}{f_2} \\ \frac{1}{f} &= \frac{1}{f_1} + \frac{1}{f_2} \end{aligned}$$

where f is the combined focal length of the system.

5.4 Human eye

Vision is our most important source of information about the external world. The components of vision are

Stimulus Visible light source

Optical components of the eye Imaging the light

- Cornea, iris and lens

Nervous system Processing and interpreting the visual images

- Retina – photoreceptor cells: *cones* and *rods*

Human eye is roughly a sphere, approximately 2.4 cm in diameter.

Cornea A transparent section in the outer cover of the eyeball. It is the **main focusing device** but the focal length is fixed.

Iris The function is similar to the aperture of a camera to control the size of the pupil ranging from 2 to 8 mm, depending on the intensity of light.

Lens Its curvature is alterable by the *ciliary muscle* to change the focal length.

Cavities of the eye Between the lens and the cornea is filled with a watery fluid called the *aqueous humor*. The space between the lens and the retina is filled with the gelatinous *vitreous humor*.

Retina Light is focused by the lens system into an inverted image at the photosensitive retina, produces nerve impulses to convey information to the brain.

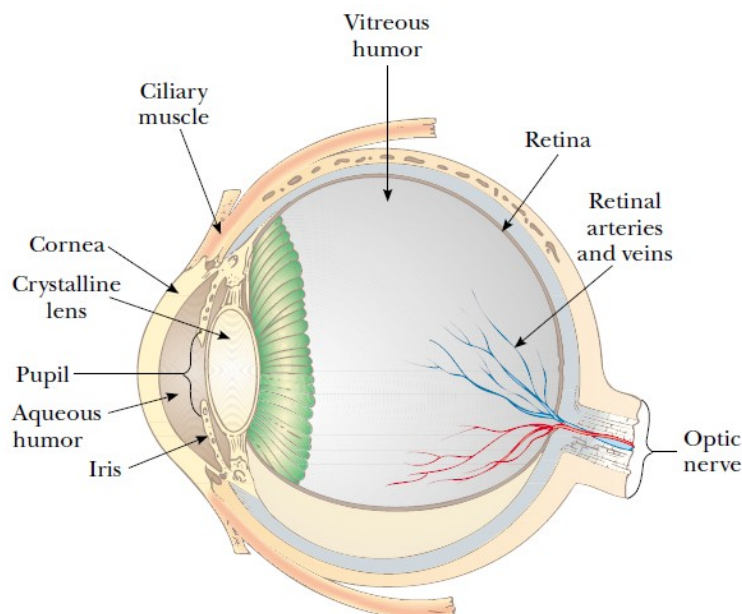


Figure 5.13: Human eye

5.4.1 Accommodation

Accommodation is a process to describe the focusing mechanism of the eye; the ciliary muscle controls the thickness and curvature of the lens to alter the focal length.

Relaxed ciliary muscle The crystalline lens is fairly **flat** for distant viewing condition, the incoming light ray can be considered parallel.

Contracted ciliary muscle The lens becomes more **spherical** to make closer object to form image on the retina. The minimum distance of sharp focus is called the *near point of the eye* as shown in Figure 5.15.

Note. The near point of the eye will be changed by ageing. At age 10, it is about 7 cm. At age 60, it is about 100 cm. This decrease in the accommodation of the eye with age is called *presbyopia*.

5.4.2 Lens system of the eye

The focusing of the light into a real inverted image at the retina is produced by refraction (see section 4.6.2) at the cornea and at the crystalline lens.

The largest part of the focusing, about two thirds, actually occurs at the cornea but the focal length is fixed. The power of the crystalline lens is small because its refractive index is only slightly greater than the surrounding fluid.

Note. The refractive power of the cornea is greatly reduced when it is in contact with water. That is why we can not see clearly without wearing suitable gears under water.

Reduced eye

Reduced eye is a model to simplify the calculation of the complicated refractions at the four surfaces of cornea and lens.

In this model, all the refraction is assumed to occur at the front surface of the cornea, and the rest parts are treated as a whole with refractive index 1.333. This model represents most closely the relaxed eye which focuses parallel light at the retina.

Nodal point

To simplify the calculation and diagram, we usually write two similar triangles where the basis lines of them represent the object and image height. The distance between the surface of cornea to the nodal point of human eye is 0.5 cm, and the distance from the nodal point to retina is 1.5 cm. The common intersection point or the vertex is called *nodal point* or *anterior focal point*.

As a pair of similar triangles, we have

$$\frac{\text{Object size}}{\text{Image size}} = \frac{\text{Distance of object from nodal point}}{\text{Distance of image from nodal point}}.$$

5.4.3 Retina

Retina consists of photoreceptor cells (cones and rods) in contact with a complex network of neurons and nerve fibers which are connected to the brain via the optic nerve. Light absorbed by the photoreceptors produces nerve impulses that travel along the neural network and then through the optic nerve into the brain.

The photoreceptors are located **behind** the neural network, so the light must pass through this cell layer before it reaches the photoreceptors.

The characteristics of the photoreceptor cells:

Cones For sharp colour vision in daylight

Rods For vision in dim light

5.5 Lens and glasses

Focal length is defined by the distance of the lens to the position of image for infinite far object, i.e. parallel incident light ray. A lens which has two sides is equivalent to two thin lenses in contact as stated previously. Thus, the focal length f of the combination is

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \quad (5.6)$$

where f_1, f_2 are their focal lengths.

For a thin lens, the distance from the object p to the lens and the image distances q from the lens is given by

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}. \quad (5.7)$$

By convention q here is taken as positive if the image is formed on the side of the lens opposite to the object and negative if the image is formed on the object side.

There are three common defects in vision associated with the focusing system of the eye:

Myopia (nearsightedness) Caused by an elongated eyeball or an excessive curvature of the cornea, or the incapability of lens due to the ciliary muscle

Hyperopia (farsightedness) The opposite case of myopia, imaged focused behind the retina due to not enough of curvature of the cornea or lens, hyperopia is similar to *presbyopia*, the latter is due to aging.

Astigmatism Caused by a nonspherical cornea

5.5.1 Lens for myopia

Far point of the eye is a maximum distance that a myopic eye can see clearly. Light from object farther away than the far point is focused in front of the retina (Figure 5.14).

To correct this defect, we may use a concave lens to dilute the excess curvature of the eye to bring the formation of image back on the retina.

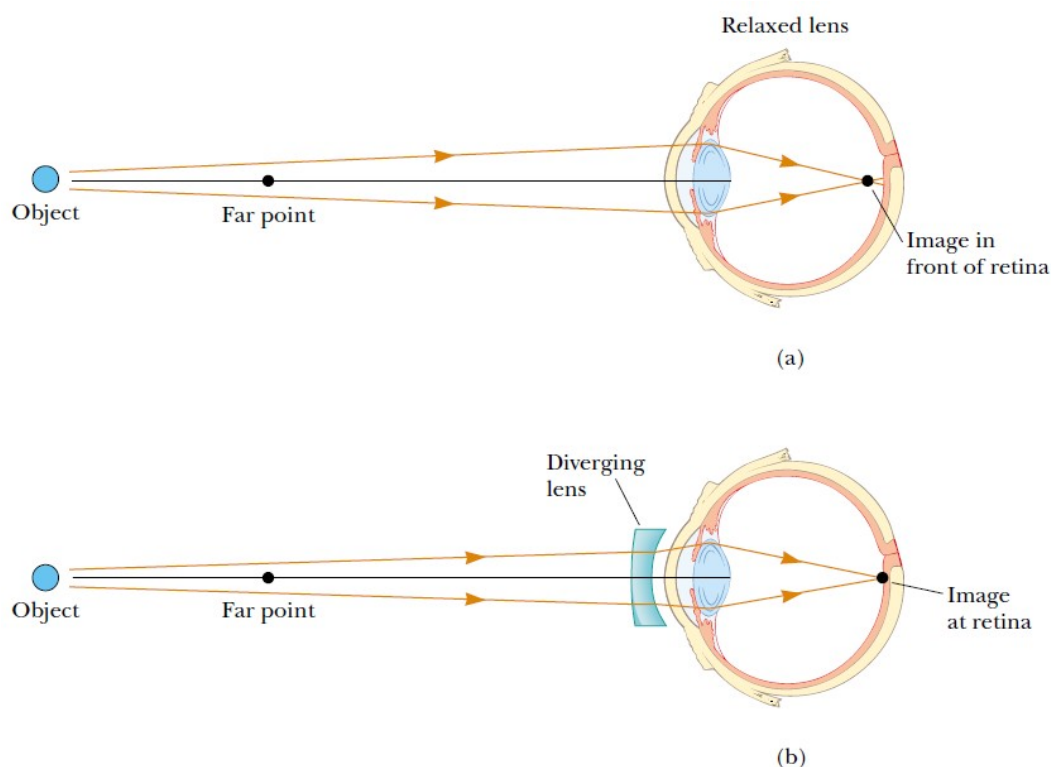


Figure 5.14: Concave lens always diverges light rays. A diverging lens helps to bring the image correctly on the retina.

Example 5.1. If a person only see a clear object at 200 cm from his eyes, a pair of concave glasses can fix this situation. Putting the object distance $p = \infty$ and the virtual focused imaged distance q to be -200 cm into (5.7), the focal length of the concave lens should be -200 cm.

5.5.2 Lens for presbyopia and hyperopia

The opposite case of myopia, we can use a convex lens to correct the position of the image formation.

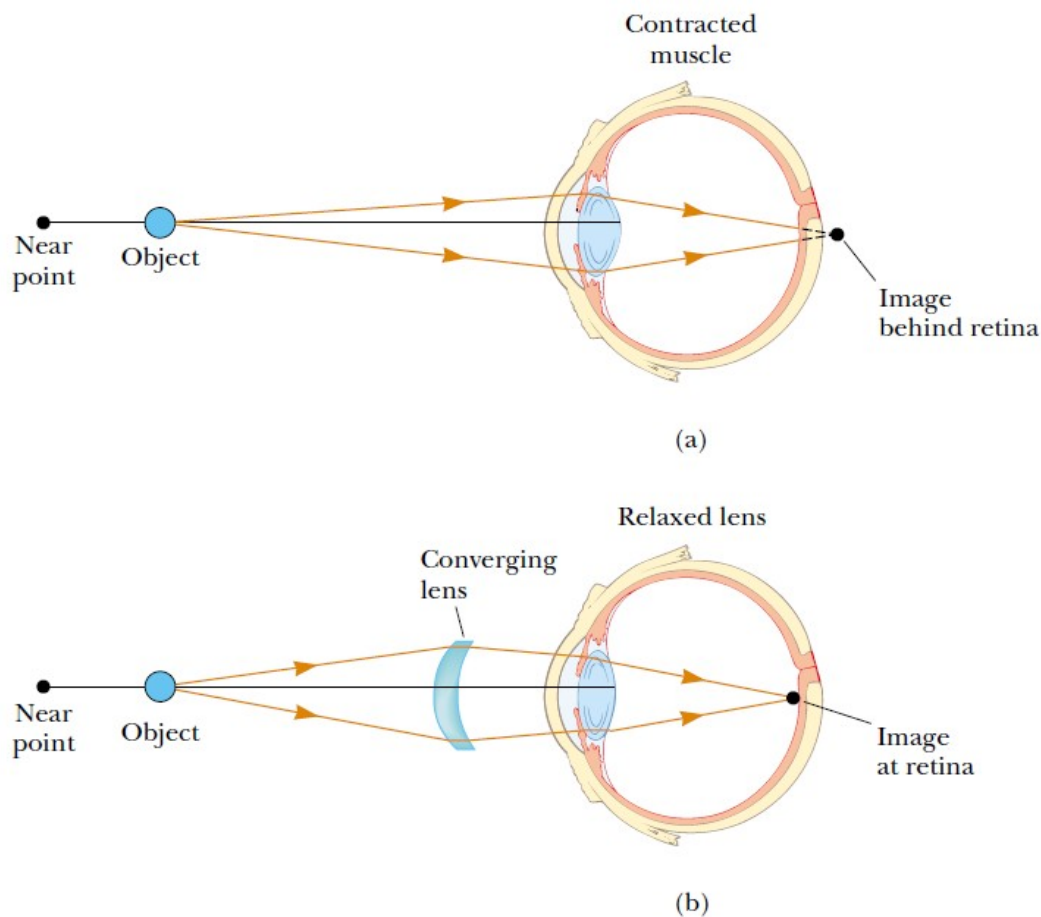


Figure 5.15: (a) Object position lying between near point and eye causes the image formation behind the retina. (b) A converging lens alters the focal length of the eye.

Example 5.2. Supposed that the near point distance of a hyperopic eye is 150 cm, because this distance is the virtual image in front of the lens we should put a minus sign here, i.e. -150 cm to be the image distance q . To view of a closed object at 25 cm from the eye, using (5.7)

$$\begin{aligned}\frac{1}{p} + \frac{1}{q} &= \frac{1}{f} \\ \frac{1}{25} + \frac{1}{-150} &= \frac{1}{f} \\ f &= 30.\end{aligned}$$

Thus, the focal length of the convex lens should be 30 cm.

5.6 Optics devices

Telescope

The basic structure of a telescope consists of an *objective lens* and an *eyepiece*.

Parallel light from a distant object enters the objective lens, which forms a real inverted image of the object. The lens of eyepiece magnifies and inverts the real image for our

viewing (Figure 5.16). The total magnification power of the telescope is:

$$\text{Magnification} = -\frac{\text{Focal length of objective lens}}{\text{Focal length of eyepiece}}$$

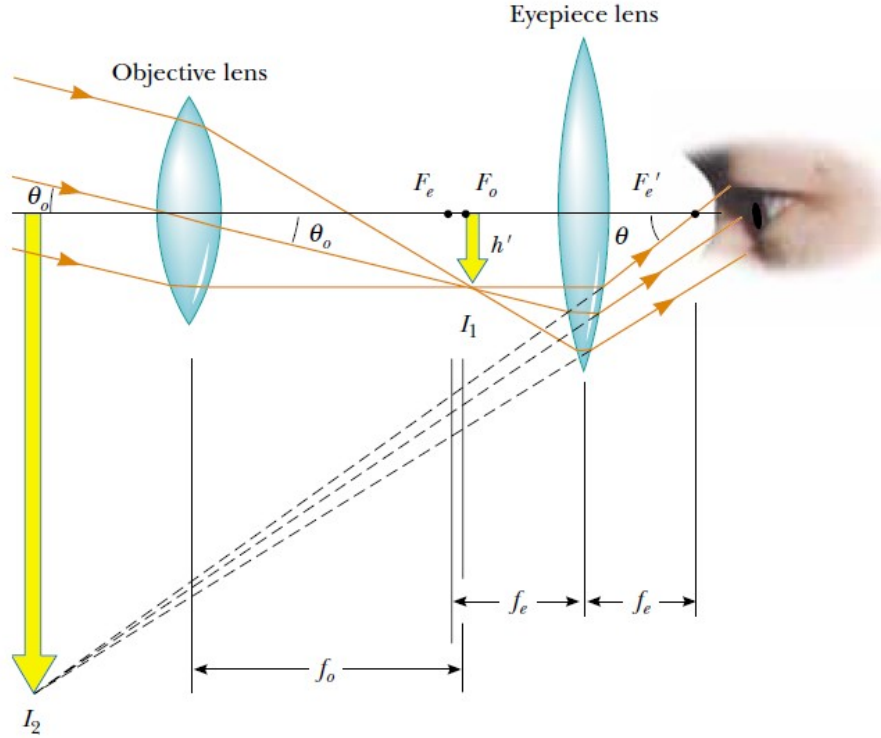


Figure 5.16: A distant object is magnified by two pieces of lens.

Note. Both the objective lens and eyepiece are convex lens.

Microscope

Similar to a telescope, there are two convex lens inside a microscope. But the focal length of the objective lens is very small for the convergent power on close object (Figure 5.17).

Fibre optics

Owing to the property of total internal reflection as the incident angle is bigger than the threshold angle of refraction from a medium with a high refractive index entering to a medium with a low refractive index, a light ray is able to be confined to propagate along a cylindrical transparent material (glass) as in Figure 5.18. The applications and technology of fibre optics are enormous and only widely used in these few decades.

Fibrescope

Fibrescope (Figure 5.19) is one of the medical applications to view the internal structure of body. It can also be used for the measurement of pressure in arteries, bladder, and uterus using optical sensors and laser surgery where powerful laser is directed through one of the bundles to the tissue which is selectively destroyed.

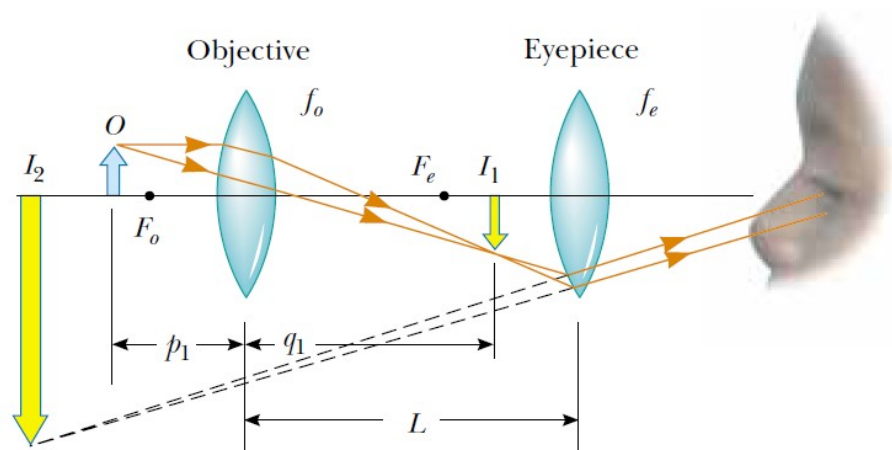


Figure 5.17: The focal length of the objective lens of a microscope is very short for near object.



Figure 5.18: Light ray travels through a bended glass fibre.

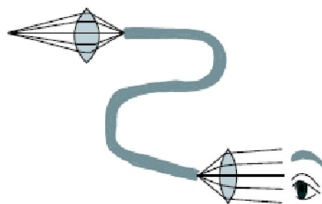


Figure 5.19: A simplified structure of fibroscope

Chapter 6

Electricity and magnetism

6.1 Electricity and the atom

Charge is one of the most fundamental properties of matter. We believe that there are two types of charges; simply called positive and negative charges. Matter possesses either of the charges or the mixture of them with the strength of magnitude in an integral multiple of some elementary unit.

6.1.1 Atom

Atom, which consists of electrons and nucleus, is the fundamental building block of matter.

Electron carrying negative electric charge $-e$ is assumed being a particle with a point-like structure, where e is a fundamental constant for the smallest electric charge outside nucleus.

Nucleus is composed of neutrons and protons, which are not fundamental particles and in general called *nucleons*. Proton carries positive electric charge with magnitude e and neutron is neutral.

Quarks Proton and neutron are hypothetically made up of (by three) quarks those are believed being the most fundamental particles of the composition of matter as well electrons. Quarks carry charges with magnitude either in $e/3$ or $2e/3$.

The number of electrons is equal to the number of protons in an atom. The numbers of neutrons of most of light elements are the same or close to the numbers of their electrons and protons. The same kind of elements with different numbers of neutrons (occur most in heavy elements) are called *isotopes*.

6.1.2 Interactions between charges

One of the basic properties of charges are matter carrying the same kind of charges repelling, carrying opposite kind of charges attracting. The magnitude of the *electric force* in an electromagnetic interaction is expressed by *Coulomb's law*.

Definition 6.1 (Coulomb's law). The magnitude of the force acting between pointlike charged objects with a separation r is given by

$$F = k \frac{q_1 q_2}{r^2}$$

where q_1, q_2 are electric charges, the constant k equals $8.99 \times 10^9 \text{ Nm}^2\text{C}^{-2}$.

The unit of charge which is also one of the fundamental physical quantities is the ‘Coulomb’ (C). One Coulomb is the amount of charge such that a force of $8.99 \times 10^9 \text{ N}$ occurs between two point-like objects with charges of 1 C separated by a distance of 1 m. Electric force between charges is like gravity obeying the inverse square law.

Example 6.1. The electron and proton of a hydrogen atom are separated (on the average) by a distance of approximately $5.3 \times 10^{-11} \text{ m}$. Find the magnitudes of the electric force and the gravitational force between the two particles.

Solution. The attractive electric force has the magnitude:

$$\begin{aligned} F_e &= k \frac{|e|^2}{r^2} = 8.99 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} \\ &= 8.2 \times 10^{-8} \text{ N}. \end{aligned}$$

The gravitational force has the magnitude

$$\begin{aligned} F_g &= G \frac{m_e m_p}{r^2} = 6.7 \times 10^{-11} \times \frac{9.11 \times 10^{-31} \times 1.67 \times 10^{-27}}{(5.3 \times 10^{-11})^2} \\ &= 3.6 \times 10^{-47} \text{ N}. \end{aligned}$$

The ratio of the forces are

$$\frac{F_e}{F_g} \approx 2 \times 10^{39}.$$

Thus, the gravitational force between charged atomic particles is negligible comparing to the electric force.

Conservation of charge

Electric charge is conserved in classical electricity. In modern physics, the **net** electric charge is conserved. For example, a photon can turn into an electron and a *positron* (positive charge electron), giving a pair of the same magnitude but in opposite charges. The net electric charge is still zero, however.

6.2 Electricity

Electricity is a subject to study the interactions between charges. It is a partial theory of the broader *electromagnetism*.

The interaction between charged objects is the Coulomb force. Similar to gravitational potential energy, *electric potential* is defined to describe electric energy. On the other hand, *electric field* E is defined to describe the strength (lines of flux) of a field created by a charge object to affect a test particle at various position in spacetime.

6.2.1 Electric field

Electric field may be regarded as a vector field. The directions of the arrows leaving from a positive charge and going into a negative charge as a convention are shown in Figure 6.1.

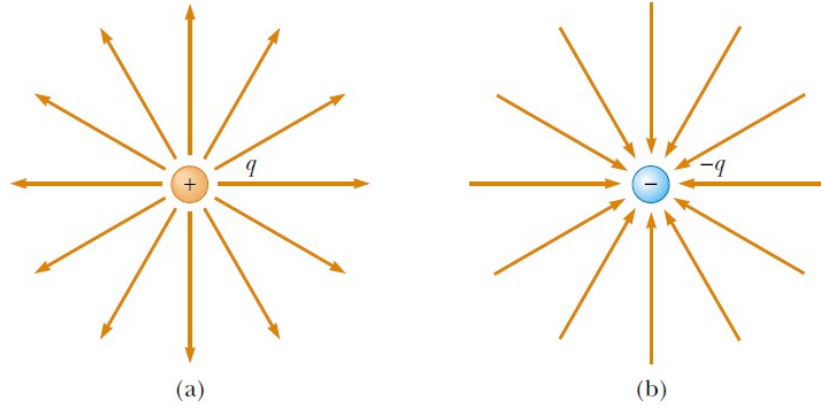


Figure 6.1: The electric field lines for a point charge. (a) For a positive point charge, the lines are directed radially outward. (b) For a negative point charge, the lines are directed radially inward.

The variation of an electric field, i.e., changing the distance separation between charged objects (supposed charges being constant), releases *electromagnetic radiation* (wave).

The definition of an electric field is similar to a gravitational field, which is defined as gravitational force per unit mass, i.e.

$$g = \frac{F}{m} = \frac{GMm}{r^2m} = \frac{GM}{r^2}.$$

By the assumption of gravitational mass equals to the inertial mass, the gravitational field is g —the gravitational acceleration on the surface of Earth. Analogously, define electric field to be the electric force per unit charge

$$E = \frac{F}{q} = \frac{kQq}{r^2} \frac{1}{q} = \frac{kQ}{r^2}.$$

Thus, electric field also obeys inverse square law (Figure 6.2).

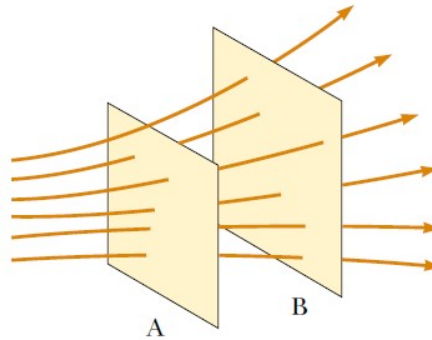


Figure 6.2: Electric field lines penetrating two surfaces. The magnitude of the field is greater on surface A than on surface B.

Example 6.2 (An accelerating positive charge). A positive point charge q of mass m is released from rest in a uniform electric field $\mathbf{E}$ directed along the x -axis, as shown in Figure 6.4. Describe its motion.

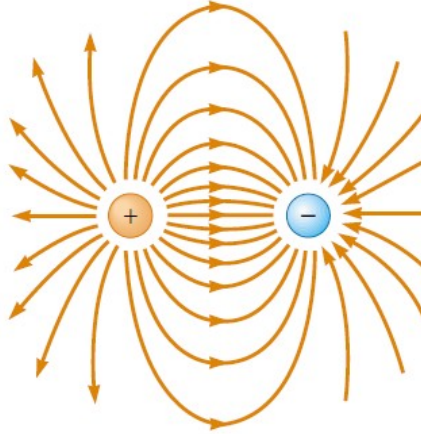


Figure 6.3: (a) The electric field lines for two point charges of equal magnitude and opposite sign. The number of lines leaving the positive charge equals the number terminating at the negative charge.

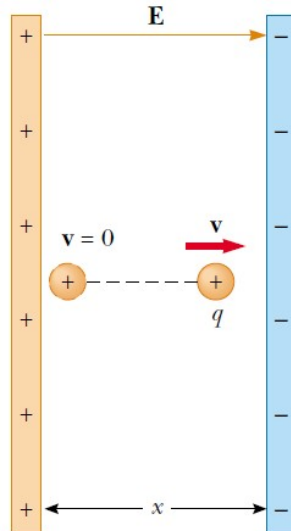


Figure 6.4: A positive point charge q in a uniform electric field $\mathbf{E}$ undergoes constant acceleration in the direction of the field.

Solution. We may ignore gravitational effect as demonstrated in Example 6.1. The acceleration by the electric field is constant and is given by $q\mathbf{E}/m$. The motion is simple linear motion along the x axis. Therefore, we can apply the equations of kinematics in one dimension:

$$\begin{aligned} x_f &= x_i + v_{xi}t + \frac{1}{2}a_xt^2, \\ v_{xf} &= v_{xi} + a_xt, \\ v_{xf}^2 &= v_{xi}^2 + 2a_x(x_f - x_i). \end{aligned}$$

As usual, the subscripts i, f stand for the initial and final states. Taking $x_i = 0$ and

$v_{xi} = 0$, we have

$$\begin{aligned}x_f &= \frac{1}{2}a_x t^2 = \frac{qE}{2m}t^2, \\v_{xf} &= a_x t = \frac{qE}{m}t, \\v_{xf}^2 &= 2a_x x_f = \left(\frac{2qE}{m}\right)x_f.\end{aligned}$$

The kinetic energy of the charge after it has moved a distance $x = x_f - x_i$ is

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{2qE}{m}\right)x = qEx.$$

We can also obtain this result from the work and energy theorem because the work done by the electric force is $F_e x = qEx$.

6.2.2 Electric current

Electric current is a local physical quantity simply to tell us how much electric charge passing through a certain region per unit time. In other words, the average current is the change of electric charge over a time interval $\Delta q/\Delta t$. The instantaneous electric current is

$$I = \frac{dq}{dt} \quad (6.1)$$

where q is the amount of charge. Historically, electric current I is defined as the **reverse** direction of the flow of electrons since people thought that the moving charges in a wire was positive. That being said, electric current I flows from “positive (high potential) to negative (low potential) if the charged objects are connected by a conductor. As *free electrons* are supposed moving freely in a metal, current is moving in the opposite direction of the flow of electrons by definition.

It should be noted that the speed of the interactions of the changing of electric charges is the speed of light or very close to it depending on the media, however the speeds of the movement of electric charges are not (e.g., the flow of electrons in a wire).

6.2.3 Magnetism

Magnetism is a subject of magnetic interaction between *magnetic poles*. Unlike the existence of separate positive and negative electric charges, there is so far no discovery of ‘magnetic charge’ and ‘monopole,’ i.e. no single magnetic pole exists in nature.

Magnetic poles **always** exist in a pair, they are called south (S) and north (N) poles. Similar to the definition of electric field, a *magnetic field* is a set of measured readings of the strength at different points of spacetime, which can be pictured as the lines of flux coming from the north and entering into the south of the poles. Because there is no magnetic object such as monopole which could change the magnetic charge per unit time, hence no magnetic current.

We shall see that electricity and magnetism are actually the same thing in two different manifestation of a single basic theory, called *electromagnetism*.

6.2.4 Quantisation of charge

The magnitude of electric charge is not continuous. Any object carrying electric charge must be in the magnitude of some integral multiples of e . Charge of electron $-e$ is the smallest value of charge outside nucleus (ignoring the quark having $e/3$ charge business at the moment). Discrete values of physical quantities in organized forms (packet of energy, lump of energy, fixed values of angular momentum, etc.) rather than continuous are categorized as *quanta*.

Electron

One of the convenient examples to demonstrate the *cathode ray* experiment that invented in hundred years ago is the Cathode-Ray-Tube (CRT) of the old fashion vacuum tube television set¹ (not the LCD display). A metal plate connecting to the negative potential called cathode is heated causing electrons to jump to higher states, another metal plate with positive potential called anode attract and collect the electrons from the cathode flying through vacuum (Figure 6.5).

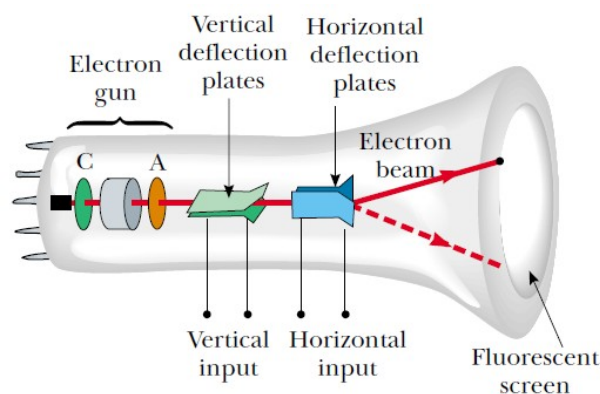


Figure 6.5: Electrons leaving the hot cathode C are accelerated to the anode A. In addition to accelerating electrons, the electron gun is also used to focus the beam of electrons, and the plates deflect the beam.

6.3 The nucleus

6.3.1 Radioactivity

Any emissions of high speed microscopic particles may be regarded as radiations. Historically, three common types of radiations have been studied:

Alpha (α) The emission of helium nuclei, i.e., He^{2+} .

Beta (β) The emission of a beam of electrons.

Gamma (γ) Apart from α and β emissions involving matters, γ radiation is in fact a beam of highly energetic electromagnetic wave.

¹The electron beams of most of the home vacuum tube television set are controlled by magnetic fields rather than electric fields as shown in the figure.

Note. High energetic EM wave does not mean the wave having a very high intensity (amplitude). Energy of EM wave only depends on its wavelength.

6.3.2 Strong interaction and nuclear fission

Proton carries positive electric charge. A number of protons in the nucleus should have repulsed to each other due to the same charge. The existence of an atom implies that there must be the existence of a larger attractive force which even greater than the electric repulsion of the like charges in a range, is simply called the *strong* force, for binding the protons and neutrons together. Nevertheless, the strong and the electric interactions may not always maintain the equilibrium state especially when the number of protons increases.

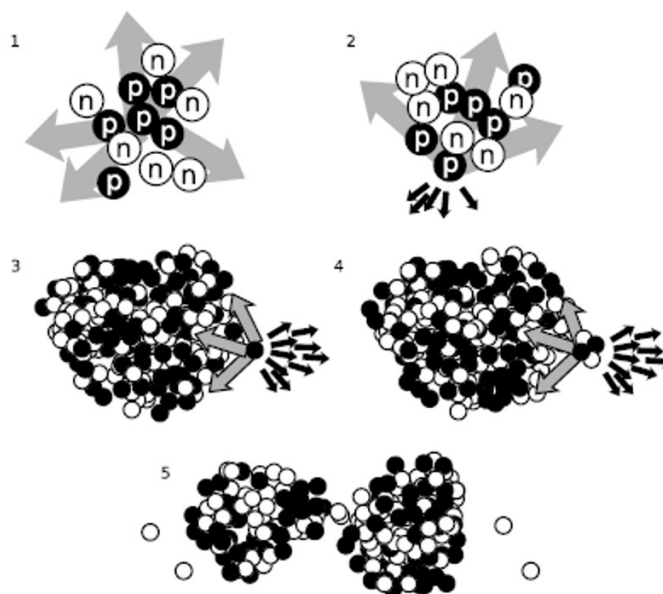


Figure 6.6: Nuclear fission

Figure 6.6 shows the stabilities of a nuclear in different scenarios:

1. The electric repulsion among the protons are not in great different in evenly distributed circumstance when the number of protons is small. The strong binding force is strong enough to hold them inward.
2. Similar to the theory of surface tensor of matter but in opposite sense (repulsion instead of attraction), the short range strong force is comparably weak to bind the edge protons which have the tendency to be repelled out.
3. The unbalance interactions between the electric repulsion and strong force on the edge protons, causing them to be expelled out easily for a large atom.
4. The comparably stable nucleus—helium nuclei are released as an α emission.
5. When the strong force is no longer strong enough to hold the electric repulsion due to the large number of protons (heavy atom), the atom will break into two pieces—*nuclear fission*.

6.3.3 Weak interaction

It is discovered that proton and neutron could be transformed to each other in some special circumstances:

$$n \rightarrow p + e^- + \bar{\nu}$$

and

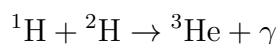
$$p \rightarrow n + e^+ + \nu$$

where e^- is the usual electron, e^+ is the anti-particle of electron called positron, ν is one of the elementary particles called neutrino, and $\bar{\nu}$ is its anti-particle. The internal binding force of these transformations is called *weak interaction*.

6.3.4 Nuclear fusion

Contrary to the fission of heavier elements, lighter atoms can be combined to form a heavier atom. This process is called nuclear *fusion*. However, it is difficult to proceed in nature since nuclear fusion requires extremely high temperature for the ignition. In addition, the most stable element in nature is iron as which is at the middle of the two extremes – fusion and fission.

One of the fusion reactions of the sun is:



The excess energy is almost all carried off by the gamma ray.

The energy released in fusion and fission mainly come from the mass difference of the atoms before and after the nuclear reaction. According to Einstein's famous equation $\Delta E = \Delta mc^2$, a small amount of lost mass can produce a tremendous energy due to the great numerical value of light speed square.

6.3.5 The fundamental interactions

We have so far mentioned four fundamental interactions in nature, they are

Electromagnetic The interaction between electric charges. Theoretically, all the daily physical phenomena except gravity and nuclear physics may be explained by electromagnetism because everything is made up of atoms. The *mediator* of the interaction is photon, i.e. light. The modern theory to explain this interaction is called *quantum electrodynamics* (QED).

Strong Neutron and proton are made up of quarks. The interaction amongst quarks is strong interaction. The theory of the topic is *quantum chromodynamics* (QCD).

Weak The interaction of the transformation between neutron and proton. The theory of weak interaction may be integrated into QED to form a single theory called *electroweak*.

Gravity is the oldest theory that we discovered and seemed having been solved. The universal attraction between matters was formally discovered and explained by Newton. A radical and revolutionary theory of gravity called *general relativity* was proposed by Einstein, in which gravity may be viewed and interpreted as the curvature of the

geometry of spacetime rather than a force. Nowadays, physicists trying to “quantize” gravity is believed to be one of the hardest problems in science. One of the most popular attempted theories of “quantum gravity” is *loop quantum gravity*.

As the success of the merge of electromagnetic and weak interactions to form a single theory known as electroweak, scientists are also trying to unify the strong, weak and electromagnetic interactions together. The proposed theory is known as *grand unification theory* (GUT). It is obvious that scientists also have a bigger prospect attempting of to unify all four interactions into a single theory. The most popular candidates of those theories are the *strings theory* and the variants.

6.4 Electricity and circuit

6.4.1 Current

Current is a quantity to measure the flow of electric charge. That is,

$$I = \frac{dq}{dt}.$$

Current is defined by the rate of change of electric charge with respect to time. The direction of the flow is assumed to be from positive to negative potential. The unit of current is Coulomb per second, i.e. C/s, however, we prefer to use Ampère (A) for convenience.

6.4.2 Voltage

Imagine before connecting the two ends of a battery by a wire, there exists a “potential” between the two poles—the tendency of the electric charges to flow from the positive to negative. When the wire is connected, the charges are continuously passing through the wire and the “potential” of the flow gradually drops until the two poles made neutral. The tendency of the charge flowing or in other words—potential can be redefined as voltage.

Voltage is an abstract idea to describe the *potential difference* of charged objects. Just similar to the gravitational potential, the potential energy will drop to zero when a rock reaches to the bottom of a cliff. In fact, voltage is defined by the electric potential energy U per unit charge:

$$\Delta V = \frac{\Delta U}{q}. \quad (6.2)$$

In practice, instead of mentioning the charge and energy, we often define the voltage in terms of the power P and the electric current:

$$P = IV. \quad (6.3)$$

The unit of voltage is J/C, or the more convenient unit Volt (V).

Proof. Power is defined as the rate of change of energy with respect to time

$$P = \frac{dU}{dt}$$

where U is the electric potential energy. By the chain rule, we have

$$P = \frac{dU}{dq} \frac{dq}{dt} = VI.$$

□

6.4.3 Resistance

When electrons and ions pass through media they must experience a resistant force analogous to the viscous force in fluid, frictional force between rigid bodies. Actually, in microscopic level, all the resistant forces (viscous and frictional forces) are arisen by the electromagnetic interactions among atoms or molecules. Electrical resistance is defined by *Ohm's law*

$$R = \frac{V}{I} \quad (6.4)$$

in which the unit of R is Ω .

Superconductors

Scientists have discovered that some materials under certain critical (usually at extremely low) temperature exhibit zero resistance. Since the resistant force of matter come from the electric repulsion among electrons (which are *fermions*). If two electrons are able to fuse together in a special way known as *Cooper's pair*, which exhibits electric neutral, i.e. no resistance, due to the pairs possessing the properties of *bosons*.

Resistors

We may add some electrical components known as *resistors* into an *electric circuit* to alter the current and voltage. There are two types of connection of resistors in an electric circuit:

Series The total resistance of resistors connected in series is

$$R = R_1 + R_2 + \dots + R_n \quad (6.5)$$

where $R_1, \dots, R_n$ are the individual resistors.

Parallel Since the potential difference V of each resistor is exactly the same, the total current is

$$I_1 + I_2 + \dots + I_n = \frac{V}{R_1} + \frac{V}{R_2} + \dots + \frac{V}{R_n} = \frac{V}{R},$$

thus,

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}. \quad (6.6)$$

6.4.4 Capacitance

The *capacitor* is a circuit element that stores electric charges. In its simplest form it consists of two conducting plates separated by an insulator. Figure 6.7 shows that a parallel-plate capacitor consists of two parallel conducting plates with area A each, separated by a distance d . When the capacitor is charged, the plates carry equal amounts of charge. One plate carries positive charge, and the other carries negative charge. *Capacitance* C is measured in Faraday (F). In practice, sometimes the unit F is too big to use, we often use the microfarad (μF) instead (i.e., $1\text{ F} = 10^6 \mu\text{F}$). The relation between the stored charge Q , and the voltage across the capacitor is given by

$$Q = CV. \quad (6.7)$$

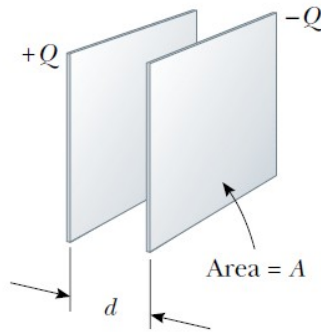


Figure 6.7: A parallel-plate capacitor with the plates separated by a distance d .

In a charged capacitor, positive charges are on one side of the plate, and negative charges are on the other. The amount of energy U stored in such a configuration is given by

$$U = \frac{1}{2}CV^2. \quad (6.8)$$

Proof. Recall that the rate of change of electric energy U with respect to charge Q is the voltage V :

$$\frac{dU}{dQ} = V.$$

Taking the differential of (6.7) gives

$$dQ = CdV$$

since the capacitance C is supposed to be a constant. Therefore, we have

$$dU = CVdV.$$

Integrating over the intervals $[0, U]$ and $[0, V]$, respectively, we have

$$\int_0^U dU = \int_0^V CVdV \Rightarrow U = \frac{1}{2}CV^2.$$

□

Combinations of capacitors

Similar to resistors, There are parallel and series connections for capacitors (Figure 6.8 & 6.9). The equations for these two types of connections are just in the opposite senses of the combination of resistors.

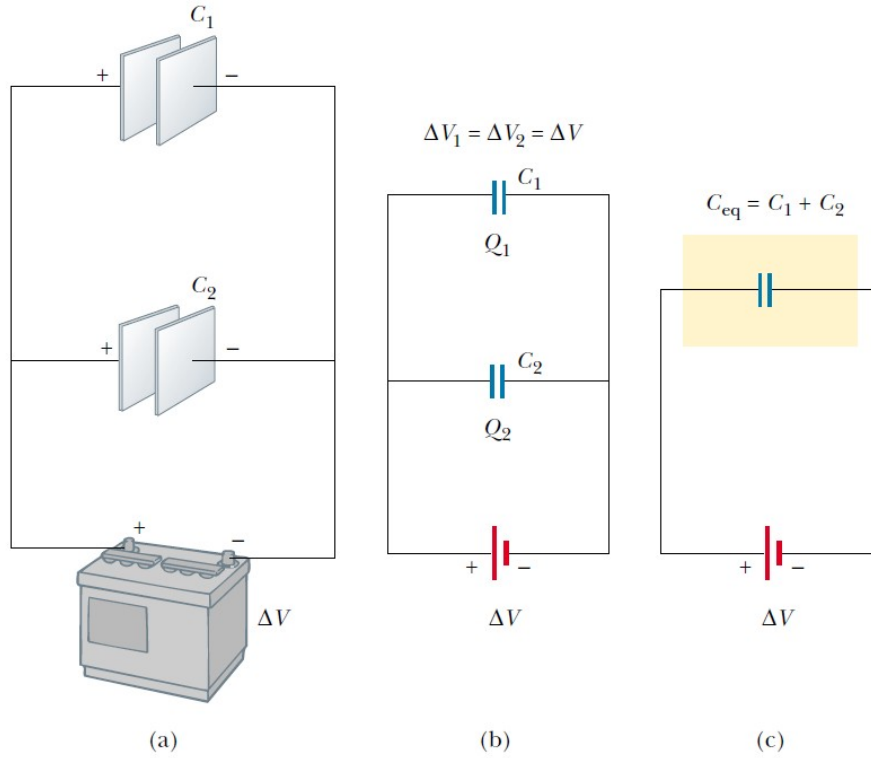


Figure 6.8: A parallel combination of two capacitors gives the equivalent capacitance $C_{eq} = C_1 + C_2$.

6.5 Electromagnetism

6.5.1 Electric and magnetic fields

Definition 6.2. The electric field vector, $\mathbf{E}$, at any location in space is found by placing a test charge q at that point. The electric field vector is then given by

$$\mathbf{E} = \frac{\mathbf{F}}{q}$$

where $\mathbf{F}$ is the electric force on the test charge.

Definition 6.3. The magnetic field vector, $\mathbf{B}$, at any location in space is defined by observing the torque exerted on a magnetic test dipole D consisting of a square current loop. The field's magnitude is

$$B = \frac{\tau}{D \sin \theta}, \quad (6.9)$$

where θ is the angle by which the loop is misaligned. The direction of the field is perpendicular to the loop; of the two perpendiculars, we choose the one such that if we look along it, the loop's current is counter-clockwise. The magnetic dipole moment by definition is given by the product of the electric current and the enclosed area looped by the current. We shall have more of it later.

Force on a charge moving through a magnetic field

The magnetic force on a moving charged particle is given by

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$$

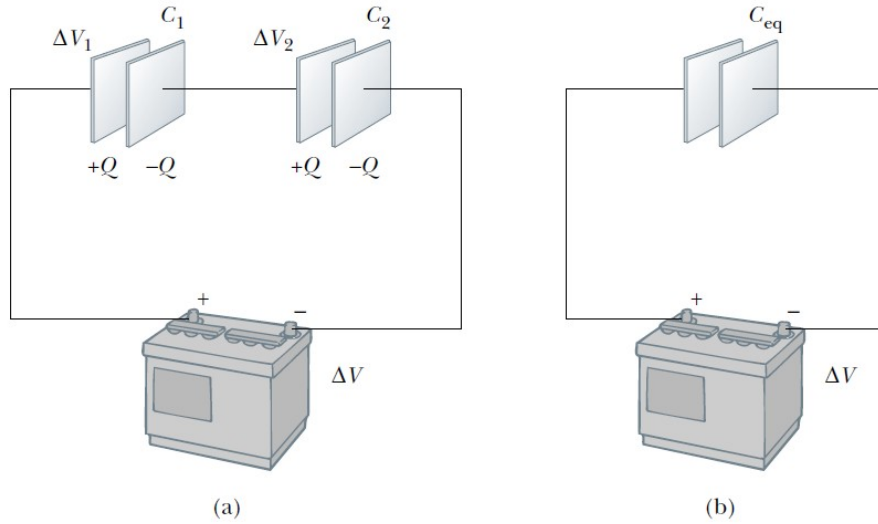


Figure 6.9: (a) A series combination of two capacitors gives the equivalent capacitance $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$.

where $\mathbf{v}$ is the velocity vector of the particle. To summarise:

- The magnitude F of the magnetic force exerted on the test particle is proportional to the charge q and to the speed v of the particle.
- The magnitude and direction of $\mathbf{F}$ depend on the velocity of the particle and on the magnitude and direction of the magnetic field $\mathbf{B}$.
- When a charged particle moves parallel to the magnetic field vector, the magnetic force acting on the particle is zero.
- When the particle's velocity vector makes any angle $\theta \neq 0$ with the magnetic field, the magnetic force acts in a direction perpendicular to both $\mathbf{v}$ and $\mathbf{B}$; i.e., $\mathbf{F}$ is perpendicular to the plane formed by $\mathbf{v}$ and $\mathbf{B}$.

The directions of the vectors can be determined by the right-hand rule (Figure 6.10). The right-hand rule for determining the direction of the magnetic force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ acting on a particle with charge q moving with a velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$. The direction of $\mathbf{v} \times \mathbf{B}$ is the direction in which the thumb points. (a) If q is positive, $\mathbf{F}$ is upward. (b) If q is negative, $\mathbf{F}$ is downward, antiparallel to the direction in which the thumb points.

Figure 6.11 shows the motion of a charge particle in a uniform magnetic field $\mathbf{B}$. When the velocity of the positive charge particle q is perpendicular to the field, it moves in a circular path in a plane perpendicular to $\mathbf{B}$. The magnetic force $\mathbf{F}_B$ acting on the charge is always directed toward the centre of the circle.

Lorentz force

Because a moving charge particle experiences induced forces in both of electric and magnetic fields, we may combine these two induced forces together by a single force equation:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

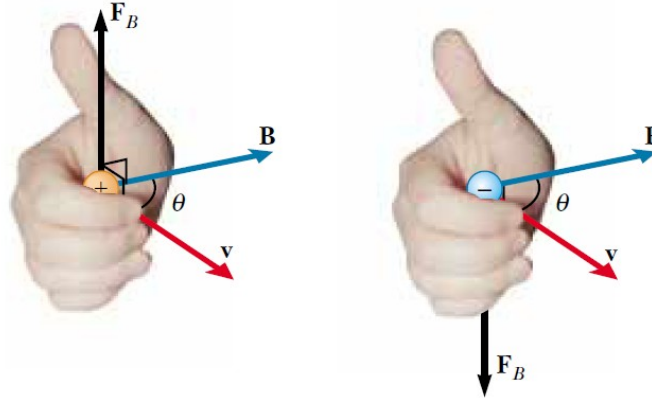
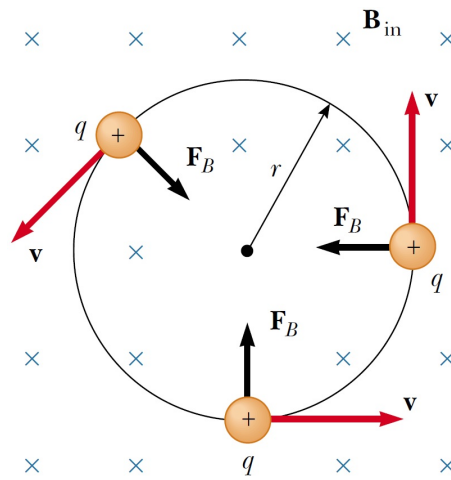


Figure 6.10: The right-hand rule

Figure 6.11: Motion of a charge in a uniform $\mathbf{B}$ field.

Since velocity is relative, we may keep the charge particle motionless and change the magnetic field $\mathbf{B}$ in the direction transversely to the magnetic field lines to produce the similar Lorentz force.

6.5.2 Multi-variable calculus (Supplementary notes)

The derivative of a function $f(x) := y$ with respect to x by definition is given by

$$\frac{df(x)}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

which measures the ratio of how much change of $f(x)$ to the variation of x infinitesimally. It is called as “the rate of change of $f(x)$ with respect to x ”. If you draw the function on a 2-dimensional x, y coordinate system, dy/dx is the slope of the tangent line at x where we are interested in.

In a similar fashion, we can define derivatives for *multi-variable function*. First of all, what is multi-variable function? It is simply

$$f(x, y, z, \dots),$$

a function depends on variables more than a single one. All the $x, y, z, \dots$ are *independent variables*. The term “independent” is crucial. It tells us that **all the variables cannot**

affect to each other. It means that we may differentiate the function f with respect to x only by treating all the rest of the variables as constants. Similarly, we may also differentiate f with respect to y by keeping others as constants.

Introduce a new “differential operator” symbol “ ∂ ” for this type of differentiation, instead of the usual single variable differential operator “ d ”. As an example, suppose that a function $f(x, y)$ depends on x and y independent variables. The possible derivatives of f are:

$$\frac{\partial f}{\partial x} \quad \text{and} \quad \frac{\partial f}{\partial y}.$$

This type of differentiation is called *partial differentiation*, and the derivatives are *partial derivatives*. The formal definition is

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

where y in the equation is regarded as a constant. And we have a similar formula for the variable y .

Example 6.3. A simple sinusoidal wave equation $\psi(x, t) = \psi_0 \sin(kx - \omega t)$ where k and ω are the wave number and angular frequency which are supposed to be constants. We would like to find the derivatives of ψ with respect to x and t .

Solution. Differentiate ψ with respect to x , remember to keep the t as a constant, then we have

$$\frac{\partial \psi}{\partial x} = \frac{\partial}{\partial x} [\psi_0 \sin(kx - \omega t)] = \psi_0 k \cos(kx - \omega t). \quad (6.10)$$

Similarly, the derivative of ψ with respect to t :

$$\frac{\partial \psi}{\partial t} = \frac{\partial}{\partial t} [\psi_0 \sin(kx - \omega t)] = -\psi_0 \omega \cos(kx - \omega t). \quad (6.11)$$

Second partial derivatives

Define the *second partial derivatives* with the properties exactly like those in the ordinary differentiation. It means that we partial differentiate the function $f(x, y)$ twice:

$$\frac{\partial}{\partial x} \frac{\partial f}{\partial x} = \frac{\partial^2 f}{\partial x^2} \quad \text{and} \quad \frac{\partial}{\partial y} \frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial y^2}.$$

Nevertheless, you may notice that we may differentiate f with respect to x first, then y , or vice versa:

$$\frac{\partial}{\partial y} \frac{\partial f}{\partial x} = \frac{\partial^2 f}{\partial y \partial x} \quad \text{or} \quad \frac{\partial}{\partial x} \frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial x \partial y}.$$

Luckily, it can be proved that

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Example 6.4. Using the result (6.10), the second derivative of ψ with respect to x is

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{\partial}{\partial x} [\psi_0 k \cos(kx - \omega t)] = -\psi_0 k^2 \sin(kx - \omega t) = -k^2 \psi.$$

The second derivative of ψ with respect to t is

$$\frac{\partial^2 \psi}{\partial t^2} = \frac{\partial}{\partial t}[-\psi_0 \omega \cos(kx - \omega t)] = -\psi_0 \omega^2 \sin(kx - \omega t) = -\omega^2 \psi.$$

Both equations have function ψ on the right hand sides, equating both equations with ψ we obtain

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{k^2}{\omega^2} \frac{\partial^2 \psi}{\partial t^2}. \quad (6.12)$$

This is the one-dimensional *wave equation*.

Partial differentiation in 3-dimension

It is unavoidable to use vector analysis in 3-dimensional space, or in 4-dimensional space and time. Recall the dot product

$$\mathbf{u} \cdot \mathbf{v} = \text{number}$$

where vector $\mathbf{u}$ dots vector $\mathbf{v}$ always giving a number (scalar). Considering 1-dimensional case, imagine that the scalar $\partial F / \partial x$ is a result of the dot product of two vectors:

$$\frac{\partial F}{\partial x} := \frac{\partial}{\partial x} \cdot \mathbf{F} = \text{number}$$

Thus, **both $\partial / \partial x$ and $\mathbf{F}$ are vectors**. In this new definition, $\partial / \partial x$ is a vector in x -direction. In general, in 3-dimension we have the three *basis vectors*

$$\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

in x, y, z directions. Don't be scared by these symbols, they have exactly the same meaning as

$$(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z) \quad \text{or} \quad (\mathbf{i}, \mathbf{j}, \mathbf{k})$$

unit vectors as we learned previously but just written in the form of differential operators.

It is convenient to define a new symbol for this set of operators:

$$\nabla := \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

Example 6.5. The dot product of the differential operator ∇ and the vector $\mathbf{F} = (F_x, F_y, F_z)$:

$$\nabla \cdot \mathbf{F} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (F_x, F_y, F_z) = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Sometimes people prefer to write

$$\nabla \cdot \mathbf{F} = \text{div} \mathbf{F}$$

where “div” is known as *divergence*.

Example 6.6. The differential operator ∇ acts on a function (scalar) ψ :

$$\nabla\psi = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \psi = \left(\frac{\partial\psi}{\partial x}, \frac{\partial\psi}{\partial y}, \frac{\partial\psi}{\partial z} \right)$$

resulting a vector. People also like to write the expression in this way:

$$\nabla\psi = \text{grad}\psi$$

where “grad” means *gradient*. The second derivative of ∇ is defined as ∇^2 . That is,

$$\nabla^2 := \left(\frac{\partial^2}{\partial x^2}, \frac{\partial^2}{\partial y^2}, \frac{\partial^2}{\partial z^2} \right).$$

Thus, the second derivative of ψ is

$$\nabla^2\psi.$$

Now, with this new notation we may directly write down the 3-dimensional version of the wave equation (6.12) without proof in this way:

$$\nabla^2\psi = \frac{k^2}{\omega^2} \frac{\partial^2\psi}{\partial t^2}$$

The last example is about the cross product operation. This would look a little bit complicated.

Example 6.7. The cross product of ∇ and the vector $\mathbf{F}$:

$$\begin{aligned} \nabla \times \mathbf{F} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \times (F_x, F_y, F_z) \\ &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \mathbf{e}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \mathbf{e}_y + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \mathbf{e}_z \end{aligned}$$

resulting a vector. As you expected, another name for this expression:

$$\nabla \times \mathbf{F} = \text{curl}\mathbf{F}$$

where “curl” stands for *curl*.

6.5.3 Maxwell’s equations

Electricity and magnetism are the manifestations of a single theory known as *electromagnetism*.

The dynamics of charges and current, the relation between electric and magnetic fields strength are expressed by *Maxwell’s equations*:

$$\nabla \cdot \mathbf{B} = 0, \tag{6.13}$$

$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0, \tag{6.14}$$

$$\nabla \cdot \mathbf{E} = \rho, \tag{6.15}$$

$$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = \mathbf{j}. \tag{6.16}$$

where $\mathbf{E}, \mathbf{B}$ are electric and magnetic field strengths, $\rho, \mathbf{j}$ are charge density and current density, c is the speed of light in vacuum. The interpretation of Maxwell’s equations:

- (6.13) tells us there are no magnetic charges.
- (6.14) is Faraday's law; a changing magnetic field produces an electric field.
- (6.15) is Gauss' law; the total charge inside a closed surface may be expressed by the integration of the normal component of $\mathbf{E}$ over the surface.
- (6.16) is Ampère's law, $\nabla \times \mathbf{B} = \mathbf{j}$, with Maxwell's additional term $\partial\mathbf{E}/\partial t$, stating that the changing of electric field produces magnetic field.

6.5.4 Magnetic dipole

Where does the magnetic field of a magnet come from?

We have discovered that moving electric charge (i.e. current) generates magnetic field, which is always perpendicular to the direction of the moving charge. From equation (6.16) we see that if electric field is constant, i.e., $\partial\mathbf{E}/\partial t = 0$ we have Ampère law

$$\nabla \times \mathbf{B} = \mathbf{j}$$

where $\mathbf{j}$ is the current density. In other words, $\nabla \times \mathbf{B}$ sometimes written as $\text{curl}\mathbf{B}$, it means that the directions of magnetic field and current are “curl” (perpendicular) to each other.

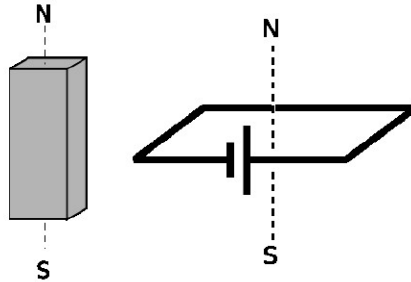


Figure 6.12: Induced magnetic dipole acts much like a bar magnet.

In classical physics, electrons circulate around nuclei like planets circulating around the sun. Various direction of magnetic fields inside a material are generated due to the moving electrons analogous to Figure 6.12. Nevertheless, the net effect of the magnetic fields of ordinary material are approximately zero because the fields tend to cancel out by different orientations of the circular motion of electrons. Magnet, or for example, the *magnetised* iron have aligned *magnetic dipoles* in atomic level as shown in Figure 6.13, reinforcing the resultant of magnetic effect.

Magnetic dipole moment

A dipole tends to align itself to the surrounding magnetic field (Figure 6.14). This rotation creates a torque τ as shown in (6.9), where D is magnetic dipole moment.

Magnetic dipole moment can be defined in terms of current I and area of a square loop A by

$$D = IA.$$

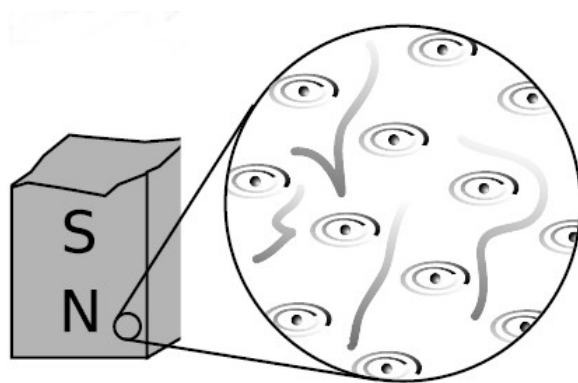


Figure 6.13: Atomic level of a magnet

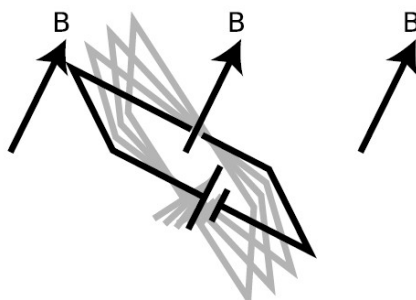


Figure 6.14:

By definition, the unit of magnetic moment is Ampère-metre² (Am²). In classical mechanics, the unit of torque is Newton-metre. Therefore, from (6.9) the unit of magnetic field B is

$$\frac{[N][m]}{[A][m^2]} = \frac{[N]}{[A][m]},$$

or “Newton per Ampère per metre.” As usual, we have invented a new unit for this:

$$1\mathbf{T} = 1 \text{ NA}^{-1}\text{m}^{-1}$$

where T is known as “Tesla.”

Solenoid

Electric current flowing around a loop produces magnetic field. We may make a *solenoid* which is a series of loops made by a wire to reinforce the magnetic field strength as shown in Figure 6.15.

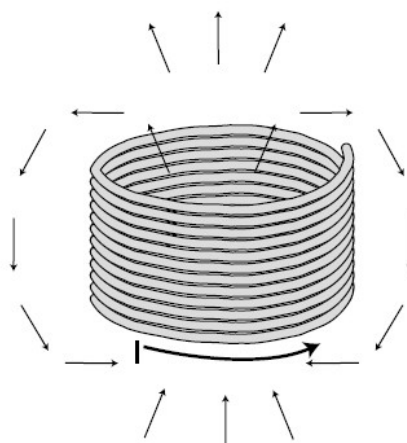


Figure 6.15: Current is flowing from left to right (in the readers point of view), magnetic field is produced passing through the interior of the loops from the bottom to the top. The directions of the magnetic field and current obey the right hand rule as well.

Chapter 7

Relativity and quantum physics

7.1 Special relativity

In 1905 Albert Einstein published three papers. The first (photoelectric effect) paper made him to be the Nobel prize winner in 1921. The second paper discussed about the movement of small particles in a fluid (Brownian motion). The third paper was called *On the electrodynamics of moving bodies*, i.e., Special Relativity (SR), it has changed the foundation of physics and the way we understand nature.

7.1.1 Coordinate transformation

To describe the motion a body, one of the contemporary ideas is that one first needs to know **where** the object is at a given **time**. Nevertheless, in order to state where the object is, one needs to measure its position relative to somewhere. Thus, one needs a *point of reference* from which to define the position of the object. The chosen point of the reference that we simply call the origin. One also needs a **clock** so that one can specify the time t for the evolution of the object at that given position. When one has the origin of reference and the frame itself, with a clock to measure the time and time interval, we say that we have a *frame of reference* or simply a *frame*.

Newton's first law states that

- if an object is at rest it will stay at rest if no external force acts upon it, or
- if an object is moving it will keep on moving at constant velocity if no external force acts upon it.

Newton's first law depends on some chosen frame to describe the motion of an object. For instance, if we are using the moving object itself as the origin of the frame of reference, the object is always at rest relative to its own frame no matter what forces are acting on it. A similar example is an rest object inside an orbiting space shuttle. It does not move relative to the space shuttle even though gravity is acting on them. An opposite example is driving. When a car accelerates we lead backward and when it is braked we lead forward even though there are no external force acting on it along the moving direction.

Galileo transformation

Most of the contemporary physical theories are assumed to be valid in a very special reference system—the *inertial system*. If one such inertial frame of system exists, an

infinite number of other inertial frames also exist since any frame that is moving at a constant velocity relative to the first inertial frame is also an inertial frame. The frames in which the law of inertia does not hold are those that are accelerating with respect to inertial frames. They are called non-inertial frames of system. The requirement that the laws of physics must be the same no matter which inertial frame is used to describe motions of objects can be restated as:

The laws of physics must be invariant under the Galileo transformation.

$$x = x' + vt, \quad y = y', \quad z = z', \quad t = t'. \quad (7.1)$$

This set of equations relates the coordinate systems K to K' with the relative velocity v in x -direction.

Note. The time interval in the first equation is t instead of t' since $t = t'$ given by the fourth equation. Time is unique and **absolute** as stated in Newtonian physics. All observers agree the same time interval according to Newton's theory.

7.1.2 Lorentz transformation

In the later half of the 19th century, Maxwell discovered the equations describing the properties of electricity and magnetism. His equations predicted that electromagnetic waves propagate at the speed of light $c = 3 \times 10^8$ m/sec, establishing that light is in fact an electromagnetic radiation with a particular frequency interval of the spectrum.

Maxwell's equations is not invariant under Galileo transformation (7.1) (i.e physical properties namely, electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ are changed under the transformation). This is evident from the fact that Maxwell's equations predicted the speed of light. Since in the Newtonian mechanics the speed of an object depends on the inertial frame where we observe it from. Maxwell's equations could not be correct in one particular inertial frame. This contradicts the belief that the laws of physics should be the same regardless of from what inertial frame we make the observation.

To overcome this difficulty, physicists thought that light had to travel via a hypothetical medium called *ether*, just like sound has to travel via air or other sort of media. The speed of light predicted by Maxwell's equations was interpreted as that relative to ether. So if the speed of light one observes is different from that predicted by Maxwell, it would **not** mean that the laws of physics is different in our inertial frame from others. Our inertial frame could be simply moving relative to ether!

The constancy of speed of light

In the beginning of the 20th century, an experiment which was conducted by Michelson and Morley measured the dependence of speed of light relative to the speed of observer. To everyone's surprise, it showed that the speed of light was always the same no matter what the motion of the observer was!

Einstein believed the prediction of Maxwell's equations and respected the result of the constancy of speed of light from the equations (It seems that he did not know about the Michelson-Morley experiment). He has postulated:

- All the laws of physics are the same in every inertial frame of reference.

- This postulate implies that there is no experiment whether based on the laws of mechanics or the laws of electromagnetism from which it is possible to determine whether or not a frame of reference is in a state of uniform motion.
- The speed of light is independent of the motion of its source.

Lorentz transformation

Maxwell's equations are invariant under the *Lorentz transformation*. Einstein extended the idea to all physics **in inertial system**. In special relativity, physics is invariant under

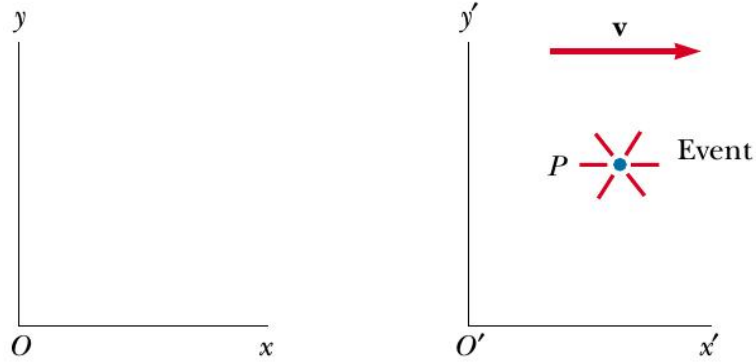


Figure 7.1: An event that occurs at some point P is observed by two observers, one at rest in the K frame and the other in the K' frame, which is moving in the right with a velocity v .

the transformation (Table 7.1):

$$x = \gamma(x' + vt'), \quad y = y', \quad z = z', \quad t = \gamma\left(t' + \frac{v}{c^2}x'\right) \quad (7.2)$$

where

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

In short, equation (7.2) is just for relating the two sets of coordinates (x, y, z, t) and (x', y', z', t') reported by the two inertial observers. Physics satisfies Lorentz transformation in inertial system:

- Time is no longer absolute, where t' is the time measured in K' frame.
- The distance and time measured in K frame now have to be multiplied by a factor γ , i.e. depending on the relative velocity v of the frames.

When v is small compared with c ,

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \approx 1.$$

The Lorentz transformation (7.2) becomes

$$x = x' + vt', \quad y = y', \quad z = z', \quad t = t' + \frac{v}{c^2}x'$$

for the Newtonian approximation in practice.

In fact, v is much smaller than c in our daily situations, $v/c^2 \rightarrow 0$ in the fourth equation becomes the Galileo transformation (7.1).

7.1.3 Space and time

Time dilation

We would like to relate the time intervals of two successive events measured by two different inertial observers.

Consider the situation that two observers O and O' in different inertial systems measure the time interval of a photon traveled as shown in Figure 7.2. O is at rest relative to the ground and O' staying in the train which is moving at a speed v . The experiment is performed in the train measured by the local observer O' .

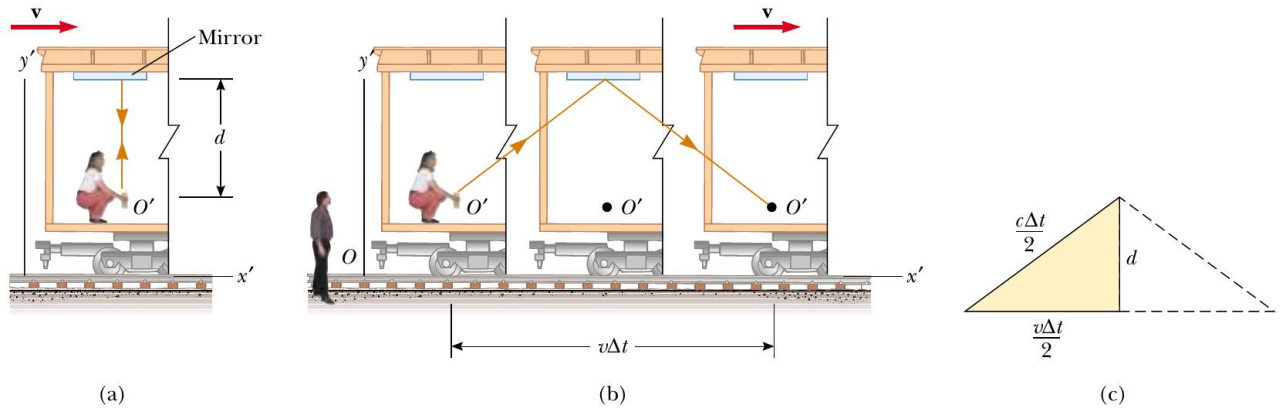


Figure 7.2: (a) A mirror is fixed to a moving vehicle, and a light pulse is sent out by observer O' at rest in the vehicle. (b) Relative to a stationary observer O standing alongside the vehicle, the mirror and O' move with a speed v . (c) The right triangle for calculating the relationship between Δt and $\Delta t'$.

Remember speed of light being a constant c in all inertial systems, the Pythagorean theorem gives

$$\left(\frac{c\Delta t}{2}\right)^2 = \left(\frac{v\Delta t}{2}\right)^2 + d^2.$$

Solving for Δt gives

$$\Delta t = \frac{2d}{\sqrt{c^2 - v^2}} = \frac{2d}{c\sqrt{1 - v^2/c^2}}.$$

Because $\Delta t' = 2d/c$, the expression becomes

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - v^2/c^2}} = \gamma \Delta t' \quad (7.3)$$

where $\gamma = (1 - v^2/c^2)^{-1/2}$ as usual.

Since $\gamma \geq 1$, (7.3) implies to $\Delta t \geq \Delta t'$. Clock that is moving with the system always being observed running slower by other inertial observers O not moving together. Call $\Delta t'$ the *proper time* interval in (7.3) in which the time interval is measured by O' . In other words, proper time is simply the time measured by a clock which is sticking together with the system.

Lorentz contraction

To measure the length of a rigid rod we in principle need to carry a clock to run from one end to the other end of the rod with a uniform speed v relative to the rod, and to see the elapsed time. The length of the rod observed by O' is simply

$$l = v\Delta t'$$

where $\Delta t'$ is the proper time as the observer O' carrying the clock to travel. A second observer O who is at rest relative to the rod can know the reading of the length by

$$l_0 = v\Delta t$$

where Δt is the elapsed time that O observes the time interval of O' moving with a relative speed v . Since $\Delta t = \gamma\Delta t'$ from (7.3), combining the above two equations we have the relation of the lengths

$$l = \frac{l_0}{\gamma} \quad (7.4)$$

where l_0 is the *proper length* of the rod. As $\gamma \geq 1$, the longitudinal length of an object is always measured shorter than in at rest.

For a rigid body, since the transverse dimensions (i.e. the y and z directions) of the body have no change according to Lorentz contraction, the volume $\mathcal{V}$ of the body decreases according to

$$\mathcal{V} = \mathcal{V}_0 \sqrt{1 - \frac{v^2}{c^2}},$$

where $\mathcal{V}$ is the *proper volume* of the body.

Invariances

Some physical quantities such as energy or “length” are supposed to be no change in any situations. We have the postulate:

Physics is required and must be invariant in any inertial systems (in fact, could be applied in any systems).

Example 7.1. The length of a rod should be a constant in any inertial systems. However, according to special relativity different observer in different system will observe different length. Does special relativity conflict to itself?

The usual concept of length is defined by the Pythagorean theorem

$$\begin{aligned} \Delta l &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\ &= \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} \end{aligned}$$

in a three-dimensional space and a one-dimensional **absolute** time (flat) geometry. It is invariant under Galilean transformation:

$$\Delta l = \Delta l'.$$

We have got to define a new quantity of “length” to be invariant under Lorentz transformation.

It is suggested to check that the defined quantity

$$\Delta s^2 = -c^2 \Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$$

is invariant under Lorentz transformation, where c is the speed of light. Thus

$$\Delta s^2 = \Delta s'^2. \quad (7.5)$$

Clearly, this quantity is defined in four-dimensional “spacetime” geometry. Time is no longer a fixed and absolute physical quantity in this formalism. Δs is called *spacetime interval* rather than length. Indeed, the **infinitesimal** interval is known as *metric*:

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad (7.6)$$

It is perfectly fine to define as $ds^2 = c^2 dt^2 - (dx^2 + dy^2 + dz^2)$, as merely a matter of convenient preference so long as satisfying (7.5). Equation (7.6) which in fact is a special form of metric only existing in inertial system, is called *Minkowski metric*.

Suppressing one of the x, y, z coordinate axis in 4-dimensional spacetime and rescaling $t \rightarrow \sqrt{-1}t$, we may “draw” a cone-like diagram as in Figure 7.3 for (7.6) to represent *causality*. Any metrics have these three cases of possibility:

$ds^2 > 0$	space-like
$ds^2 = 0$	light-like or null
$ds^2 < 0$	time-like

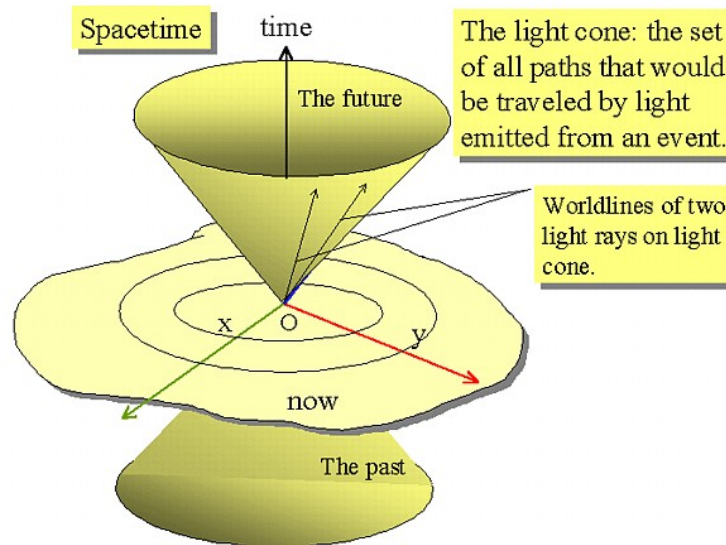


Figure 7.3: It is impossible to draw 4D diagram on a 2D plane. The light ‘cone’ in this diagram is just a convenient way to represent 4D spacetime. In fact, the circles along the T -axis are spheres.

✎. Our any daily activities are inside of the light cone (time-like). Nevertheless light travels on the surface of a light cone (light-like). Event in the space-like region is regarded as unphysical.

7.1.4 Relativistic momentum

The generalization of momentum in special relativity must satisfy:

- Relativistic momentum must conserve in all frames.
- It must reduce to Newtonian momentum at low speeds.

Define

$$\mathbf{p} = \frac{m_0 \mathbf{v}}{\sqrt{1 - \frac{v^2}{c^2}}} = m \mathbf{v} \quad (7.7)$$

where m_0 is the *rest mass*. We should interpret the equation as the mass of a body m increasing as its relative velocity increases.

Einstein's postulate for a system of particles:

The total momentum of a system of particles is always conserved in all frames of reference, whether or not the total number of particles involved is constant.

That is, particles can be created or annihilated.

It turns out that this postulate as well as the definition of relativistic momentum have been experimentally verified.

4-velocity

Velocity is defined by the rate of change of displacement, i.e, in some sense the spatial distance divided by time. In special relativity, space and time are merged into 4-dimensional spacetime with a frame dependent time coordinate. It makes sense to redefine velocity.

Recall the function of the spacetime interval

$$\Delta s^2 = -c^2 \Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2.$$

Dividing above by $\Delta \tau^2$ gives

$$\begin{aligned} \frac{\Delta s^2}{\Delta \tau^2} &= -c^2 \frac{\Delta t^2}{\Delta \tau^2} + \frac{\Delta x^2}{\Delta \tau^2} + \frac{\Delta y^2}{\Delta \tau^2} + \frac{\Delta z^2}{\Delta \tau^2} \\ &= \left(c \frac{\Delta t}{\Delta \tau}, \frac{\Delta \mathbf{x}}{\Delta \tau}, \frac{\Delta \mathbf{y}}{\Delta \tau}, \frac{\Delta \mathbf{z}}{\Delta \tau} \right) \cdot \left(-c \frac{\Delta t}{\Delta \tau}, \frac{\Delta \mathbf{x}}{\Delta \tau}, \frac{\Delta \mathbf{y}}{\Delta \tau}, \frac{\Delta \mathbf{z}}{\Delta \tau} \right) \end{aligned}$$

where τ is the *proper time*, measured by the carrying clock. In the limit $\Delta \tau \rightarrow 0$, we define the *4-velocity*:

$$u := \frac{ds}{d\tau} = \left(c \frac{dt}{d\tau}, \frac{d\mathbf{x}}{d\tau}, \frac{d\mathbf{y}}{d\tau}, \frac{d\mathbf{z}}{d\tau} \right), \quad (7.8)$$

or written in terms of the usual 3-velocity:

$$u = \left(c\gamma, \gamma \frac{d\mathbf{x}}{dt}, \gamma \frac{d\mathbf{y}}{dt}, \gamma \frac{d\mathbf{z}}{dt} \right) = (c\gamma, \gamma \mathbf{v}) \quad (7.9)$$

where $\gamma = dt/d\tau$ is in fact the time dilation formula defined in (7.3) if we identify $\Delta \tau = \Delta t'$.

4-momentum

Having defined the 4-velocity, it follows the definition of *4-momentum*:

$$p := m_0 u = (\gamma m_0 c, \gamma m_0 \mathbf{v}) := \left(\frac{E}{c}, \mathbf{p} \right). \quad (7.10)$$

Noted that a new physical quantity is defined for the time component, namely, $E = \gamma m_0 c^2$. We require the 4-momentum being conserved, the (Lorentz) norm gives

$$p^2 = \left(\frac{E}{c}, \mathbf{p} \right) \cdot \left(\frac{E}{c}, -\mathbf{p} \right) = \frac{E^2}{c^2} - \mathbf{p}^2 \quad (7.11)$$

which ought to be invariant. This is the relativistic equation for the motion of a particle that we expect all particles (with or without mass) obeying it. In view of the time component we has just defined

$$\begin{aligned} E &= \gamma m_0 c^2 \\ &= \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &\approx m_0 c^2 + \frac{1}{2} m_0 v^2 \end{aligned}$$

when $v \ll c$. This gives the famous formula

$$E = m_0 c^2 \quad (7.12)$$

for a particle's energy in the **rest frame**. As an example for the *nuclear fission reaction* we understand that the losing mass is converted into energy via the formula $m_0 c^2$, and all the kinetic energies of the releasing particles are described by the second term $m_0 v^2/2$ of the equaton (except massless particles).

Momentum of a photon

Recall the relativistic momentum conservation law (7.11):

$$p^2 = \frac{E^2}{c^2} - \mathbf{p}^2.$$

In the case of a photon, having no mass $m_0 = 0$ just gives

$$p^2 = \frac{E^2}{c^2}.$$

Using the fact that $E = h\nu$ where h is the Planck's constant, we obtain the momentum for a photon

$$p = \frac{h}{\lambda} \quad (7.13)$$

depending on the wavelength λ of it.

7.2 Atomic and nuclear physics

Atom is considered as the smallest unchangeable component of a chemical element. Unchangeable here means in this case by chemical means; i.e., by reactions with acids or bases or the effect of moderate temperatures, atoms may only be slightly changed, namely, in their degree of *ionization*.

7.2.1 Isotopes

The numbers of neutrons of most of light elements are about the same as their electrons and protons, but sometimes the elements have more or less neutron numbers. Elements with the same nuclear charge number Z and differing atomic mass numbers A are called *isotopes*.

The difference of the atomic number is due to the extra number of neutrons. For example, ${}^{12}_6\text{C}$ and ${}^{14}_6\text{C}$. Two extra neutrons in ${}^{14}_6\text{C}$ atom causes the atom radioactively unstable — a tendency to release neutrons.

The physical techniques for exact measurement of atomic masses and separation of atoms from different masses are called *mass spectroscopy*. These methods usually measure the ratio of charge to mass e/m .

Example 7.2. An isotope of chlorine with mass number $A = 35$ occurs with an abundance of 75.4%; the other stable isotope with $A = 37$ has an abundance of 24.6%. The resulting *relative atomic mass* of the isotope mixture is

$$A_{\text{rel}} = 35.457.$$

Another example, the abundance of hydrogen: 99.986% for ${}^1_1\text{H}$, and 0.014% for ${}^2_1\text{H}$.

Example 7.3. Which pair of the isotopes

$${}^1_1\text{H}, {}^2_1\text{H} \quad \text{and} \quad {}^{235}_{92}\text{U}, {}^{238}_{92}\text{U}$$

are easier to separate?

Solution. The mass difference of the hydrogen pair is 100%, but the uranium pair is 1.25%. Separating isotopes with higher mass ratio is relatively easier than the lower. The hydrogen isotopes are the easiest to be separated.

7.2.2 Photon

Light, which is regarded as a wave, the phenomena were experimentally shown in 17th and 18th centuries. The examples are:

Young's two slits experiment ¹ The interference pattern (constructive and destructive) shows the wave properties (Figure 7.4).

Diffraction experiment Newton's ring

¹It has been ranked the greatest experiment in human science.

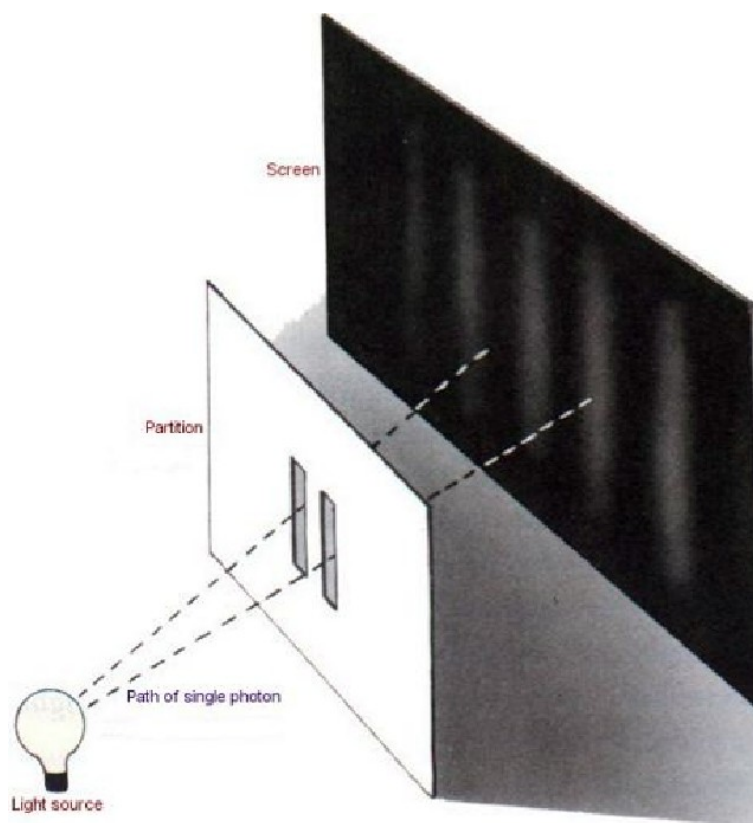


Figure 7.4: The interference pattern formed on the screen must be produced by *coherence* source.

Black body radiation

Quantisation of energy in the interaction of light with matter was postulated for the first time by Planck in 1900, in his theoretical analysis of the spectral distribution of the light emitted by a black body radiator.

☞. Hot objects emit EM radiation as a result of their temperature. The colour which we see in a thermal radiator (furnace) changes from dark red to bright red to yellow to white as the temperature of the furnace is increased.

Black body radiation occurs in a **constant** temperature enclosure. It is **identical** in all respects with the radiation emitted by a body at the same temperature, and which absorbs all the radiation falling on it.

☞. A black body source can be constructed by making a small hole in an enclosure and thereby make measurements of the intensity of the emitted radiation as a function of wavelength and temperature (Figure 7.5).

At thermal equilibrium:

$$\text{radiation emission} = \text{radiation absorption}$$

on the wall of the cavity.

All classical physics failed to produce a formula which would give satisfactory agreement with the black-body radiation experiment. That is, the classical theory would always give an expression **without** a maximum.

In fact, classical physics predicts that the intensity of radiation would increase to infinity as the wavelength going to zero.

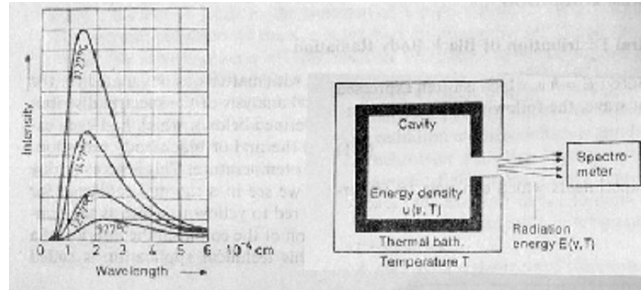


Figure 7.5: Typical measured curves of the intensity distribution in the black-body radiation at various temperature.

Planck's constant

Planck assumed that, at any frequency ν , the energy of the radiation at that frequency could only change by an *integral multiple* of the quantity $h\nu$ where h is a new fundamental constant of nature. He derived

$$\lambda_{\max} \frac{kT}{c} = \frac{h}{4.965}$$

to satisfy the black-body radiation experiment, where the Planck's constant

$$h = (6.626196 \pm 0.000050) \times 10^{-34} \text{ Js.}$$

These discrete packets of energy are called *quanta of energy*.

Photoelectric effect

Einstein assumed that light consists of particles² – light quanta or photons. Each light wave of frequency ν corresponds to a number of photons. In 1905, Einstein proposed that the reason of electrons going out from a metal plate by shining on light is due to the particle nature of light—photon. Photon is treated as a billard ball to knock an electron out from the surface of metal if the energy is strong enough.

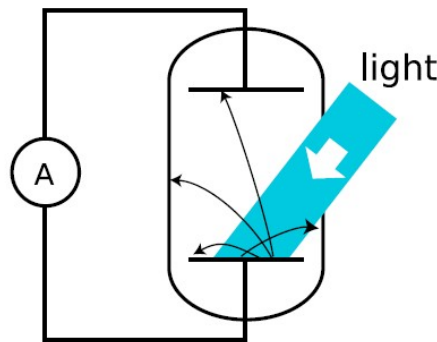


Figure 7.6: Electrons are knocked out by the incident photons.

The energy conservation equation of the system is:

$$h\nu = W + \frac{1}{2}m_e v^2 \quad (7.14)$$

²In fact, Newton postulated the particle version of light in 300 years ago.

where W is the *work function* of the metal plate, ν the frequency of the incident light, m_e and v are the mass and velocity of the knocked out electron.

It should be noted that if the energy of the incident photon is less than the work function W of the metal, no electron emits.

✎. The most important notion of the photoelectric effect is that light possesses **particle nature**. The electrons go out from the metal plate **almost at once** when light is shone on it. The emission of electrons in this extremely short time cannot be explained by the wave version of light.

7.2.3 Duality

The motion of electrons in electric and magnetic fields may initially be understood as a particle motion. However, a number of different experiments in which electrons and other particles show **interference** and **diffraction** phenomena. These imply that electrons and other quantum particles exhibit wave character.

de Broglie's hypothesis

Prince de Broglie suggested that electrons must have a wave character. He postulated that

$$\lambda = \frac{h}{p} \quad (7.15)$$

where p is the momentum of the electron, λ is the corresponding wavelength of the electron, with the Planck's constant h .

Example 7.4. Equation (7.15) is quite similar to (7.13) in special relativity, in fact, the same mathematical formula. Do these equations have the same physical and mathematical foundation?

Solution. Equation (7.15) cannot be mathematically proved. It is a postulate. Unlike the similar looking equation (7.13)

$$p = \frac{h}{\lambda}$$

is derived from special relativity.

Experiments have shown that **all** particles can be made to show wavelike behaviour under suitable circumstances and the wavelength is **always** given by the de Broglie equation (7.15).

✎. The momentum of any macroscopic object is too large comparing to the Planck's constant, it implies that the wavelength would be too small to be observed.

Particle can be viewed as a wave, which in the opposite can be treated as particle. This basic property of matter (everything) is called *wave-particle duality*. Wave is easy to be analysed in classical mechanics but the particle version is comparably difficult and abstract. We require *quantum mechanics* to interpret the behaviour of particle.³

³The basic idea is that quantum means a packet of fixed amount of quantity. In quantum world, anything is discrete.

Electron microscope

Electron has wave properties with wavelengths smaller than visible light.

Electrons with shorter wavelengths can probe a smaller scale of microscopic world that visible light cannot attain. Sharing the same properties of visible light, optical systems like the *electron telescope and electron microscope* can be designed for “electron wave.” To build a “focusing” system of an electron microscope we can use magnetic field to alter the path of electron beams similar to the accelerators in high energy physics. For example, the electrons (from cathode) emitted by the electron gun of the CRT are focused and accelerated by the magnetic field to collide the screen (anode) (Figure 6.5).

7.2.4 Bohr atom

Different elements have different configurations of *energy levels*. Different elements can absorb or emit different sets of photons with particular frequencies according to the elements. Each set of these frequencies is called *spectrum*. For example, spectrum of hydrogen atom is different from the spectrum of helium atom.

✎. We may know the ingredients of some unknown substance from the emission of spectrum. This study is called *spectroscopy*.

Classical physics cannot explain the discrete energy values of the spectral lines. The most difficult one is that orbiting electrons are accelerated charges, they should radiate electromagnetic waves with frequencies equal to their orbital frequencies, $\nu = \omega/2\pi$. They would thus lose energy continuously, i.e., their orbits are unstable and they would be revolving around the nucleus with decreasing radius and finally to collide into the nucleus. Of course, we know that it never happens.

Bohr’s postulates

The electron of a hydrogen atom is restricted to distinct orbits around the nucleus. The perimeters of the orbits are multiple of the wavelength of the electron in some discrete values:

$$\text{Perimeter of the orbit} = n\lambda_e.$$

This atomic model of hydrogen atom is called *Bohr atom*.

Bohr has the postulates for the stability of atom:

- Only certain discrete orbits with the energies E_n of the electrons are allowed.
- The motion of the electrons in these quantised orbits is *radiationless*. An electron can be transferred from an orbit with lower (negative) binding energy E_n (larger r) to an orbit with higher (negative) binding energy $E_{n'}$ (smaller r), emitting radiation by

$$E_n - E_{n'} = h\nu.$$

Light absorption is the reverse process.

- With increasing orbital radius r , the laws of quantum atomic physics become identical with those of classical physics – *correspondence principle*.

Electromagnetic radiation

Energy level of electron:

- The actual structure of an atom is far more complicated than Bohr atom. Only the electron of hydrogen atom at its **lowest** energy state (the orbit closest to the nucleus) forms a shape of sphere.
- Electron can jump to other orbits (states) provided that it absorbs or emits a definite amount of *quantum of energy*.
- Electron at a particular state is responsible of a certain amount of energy level.
- If an electron at a state is capable to escape from the atom, that state is called *ionization state* and the energy level of the state is called *ionization energy*.

An atom with two energy levels E_1, E_2 may interact with EM radiation in three different ways:

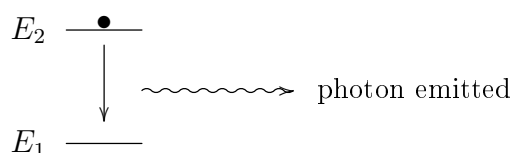
- Absorption of a photon takes the atom from E_1 into the energetically higher level E_2 . A photon of energy

$$\Delta E = E_2 - E_1 = h\nu$$

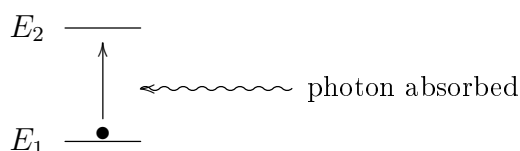
is absorbed from the radiation field.

- Emission occurs from the level E_2 spontaneously within a time (mean lifetime). A photon of energy ΔE is thereby released into the radiation field.
- Photons can also stimulate the emission of further quanta when the atom is in the higher level E_2 .

The idea of absorption and emission of photon:



A photon emitted with energy and frequency $E_2 - E_1 = h\nu$.



$E_2 - E_1$ energy is required for the incoming photon to raise the electron to a higher energy level (state).

Example 7.5. Figure 7.8 shows eight of the possible ways in which an electron in a hydrogen atom could drop from a higher energy state to a state of lower energy, releasing the difference in energy as photon. Of these eight transitions, only D, E and F produce photons with wavelengths in the visible spectrum.

- (a) Which of the visible transitions would be closest to the violet end of the spectrum, and which would be closest to the red end? Explain.

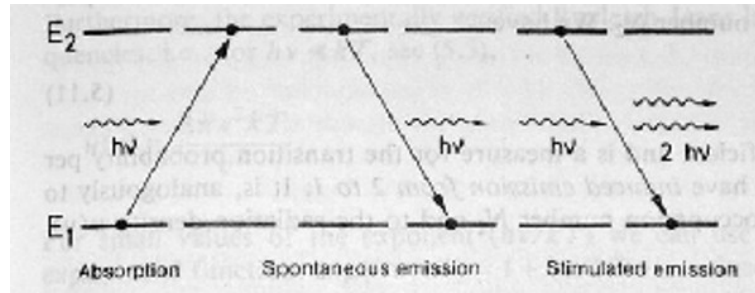


Figure 7.7: Absorption, spontaneous and stimulated emission of radiation with the quantum energy $h\nu$ between two energy levels E_1 and E_2 .

- (b) In what part of the electromagnetic spectrum would the photons from transitions A, B, and C lie? What about G and H? Explain.
- (c) Is there an upper limit to the wavelengths that could be emitted by a hydrogen atom going from one bound state to another bound state? Is there a lower limit? Explain.

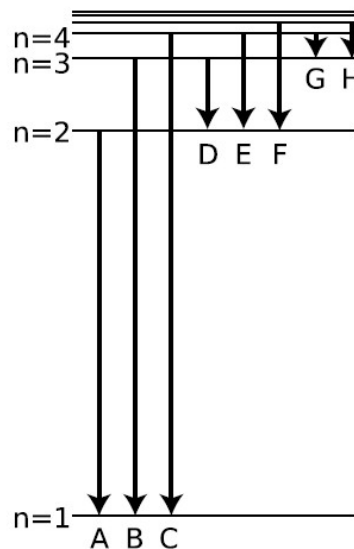


Figure 7.8: Electron transitions of hydrogen atom

- Solution.* (a) In the transitions D, E, and F. The energy difference of the transition F is the highest. It would be closest to the violet end of the spectrum. The transition D possesses the least energy, it would be closest to the red end.
- (b) Transition A could be ultraviolet. B and C could be X-ray and γ -ray if the energies are high enough. The transitions H and G could be infra-red and thermal radiation.
- (c) The upper bound state of the electron is the energy state of ionization of hydrogen atom. The lower limit on the other hand is the state that the electron is closest to the nucleus. Therefore, both upper and lower limits exist, it implies that the wavelengths do not approach to infinitesimal small (or approaching to infinitely high energy).

7.3 Quantum physics

Contrary to the common philosophical thinking, nature always reveals its **discrete** form rather than continuous. The fundamental properties of nature:

Mass is the basic property of matter, depending on the atoms, hence discrete.

Charges The magnitude of any charges would be the integral multiples of the fundamental charge $e/3$.

Spin is similar to the angular momentum of a spinning particle. The magnitude of the angular momentum is also a half-integral or integral multiples of a definite number—Planck's constant.

EM, weak and strong EM radiation is emission of photon. The energy of photon is also discrete in nature. The weak and strong interactions share similar properties of light, the mediators are the Zs-bosons and gluons, respectively. They are discrete as well.

Space, time, and gravity In general relativity, gravity is the curvature of spacetime (connection) . There is not a well established theory or experiment to demonstrate the existence of *quantum* nature of gravity so far. Since everything is in quantum form, it is “reasonable” to assume gravity having quantum nature.

An interesting question that scientists like to ask is why atom is so big.

The diameters of an atom and a nucleus are roughly ‘ 10^{-8} cm and 10^{-13} ’, respectively. It is big because too much space of an atom is empty. It seems that too much space were “wasted” for those tiny electrons and nucleus. On the other hand, electrons and protons carry negative and positive charges, respectively, what keeps the electrons from simply falling into the nucleus?

7.3.1 Heisenberg's uncertainty principle

We have no way to measure the position and momentum of a particle with absolute accuracy simultaneously.

The uncertainty of the momentum and the uncertainty of the position are **complementary**, the product of the two is always larger than a constant:

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} \quad (7.16)$$

where $\hbar = h/2\pi$. The momentum and the position of a particle in an enclosed system is called *conjugate pair*.

Example 7.6. Explain why it is impossible to measure the position and momentum of an electron at the same time (simultaneously) without uncertainty.

Solution. Interaction between *observer* and *observable* must exist for getting information from the observable to the observer. Suppose we want to measure the position of an electron, the most simplest idea is to let light shine on the electron, then the reflected light would give us some information of the states of the electron such as the position. To perform the experiment, at least one photon is required to collide on the electron,

hopefully we could receive the reflected photon for the information. Nevertheless, the “actual” position or the momentum of the electron must be altered during the occurrence of the collision to give us uncertainty.

For obtaining a better measurement of the position, we have to use a shorter wavelength of photon to increase the resolution. Higher energy of the photon by a shorter wavelength easily kicks the electron off from its original position, or resulting to alter the momentum of the electron. On the opposite, in order to not trying to affect the electron momentum by using a longer wavelength photon, we would obtain an unclear vision of image as stated above. Thus, there is no way to measure the position and momentum of an electron precisely at the same time.

The typical examples of conjugate pairs are

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}, \quad \Delta t \Delta E \geq \frac{\hbar}{2}, \quad \Delta \phi \Delta L_z \geq \frac{\hbar}{2}$$

where the pairs t and E are the time and energy, ϕ and L_z are the variation of the angle about the z -axis and the z -direction angular momentum.

Complementarity principle

Bohr introduced the *complementarity principle* in 1928.

The principle states that atomic phenomena cannot be described with the completeness demanded by classical dynamics; some of the elements that complement each other to make up a complete classical description are actually mutually exclusive, and these complementary elements are all necessary for the description of various aspects of the phenomena.

☞. From the point of view of the experimenter, the complementarity principle asserts that the physical apparatus available to him has such properties that more precise measurements than those indicated by the uncertainty principle cannot be made.

7.3.2 Quantum mechanics

Quantum mechanics (QM) is one of the two most fundamental theories in nature, the other is general relativity.

Quantum mechanics is a theory to describe and “predict” the physical behaviours of particles. However, unlike the classical mechanics, the prediction by quantum mechanics is only *probabilistic*.

All the experimental results and applications have been supporting quantum mechanics, demonstrating it to be the most favourable theory to describe particle nature. That is, nature is *undeterministic* and discrete. The two most prominent formalisms of quantum mechanics are

- Wave mechanics (Schrödinger)
- Matrix formalism of quantum mechanics (Heisenberg)

Wave equation

In the wave interpretation, we may formulate a wave equation to describe the behaviour of a particle. For a continuous traveling harmonic wave, the wavelength λ and momentum p , the energy E and frequency ν are related by

$$\begin{aligned} p &= \hbar k, & k &= \frac{2\pi}{\lambda}, \\ E &= \hbar\omega, & \omega &= 2\pi\nu. \end{aligned}$$

The basic requirements to construct the wave equation:

- It must be linear.
- The coefficients of the equation must involve only constants such as $\hbar$ and the mass and charge of the particle.

Schrödinger equation

In QM, a non-relativistic particle can be described by the *Schrödinger equation*:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r}, t)\psi = i\hbar\frac{\partial\psi}{\partial t} \quad (7.17)$$

where

- ψ is called *wavefunction*, it is the amplitude of the wave equation.
- m is the mass of the particle, V is the external potential of the particle encountered. For example, the electromagnetic attraction or repulsion for charge particle. The situation $V = 0$ implies the particle is free.
- ∇ is the 3-D differential operator. For the simple 1-D case, we have

$$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi + V(x, t)\psi = i\hbar\frac{\partial\psi}{\partial t}.$$

⚠. The schrödinger equation is a postulate rather than a deduction.

7.3.3 Schrödinger's cat

There is a very famous thought experiment (Figure 7.18):

1. If we put a cat into a room and locked her alone. It is supposed that there is no window or device to observe the inside.
2. Now if there is a device in the room which can randomly release poisoning material (e.g. radioactive substance) to kill the cat instantly.
3. The point is that we do not know when the device will release poison since the ejection is totally **quantum mechanically random**.

Having set this experiment up, we lock the door and take a break outside for a cup of coffee or something, etc. Then we reckon it is the time to check the experiment, we are going to open the door to see what will happen inside...

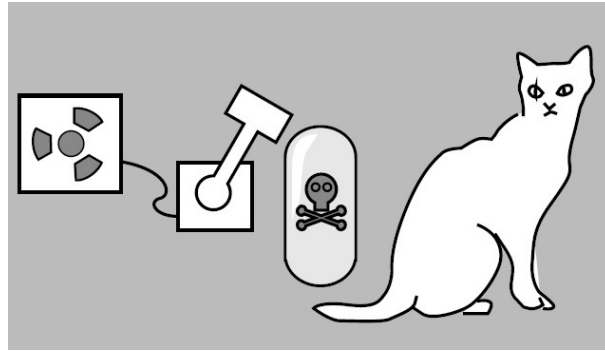


Figure 7.9: Schrödinger's cat

We would probably see either an **alive** cat or a **dead** cat. . . .

Imposing two questions for the experiment:

1. What is the situation of the cat **after** we have checked the room (have been observed)?
2. What is the fate of the cat **before** we check the room (before the observation)?

The first question is just the previous experimental result:

Either an alive cat or a dead cat

But the most interesting question is the second one, a typical answer we would guess immediately is:

Either an alive cat or a dead cat

Why do we consider this silly question?

Superposition principle

In the wave mechanics, the resultant wave amplitude is simply the vector sum of all the amplitudes of the individual waves. This is the simple mathematical idea of superposition principle. In QM, the amplitude of wave can be regarded as a vector or *state vector*, represented by $|\psi\rangle$. Thus, we have in general

$$|\Psi\rangle = a_1|\psi_1\rangle + a_2|\psi_2\rangle + \dots + a_n|\psi_n\rangle$$

for n state vectors, where $a_1, a_2, \dots$ are real numbers for the linear combination. The bracket “ $|\ \rangle$ ” with something in the slot is called ket-vector. There is another symbol called “bra” which is “ $\langle \ |$ ”, they were invented by Paul Dirac for the shorthand notations of vector and co-vector for quantum mechanics. If we put the bra and ket together, it gives a number just similar to the usual dot-product:

$$\langle\psi_1|\psi_2\rangle = \text{number}$$

In the Schrödinger cat experiment, we use $|\text{alive}\rangle$ and $|\text{dead}\rangle$ to represent the two different states⁴ of the cat in the description of QM. Mathematically, both of them are good

⁴Using these kinds of funny brackets to describe quantum states of QM are called *Dirac notations*.

solutions of our system. According to the superposition principle and the mathematical rules of differential equations we have the general solution:

$$|\Psi\rangle = |\text{alive}\rangle + |\text{dead}\rangle \quad (7.18)$$

before observation. It must be stressed that this equation does not implies either $|\text{alive}\rangle$ or $|\text{dead}\rangle$ to be exist. Equation (7.18) simply means

the **co-existence** of $|\text{alive}\rangle$ **and** $|\text{dead}\rangle$!

Schrödinger's cat experiment can be summarised as

$$\begin{aligned} |\Psi\rangle_{\text{before}} &\rightarrow |\Psi\rangle_{\text{after}} \\ |\text{alive}\rangle + |\text{dead}\rangle &\rightarrow \begin{cases} |\text{alive}\rangle \\ \text{or} \\ |\text{dead}\rangle \end{cases} \end{aligned}$$

- The **change of the state** happens at once at the moment **when** we observe the system.
- This abruptly changing of the wavefunction is called the *collapse of wavefunction*. Clearly, this experimental result is highly **subjective**, it totally depends on the experimenter (observer).

7.3.4 Quantum correlation

Two slits experiment

Now consider a Young's two slits experiemnt but this time we only allow a single electron to act as the source. By using the wave interpretation or the actual experimental result, interference pattern always appears on the screen. However, which slit has the electron passed through?

The standard QM answer is that the electron has **passed through both slits A and B at the same time** to produce the interference pattern. This can be written mathematically as

$$|\Psi\rangle = \frac{1}{\sqrt{2}}|\psi_A\rangle + \frac{1}{\sqrt{2}}|\psi_B\rangle$$

where $|\psi_A\rangle$ and $|\psi_B\rangle$ represent the electron states at the slits A and B , respectively. The numerical factors of the states tell us that the probability of the electron will pass through either slit A or B being $(1/\sqrt{2})^2 = 50\%$.

We may check whether the electron really passes through a single or both slits at the same time, or which slit it prefers to. However,

Any observation will disturb the system resulting no interference pattern!

Since when we check the system, we must obtain

$$|\Psi\rangle = |\psi_A\rangle \quad \text{or} \quad |\Psi\rangle = |\psi_B\rangle$$

resulting no interference. This abruptly changed state again is the collapse of wavefunction we have stated.

The basic requirement to do the two slit experiment for interference pattern is that the electrons or waves passing through the slits must be *coherent*. That is, their frequencies, phases of wave packets, etc are the same. That is why we always split a single source into two or more for the experiment. In other words, interference pattern cannot be created by two individual sources—not coherent, or *decoherence*.

Quantum correlation

A normal coin has 2 faces, they are “head” and “tail.” If you pick a coin from your pocket and see the “head” facing to you. What do you reckon the back side of the coin to be?

Tail !

(Unless you play cheating.) This is an example of *classical correlation*.

✎. You know one of the outcome, then you **guess** with 100% certainty to know the other outcome at once as they are *correlated*.

Consider a 2-particles 2-state *entangled system*:

$$|\Psi\rangle = a|\uparrow\rangle_1 \otimes |\downarrow\rangle_2 + b|\downarrow\rangle_1 \otimes |\uparrow\rangle_2. \quad (7.19)$$

If we take a measurement of the system knowing that the first particle (label ‘1’) is at state $|\uparrow\rangle$ then we **immediately** know that the second particle (label ‘2’) is **exactly** at the state $|\downarrow\rangle$ even we do not inspect the second particle.

The point is that this system is not a coin! It consists of two individual particles (e.g. electrons); one may be on the earth, the other could be in a many light-years away galaxy! How could the far away electron know what we got on the earth’s electron then showing the corresponding result? It seems that the electrons are connected via some non-usual-physics linkage, or again, so called *quantum correlation*, which is a typical feature of *entanglement*.

This is one of the controversies that Einstein dislikes quantum mechanics. It seems that it allows something traveling faster than light, the uncertainty somehow turns out to be certain.

Einstein disagrees quantum mechanics and said:

“God does not play dice!”

But the correlation in entangled state Einstein dislikes again and said:

“God does not play dice with electrons!”

Remark 7.1. Equation (7.19) being one of the entangled states of two particles is the fact that the equation cannot be rewritten as the product state of two separate states of particles such as

$$|\Phi\rangle = (\alpha|\uparrow\rangle + \beta|\downarrow\rangle)_1 \otimes (\gamma|\uparrow\rangle + \delta|\downarrow\rangle)_2 \neq |\Psi\rangle.$$

As it is no way to choose suitable values of α, β, γ and δ to make it compatible for (7.19).

7.4 Applications

7.4.1 X-rays

X-ray is EM radiation with very short wavelength.

By the simple formula $E = h\nu$, the high energy of X-rays is very high, with high penetration power as well due to the short wavelengths. To create X-rays, we can use

- a high energetic electron beam to bombard a target,
- X-rays to shine on target to product a new X-rays.

The applications of X-rays:

View internal body organs By injecting into the organ a fluid opaque to X-rays, the walls fo the organ then show up clearly by contrast.

Crystallography The wavelength of X-rays is about at the same order of magnitude of the spacing among atoms in molecules or crystals. We may use diffraction to probe the structure of a test substance.

Radiation therapy A typical example of this is to destroy cancerous tumors without surgery (γ -rays are also used frequently.)

Tomography

Tomography is a method by taking a lot of 2D images at different angles to form a whole 3D image.

The usual X-ray picture does not provide depth information. The image represents the total attenuation as the X-ray beam passes through the object in its path. We may use the idea to take many X-ray images at different sections of internal organ and analyzed by computer program. This is called *X-ray computerized tomography*.

7.4.2 Radioactivity

All kind of elements can release radiations. But radioactive elements implies that some nucleus are not very stable as a result to release radiations. The number of neutrons often determines the stability of the nucleus. There are three kinds of radiation; α , β and γ radiations.

Half-life is the time taken of a radioactive element to decay half of its mass.

There is a great variation in the half-life of radioactive elements. Some decay very quickly and have a half-life of only a few microseonds or less. Others decay slowly with a half-life of many thousands of years.

Example 7.7. Does half-life mean if the half-life of an element is 10 days, then after 20 days that element will disappear?

X-rays, γ -rays and the particles emitted by radioactive nuclei all have energies far greater than the energies that bind electrons to atoms and molecules. As a result, when such radiation penetrates into biological materials, it can rip off electrons from the biological molecules and produce substantial alterations in their structure. The ionized molecule may break up, or it may chemically combine with another molecule to form an undesirable complex.

4. A large dose of radiation may damages so many cells that the whole organism dies. Smaller doses may produce irreversible changes such as mutations, sterility, and cancer.

Raidation therapy

In controlled doses, radiation can be used therapeutically.

In the treatment of certain types of cancer, an ampoule containing radioactive materials is implanted near the cancerous growth, trying to destroy the cancer without greatly damaging the healthy tissue. If long-lived isotopes are used in the therapy, the material must be removed after a prescribed period.

Food preservation

Without preservation, all foods decay rather quickly. The decay is usually caused by microorganisms and enzymes that decompose the organic molecules of the food.

High-energy radiation passing through the food destroys microorganisms that cause decay. Radiation is also effective in destroying small insects that attack stored foods.

γ -rays are used most frequently in food preservation.

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